TECHNICAL SPECIFICATION · REVERSE-ENGINEERING REPORT

Staff Management Compliance Datastore Update

Legacy Excel / VBA / Power Query workflow, reverse-engineered from an operator screen recording, with a target-state architecture for its replacement

Document	Technical Specification — reverse-engineered legacy workflow (as-is) and target architecture (to-be)
Subject workflow	Monthly update of the **Staff Management** area of the Compliance Dashboard data store, via the Excel "user input tool" *Staff Management.xlsm* (VBA + cross-workbook XLOOKUPs) feeding *Compliance Data.xlsm* (Power Query)
Primary evidence	Microsoft Teams meeting-20260811_131142-Meeting Recording (1).mp4 — 1920×1080, 16 fps, 30:54 (1,854 s); operator: Andrew Biggins (Provide Wellbeing); audience: Dheeraj
Secondary evidence	…131142…(1).txt (TurboScribe transcript, untimed, cut at 30 min); …20260811 135933…(1).srt (**different meeting** — see §0.2)
Timestamps	All [HH:MM:SS] values are positions in the .mp4, re-derived by the author with an offline Whisper (small.en/base.en) transcription and frame-accurate seeks; every figure was extracted with ffmpeg at the stated second
Report date	16 October 2026
Status	Draft for business-owner review. Items in §6 and prerequisites in §7.1 must be confirmed before build starts

What the recording shows. A single operator demonstrates, end-to-end, how a new person ("Testy Tester") is onboarded into the staff-compliance data store and how compliance evidence is recorded, amended, refreshed into the dashboard model and finally rolled back. The workflow is *semi-automated*: eight worksheet tabs and four VBA buttons in one workbook read from, and write to, a second workbook through structured-table XLOOKUPs and Power Query.

- **Two workbooks, one network share.** *Staff Management.xlsm* (saved to `Z:\Performance Dashboards and Data\User input\`) is the input tool; *Compliance Data.xlsm* holds `compliance_Data_Store_Table` and the Power Query model the dashboard reads.
- **Three-stage data entry:** (1) register the person on *Names*; (2) declare which of 22 compliance types apply on *Manage Requirements* → appended to `requirements_Table`; (3) log evidence (date, status, RAG, notes) on *Manage Compliance* → written to the external data store, with due date = completion date + "Every x months".
- **Composite keys are string concatenations** (`Name & "-" & Compliance Type`), so a rename or a stray space silently breaks every lookup. There are no surrogate keys, no record IDs and no history version numbers.
- **Seven defects or fragilities are visible on screen**, including: a VBA run-time error 13 on close (29:17); a non-working "last updated" macro; Managed By resolving to `0` until a manual fix; a metric that counts onboarding rows against a 22-type denominator (23/22, 45/22); a Roles table whose header row is a data value; a stale Role field; and an entered "Core" flag that contradicts what the operator said for two compliance types.
- **The provided .srt does not belong to this video** (it covers an *Incidents*/Datix workflow from a different meeting). Elements requested in the brief that come from that workflow — Datix CSV pastes, `#REF!`/`#N/A` fixes, "Copy Data" flags — do **not** occur in this recording. This report documents what is actually on screen and flags the mismatch (§0.2, §6).
- **Build requirements (§7):** 11 business prerequisites, 20 functional and 10 non-functional requirements, technology, migration, testing, roles and phasing. Each one states why it is required and points to the evidence, and §7.9 traces every observed problem to the requirement that removes it.

- **Target design (§5):** a staged-ingest, relational model with surrogate keys, immutable versioned reference data, an append-only compliance-event ledger with derived current-state views, a review gate, and atomic publication batches replacing the save → refresh → save → refresh choreography.

Table of Contents

0. Evidence Base, Method & Reading Guide . . . 5

0.1 Source artefacts . . . 5

0.2 Critical finding: the .srt describes another workflow . . . 5

0.3 Timestamp & screenshot method . . . 5

0.4 Conventions . . . 5

1. Complete File, Workbook & Sheet Inventory . . . 6

1.1 Files and external systems . . . 6

1.2 Worksheet tabs in Staff Management.xlsm . . . 6

1.3 Cross-workbook links, Power Query and VBA . . . 7

1.4 Overview screenshots . . . 7

2. Exhaustive Column-Level Schema & Data Lineage Catalogue . . . 9

2.1 Staff Management.xlsm › Names (name_Table) . . . 9

2.2 Staff Management.xlsm › Compliance Timeframes (Compliance_Timeframe_Table) . . . 12

2.3 Staff Management.xlsm › Roles . . . 14

2.4 Staff Management.xlsm › Requirements (requirements_Table) . . . 15

2.5 Staff Management.xlsm › Manage Requirements (entry form) . . . 16

2.6 Staff Management.xlsm › Manage Compliance (entry form) . . . 17

2.7 Compliance Data.xlsm › compliance_Data_Store_Table (external, partially visible) . . . 17

3. Chronological Step-by-Step Operator Workflow . . . 19

Step 1 [00:00:00 – 00:00:56]: Start recording and share the correct screen . . . 20

Step 2 [00:00:56 – 00:02:34]: Introduce the user-input tool; clear a prior attempt . . . 20

Step 3 [00:02:34 – 00:05:28]: Tour the Names tab (master register) . . . 20

Step 4 [00:05:28 – 00:07:11]: Register new person “Testy Tester” . . . 21

Step 5 [00:07:11 – 00:11:13]: Review Compliance Timeframes (reference catalogue) . . . 22

Step 6 [00:11:13 – 00:12:09]: Explain Power Query build of timeframe tables . . . 22

Step 7 [00:12:09 – 00:12:41]: Glance at Roles . . . 22

Step 8 [00:12:41 – 00:13:11]: Introduce Requirements (per-person applicability) . . . 23

Step 9 [00:13:11 – 00:14:16]: Select the new person on Manage Requirements . . . 23

Step 10 [00:14:16 – 00:14:50]: Run VBA “List Missing” . . . 24

Step 11 [00:14:50 – 00:17:53]: Declare Required / Core for each of the 22 types . . . 25

Step 12 [00:17:53 – 00:18:29]: Run VBA “Add Requirements” . . . 26

Step 13 [00:18:29 – 00:19:14]: Fix Managed By (manual exception) . . . 27

Step 14 [00:19:14 – 00:20:03]: Inspect H–J and save the workbook . . . 28

Step 15 [00:20:03 – 00:23:24]: Enter compliance evidence on Manage Compliance . . . 29

Step 16 [00:23:24 – 00:24:30]: Run VBA “Add New Compliance” . . . 31

Step 17 [00:24:30 – 00:26:18]: Amend an existing record via “Update Existing Compliance” . . . 32

Step 18 [00:26:18 – 00:28:24]: Explain the monthly cycle and start the data refresh . . . 33

Step 19 [00:28:24 – 00:29:13]: Stop the refresh; deactivate the test person; save . . . 34

Step 20 [00:29:13 – 00:29:27]: Close the workbook: VBA run-time error . . . 35

Step 21 [00:29:27 – 00:29:45]: Decline to save; background refresh restarts . . . 36
Step 22 [00:29:45 – 00:30:54]: Stop screen share and wrap up . . . 37
4. Data Quality Checks, Manual Exceptions & Edge Cases . . . 38
4.1 Implicit business rules and keys . . . 38
4.2 Manual exceptions and workarounds observed . . . 38
4.3 Spoken vs. visual discrepancies . . . 38
4.4 Hidden failure modes in the current process . . . 39
5. Target Modern System Architecture Blueprint . . . 40
5.1 Logical flow . . . 40
5.2 Ingestion & preflight gate . . . 40
5.3 Relational data model & identity resolution . . . 40
5.4 Transformation & business-rules engine (Excel/VBA → SQL) . . . 41
5.5 Exception handling & human-in-the-loop controls . . . 41
5.6 Atomic publication & auditability . . . 42
5.7 Migration from the legacy workbooks . . . 42
6. Gap & Ambiguity Register . . . 43
7. What Is Required to Build the Modern System . . . 44
7.1 Prerequisites from the business (before any build starts) . . . 44
7.2 Functional requirements . . . 44
7.3 Non-functional requirements . . . 45
7.4 Technology components . . . 46
7.5 Data migration requirements . . . 46
7.6 Testing and acceptance requirements . . . 46
7.7 People and roles . . . 47
7.8 Delivery phases and exit criteria . . . 47
7.9 Traceability: each observed problem and the requirement that removes it . . . 47
Appendix A. Figure Index . . . 49
Appendix B. Key timestamped utterances (re-transcribed) . . . 50

0. Evidence Base, Method & Reading Guide

0.1 Source artefacts

Artefact	What it is	Used for
…20260811_131142-Meeting Recording (1).mp4	Teams screen recording, 1920×1080, 16 fps, 30:54. Git-LFS object, 130 MB	Primary evidence: all screenshots, all step timings
…20260811_131142-Meeting Recording (1).txt	TurboScribe transcript of the **same** meeting; no timestamps; stops at ~30:00 (free-tier cap)	Wording cross-check of spoken business rules
…20260811 135933-Meeting Recording (1).srt	Timed subtitles from a **different** meeting (13:59:33, ~7 min, Incidents tool)	Not used for anchoring (see §0.2)
Author-generated ASR (Whisper small.en / base.en, VAD-segmented)	Re-transcription of the .mp4 audio with start/end per utterance	Timestamp anchors for every step

0.2 Critical finding: the .srt describes another workflow

The .srt opens with “Hi Dheeraj, it’s me again. So I’m going to make a video now to show how we update the compliance for **incidents**…” and goes on to describe downloading from **Datix**, pasting the export, a category-grouping lookup column and a “hashtag REF” error (srt cues 75–98, 00:04:43–00:06:56). None of this appears in the .mp4, which covers **Staff Management** and never shows Datix, a CSV export, a paste of raw rows or a #REF! error. The file names confirm it: the .mp4 and .txt are **…131142…**, the .srt is **…135933…** (about 48 minutes later).

Consequence: timing cannot be taken from the .srt. All timestamps were rebuilt from the video’s own audio track (offline Whisper; the Hugging Face model host was blocked, so ONNX exports from GitHub were used) and checked against visible screen changes. The Incidents workflow needs its own recording and its own specification (Gap G-01).

0.3 Timestamp & screenshot method

- Frames were pulled with `ffmpeg -ss <t> -frames:v 1` at full 1920×1080 resolution, then cropped to the Excel window (1675×895) or the formula bar (1675×410) so headers and formulas stay legible on A4.
- Calibration: a 1/5 s `fps`-filter sweep sat about 2.5–5 s *later* than its nominal label. All figures were therefore re-seeked and verified frame by frame. Where a transient UI element (dropdown, taskbar preview) lasted under a second, the exact frame was found by image-difference search at 4 fps and 0.2 s steps.
- Readability gate: every figure was checked for (a) edge-energy sharpness and (b) stability against a frame 0.5 s later, so no figure catches a screen redraw, then inspected visually. Figure 1 [00:00:05] (camera) and Figure 44 [00:29:38] (a 0.3 s taskbar hover) are deliberately transient. Figures are numbered in video-time order; a figure is embedded once, and later mentions point back to it.
- Caption format: **Figure N [HH:MM:SS] — File › Tab: content / change**. “BEFORE”/“AFTER” pairs bracket each macro or edit.

0.4 Conventions

- **Table names** appear exactly as they do in formulas: `name_Table`, `requirements_Table`, `Compliance_Timeframe_Table`, `compliance_Data_Store_Table`.
- **Confidence tags** on inferred facts: **[Seen]** read directly on screen; **[Said]** stated by the operator; **[Inferred]** deduced from behaviour; **[Unknown]** not determinable, so logged in §6.

1. Complete File, Workbook & Sheet Inventory

1.1 Files and external systems

#	File / system	Format	Location (as seen)	Role	Evidence
F1	**Staff Management.xlsm**	.xlsm (macro-enabled; AutoSave Off; "Saved to this PC")	`Z:\Performance Dashboards and Data\User input\Staff Management.xlsm`	User-input tool: master lists, requirement set-up, compliance entry UI; hosts the VBA buttons	Figure 29 [00:19:53], Figure 41 [00:29:03], Figure 43 [00:29:33]
F2	**Compliance Data.xlsm**	.xlsm	Not shown; open in a second Excel window (taskbar preview). Same share assumed [Inferred]	The "data store": holds `compliance_Data_Store_Table`; Power Query builds requirements/timeframe tables from every user-input tool; feeds the dashboard	Figure 36 [00:24:03], Figure 44 [00:29:38]
F3	Other "user input tools" (e.g. *Estates*, *Medical Devices*)	.xlsm [Said]	Not shown	Sibling tools with the same Names/Requirements/Timeframes pattern; Power Query pulls them all	Said 00:03:22, 00:27:02
S1	Compliance Dashboard	BI front-end (product not named; gauge visuals, "Active = TRUE" filter)	Not shown	Consumer of the Power Query model; uses Threshold 1–5 for gauge colouring	Said 00:04:27, 00:10:30
S2	GMC register	Website	—	Source of Professional Registration evidence ("downloaded from the GMC today")	Said 00:20:58
S3	VPN	Network	—	Refresh is slow remotely ("five or six seconds in the office")	Said 00:27:19

1.2 Worksheet tabs in Staff Management.xlsm

Tab order as shown on the tab bar (Figure 2 [00:00:53], Figure 16 [00:12:23]): **Instructions · Names · Manage Requirements · Manage Onboarding · Manage Compliance · Roles · Compliance Timeframes · Requirements**. Tab colours: light blue = UI/master data, green = reference, purple = generated table.

Tab	Architectural role	Key objects	Read / written by	Shown at
Instructions	Documentation	—	Not opened	Tab bar only [Gap G-12]
Names	**Master data** (entity register: one row per person)	`name_Table` (A:Q, ~140 rows)	Manual entry; read by Requirements!G, J and by the Manage* dropdowns	02:58–07:03, 19:03, 28:53
Manage Requirements	**Staging / data-entry UI** for requirement set-up	Named cell `add_Requirements_Name` (D6); Add grid C10:F32; Update panel I10:L32; helper cols M:T; buttons *Add Requirements*, *List Missing*, *Update Requirements*	VBA writes to `requirements_Table`	13:23–18:04
Manage Onboarding	Staging UI for onboarding checklist (parallel to Manage Compliance)	Not opened	[Unknown]	Tab bar only [Gap G-12]
Manage Compliance	**Staging / data-entry UI** for compliance evidence	Add grid B8:O…; named cell `update_Co…` (R5, name truncated); Existing panel R:W (cross-workbook lookups); Update panel X:AC; buttons *Add New Compliance*, *Update Existing Compliance*	VBA writes to `Compliance Data.xlsm!compliance_Data_Store_Table`	00:53, 20:13–26:03, 29:23
Roles	Reference / lookup	Unnamed table A1:B22 (Role → Clinical/Non-clinical)	Use unknown ("I'm not sure this is actually used")	12:23
Compliance Timeframes	**Reference** (compliance-type catalogue, frequencies, grouping, thresholds)	`Compliance_Timeframe_Table` (A:O); side lists Q (named `Compliance_Type` [Inferred]) and S (Onboarding)	Manual; read by Requirements!I, Names!P, Manage Compliance (Time Frame), Power Query	07:33–12:09
Requirements	**Generated master table**: person × compliance-type applicability	`requirements_Table` (A:J, ~2,320 rows)	Appended by VBA *Add Requirements*; read by Power Query, Names!O, CT!J, Manage Requirements	12:53, 18:33–19:53

1.3 Cross-workbook links, Power Query and VBA

Mechanism	Trigger	Behaviour observed	Evidence
XLOOKUP into `'Compliance Data.xlsm'!compliance_Data_Store_Table`	Recalculation (needs F2 open)	Existing panel R:W on Manage Compliance shows the stored record for *Name-Type* keys (via `[Unique Lookup 3]`)	Figure 36 [00:24:03]
Power Query (in F2)	Data › Refresh All (twice, with saves between) [Said]	Builds the requirements and compliance-timeframe tables from every *saved* user-input tool; status bar “Running background query… (Click here to cancel)”	Figure 39 [00:27:33]; 27:27–28:35 and from 29:20
VBA *List Missing*	Button, Manage Requirements	Lists the compliance types with no requirement row for the selected name	Figure 21 [00:14:28]
VBA *Add Requirements*	Button	Appends one `requirements_Table` row per filled grid line	Figure 24 [00:18:33]
VBA *Update Requirements*	Button	Overwrites existing requirement rows (not exercised) [Said 13:50]	Figure 20 [00:14:03]
VBA *Add New Compliance*	Button, Manage Compliance	Appends rows to the external data store; clears the grid. “Doesn’t like it if you miss a row” [Said 20:45]	Figure 35 [00:23:53]
VBA *Update Existing Compliance*	Button	Overwrites the matching stored record in place (no history); supports a Delete flag (AC)	Figure 38 [00:26:03]
VBA close / last-updated prompt	Workbook close	Should ask “update the last updated date?”; instead throws **Run-time error 13: Type mismatch**	Figure 42 [00:29:23]
Auto refresh on close	Closing F1 while F2 is open [Said 29:29]	Power Query restarts; lookups “can corrupt” if F2 closes first, so the operator does not save at this point	Figure 43 [00:29:33]

1.4 Overview screenshots

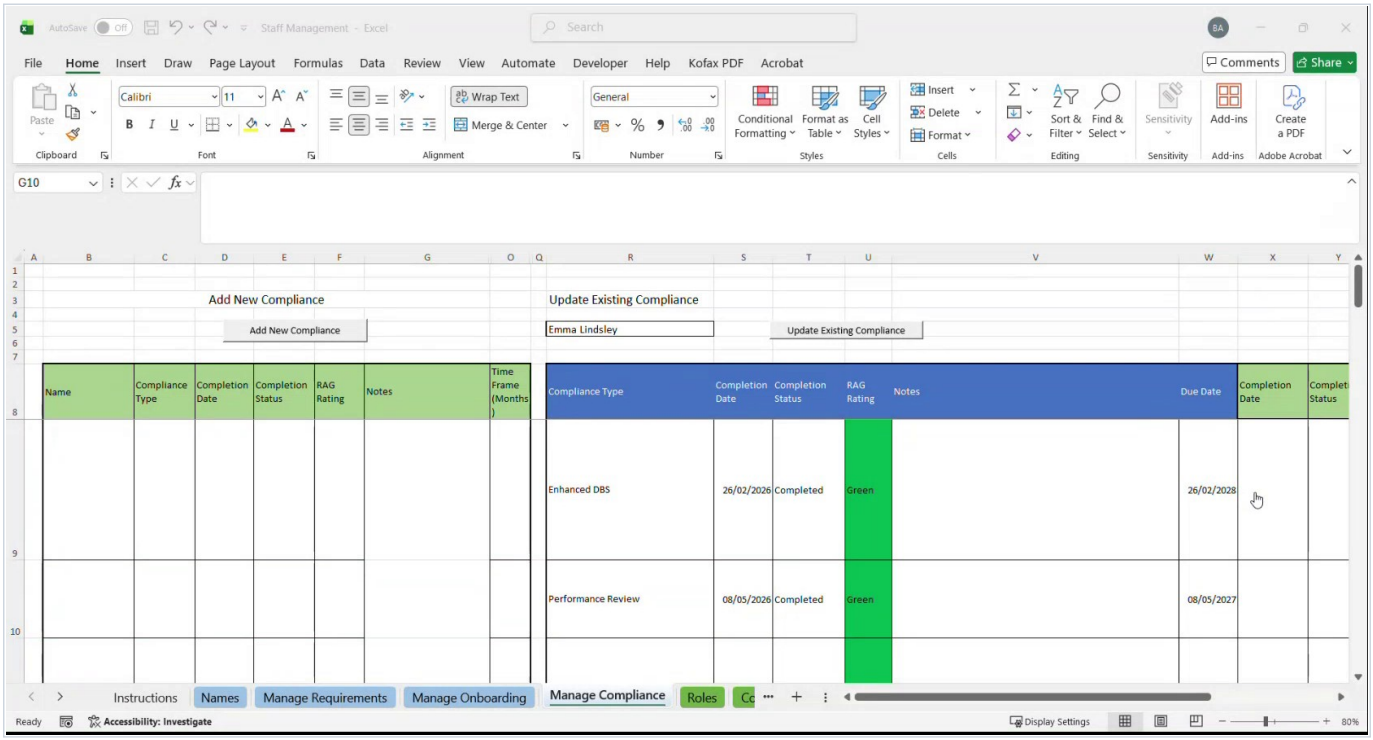


Figure 2 [00:00:53] — Staff Management.xlsm › Manage Compliance: First shared screen. Workbook open on Manage Compliance with a previous person (Emma Lindsley) loaded; full tab bar visible: Instructions, Names, Manage Requirements, Manage Onboarding, Manage Compliance, Roles, Co...

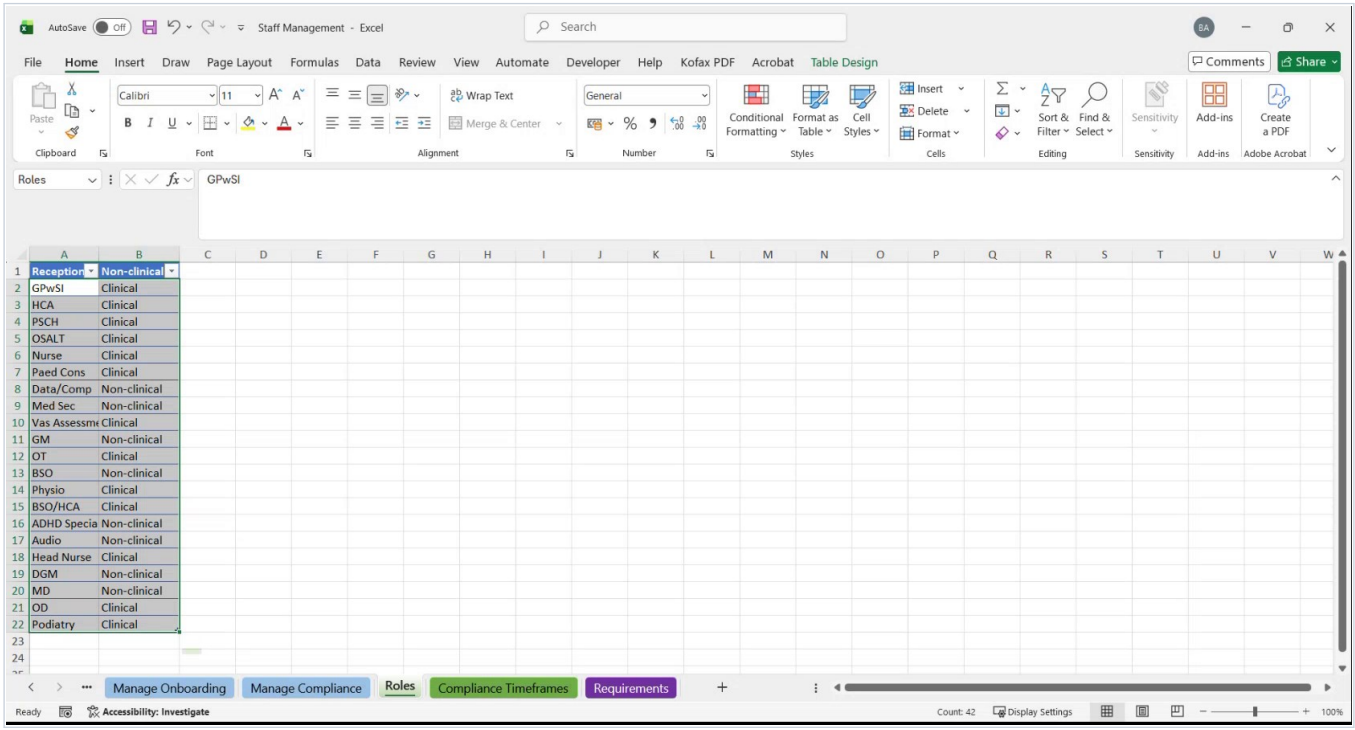


Figure 16 [00:12:23] — Staff Management.xlsm › Roles: Roles table: Role → Clinical/Non-clinical. Header row reads 'Reception | Non-clinical' – a data row has been promoted to the header

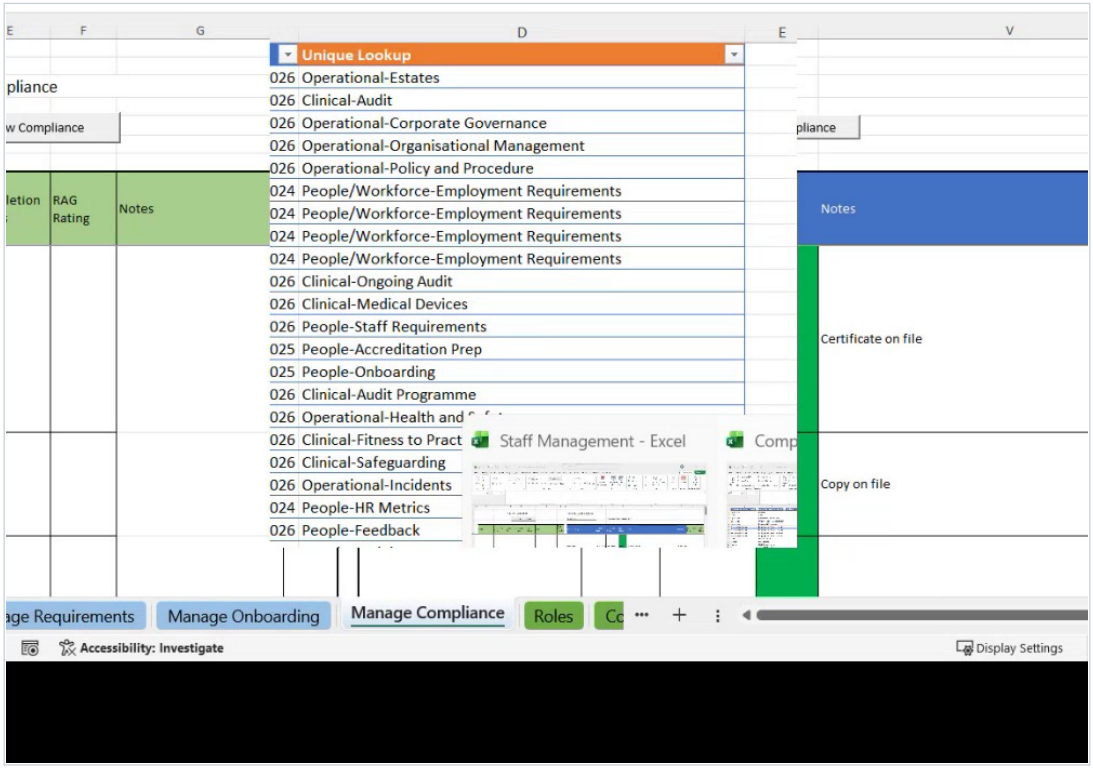


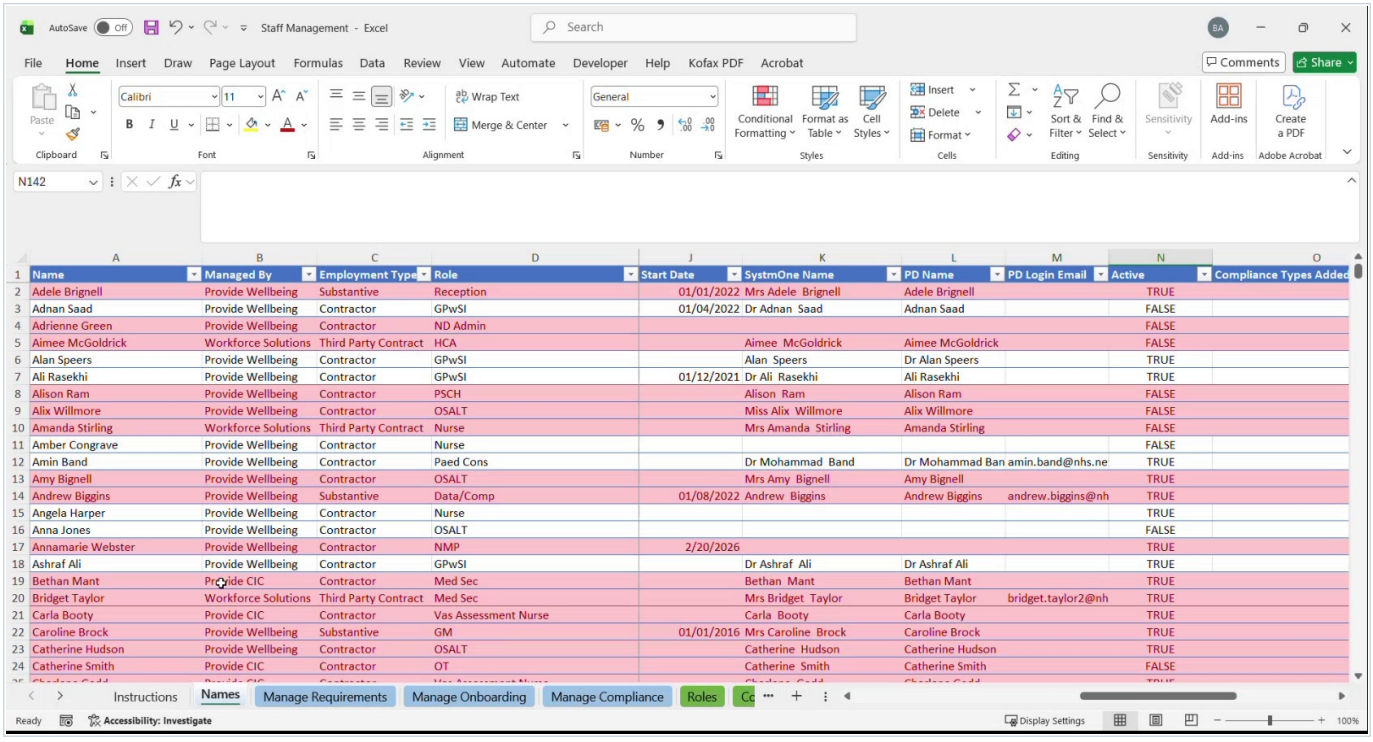
Figure 44 [00:29:38] — Compliance Data.xlsm › (taskbar preview): Brief hover over the Excel taskbar group: a second workbook ('Comp…' = Compliance Data.xlsm) is open, showing a Unique Lookup list prefixed with 024/025/026

2. Exhaustive Column-Level Schema & Data Lineage Catalogue

Origin codes: **RAW** = typed or selected by the operator from an external fact (no upstream system feed exists in this workflow); **CALC** = Excel formula; **VBA** = written by a macro; **REF** = maintained reference value; **PQ** = Power Query output.

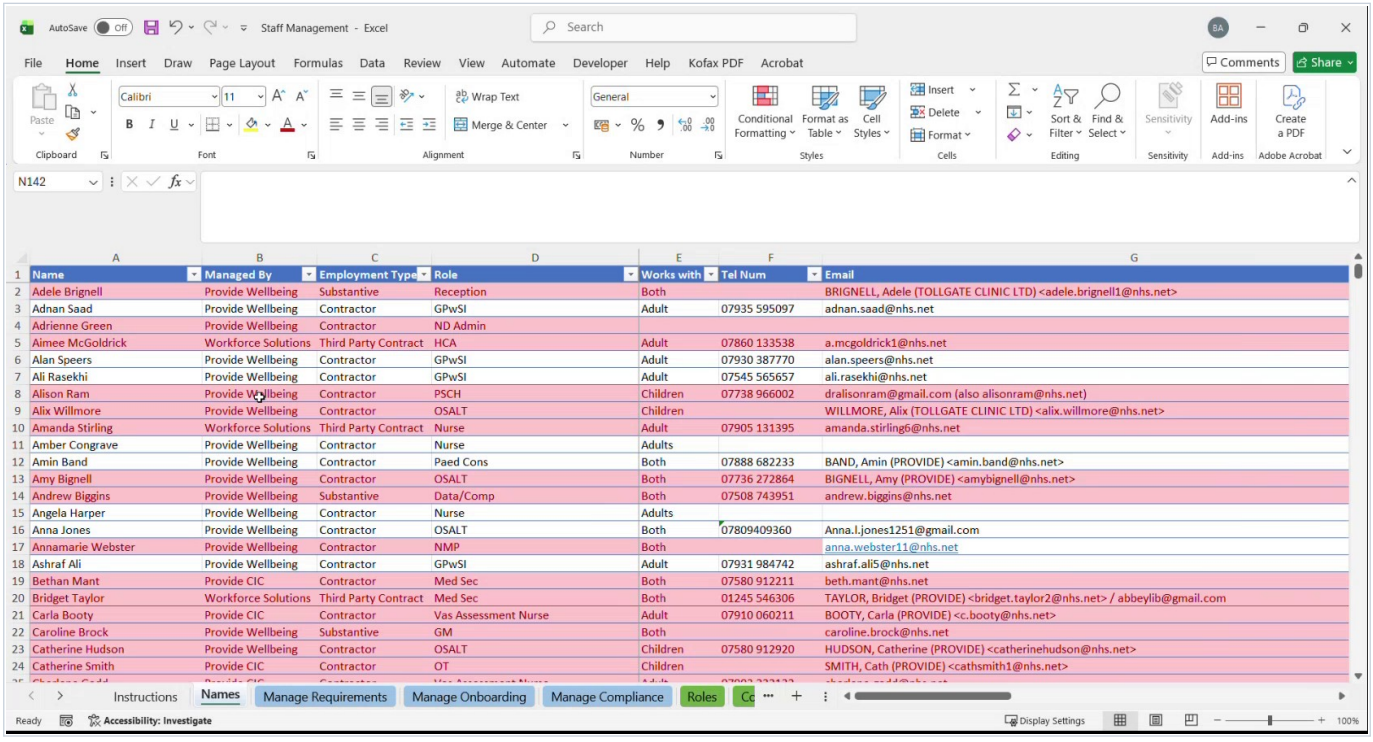
2.1 Staff Management.xlsm › Names (name_Table)

Col	Header	Type / format	Origin	Lineage, rule and evidence
A	Name	Text, free	RAW · key	Natural key used by every lookup (no ID). New rows are appended at the bottom (Testy Tester row 140 after Zoe Porter, so the list is no longer alphabetical). Spelling must match exactly across workbooks. Figure 9 [00:05:43]
B	Managed By	Text, dropdown (Provide Wellbeing / Provide CIC / Workforce Solutions)	RAW	Feeds `requirements_Table[Managed By]`. Left blank for the new starter, so Requirements showed **0** until fixed at 19:03. Figure 26 [00:19:03]
C	Employment Type	Text, list (Substantive / Contractor / Third Party Contract)	RAW	Informational [Said 04:09]
D	Role	Text, list (GPwSI, Nurse, OSALT, BSO, Med Sec, Paed Cons, …)	RAW	Informational; values match *Roles*!A (relationship not enforced)
E	Works with	Text (Adult / Children / Both / Adults)	RAW	Informational. Note “Adult” vs “Adults”: uncontrolled vocabulary. Figure 4 [00:03:13]
F	Tel Num	Text	RAW	Informational, mixed formats
G	Email	Text	RAW	Informational; some cells hold several addresses or “NAME (PROVIDE) <email>” strings
H	Contract	Text (e.g. Permanent)	RAW	Seen at 28:53 only (Figure 40 [00:28:53]); sparsely populated
I	WTE	Number	RAW	Whole-time equivalent; sparse
J	Start Date	Date	RAW	Typed for Testy Tester as 1 Aug; displays **8/1/2026** (m/d) while other rows show dd/mm/yyyy, so the format is inconsistent [Gap G-09]. Not used by any visible formula
K	SystmOne Name	Text	RAW	Alias in the clinical system (e.g. “Dr Ashraf Ali”). Informational
L	PD Name	Text	RAW	Alias in “PD” (an unnamed system). Informational
M	PD Login Email	Text	RAW	Informational
N	Active	Boolean TRUE/FALSE	RAW · **essential**	Dashboard filter: only Active = TRUE people are shown [Said 04:16]. Propagated to `requirements_Table[Active]`. Set FALSE at 29:01 to retire the test record. Figure 5 [00:04:23]
O	Compliance Types Added to Record	Integer	CALC	`=COUNTIFS(requirements_Table[Name],[@Name])`. Counts *all* requirement rows, onboarding ones included, so values of 23 and 45 appear against a denominator of 22. Figure 6 [00:04:53]
P	Total Compliance Types	Integer (22)	CALC	`=COUNTIFS(Compliance_Type,"<>")`: non-blank count of the 22-item list on Compliance Timeframes!Q. Figure 7 [00:05:03]
Q	Name if Active	Text	CALC	`=IF([@Active]=TRUE,[@Name],"")`: probably the source of the active-name dropdowns [Inferred]. Operator: “wasn’t actually needed… ignore that” [Said 05:14]. Figure 8 [00:05:23]



Name	Managed By	Employment Type	Role	Start Date	SystmOne Name	PD Name	PD Login Email	Active	Compliance Types Added
Adele Brignell	Provide Wellbeing	Substantive	Reception	01/01/2022	Mrs Adele Brignell	Adele Brignell		TRUE	
Adnan Saad	Provide Wellbeing	Contractor	GPwSI	01/04/2022	Dr Adnan Saad	Adnan Saad		FALSE	
Adrienne Green	Provide Wellbeing	Contractor	ND Admin					FALSE	
Aimee McGoldrick	Workforce Solutions	Third Party Contract	HCA		Aimee McGoldrick	Aimee McGoldrick		FALSE	
Alan Speers	Provide Wellbeing	Contractor	GPwSI		Alan Speers	Dr Alan Speers		TRUE	
Ali Rasekhi	Provide Wellbeing	Contractor	GPwSI	01/12/2021	Dr Ali Rasekhi	Ali Rasekhi		TRUE	
Alison Ram	Provide Wellbeing	Contractor	PSCH		Alison Ram	Alison Ram		FALSE	
Alix Willmore	Provide Wellbeing	Contractor	OSALT		Miss Alix Willmore	Alix Willmore		FALSE	
Amanda Stirling	Workforce Solutions	Third Party Contract	Nurse		Mrs Amanda Stirling	Amanda Stirling		FALSE	
Amber Congrave	Provide Wellbeing	Contractor	Nurse					FALSE	
Amin Band	Provide Wellbeing	Contractor	Paed Cons		Dr Mohammad Band	Dr Mohammad Ban	amin.band@nhs.ne	TRUE	
Amy Bignell	Provide Wellbeing	Contractor	OSALT		Mrs Amy Bignell	Amy Bignell		TRUE	
Andrew Biggins	Provide Wellbeing	Substantive	Data/Comp	01/08/2022	Andrew Biggins	Andrew Biggins	andrew.biggins@nh	TRUE	
Angela Harper	Provide Wellbeing	Contractor	Nurse					TRUE	
Anna Jones	Provide Wellbeing	Contractor	OSALT					FALSE	
Annamarie Webster	Provide Wellbeing	Contractor	NMP	2/20/2026				TRUE	
Ashraf Ali	Provide Wellbeing	Contractor	GPwSI		Dr Ashraf Ali	Dr Ashraf Ali		TRUE	
Bethan Mant	Provide CIC	Contractor	Med Sec		Bethan Mant	Bethan Mant		TRUE	
Bridget Taylor	Workforce Solutions	Third Party Contract	Med Sec		Mrs Bridget Taylor	Bridget Taylor	bridget.taylor2@nh	TRUE	
Carla Booty	Provide CIC	Contractor	Vas Assessment Nurse		Carla Booty	Carla Booty		TRUE	
Caroline Brock	Provide Wellbeing	Substantive	GM	01/01/2016	Mrs Caroline Brock	Caroline Brock		TRUE	
Catherine Hudson	Provide Wellbeing	Contractor	OSALT		Catherine Hudson	Catherine Hudson		TRUE	
Catherine Smith	Provide CIC	Contractor	OT		Catherine Smith	Catherine Smith		FALSE	

Figure 3 [00:02:58] — Staff Management.xlsm › Names: Names table (name_Table), columns A–D and I–N: Name, Managed By, Employment Type, Role, WTE, Start Date, SystmOne Name, PD Name, PD Login Email, Active



Name	Managed By	Employment Type	Role	Works with	Tel Num	Email
Adele Brignell	Provide Wellbeing	Substantive	Reception	Both		BRIGNELL, Adele (TOLLGATE CLINIC LTD) <adele.brignell1@nhs.net>
Adnan Saad	Provide Wellbeing	Contractor	GPwSI	Adult	07935 595097	adnan.saad@nhs.net
Adrienne Green	Provide Wellbeing	Contractor	ND Admin			
Aimee McGoldrick	Workforce Solutions	Third Party Contract	HCA	Adult	07860 133538	a.mcgoldrick1@nhs.net
Alan Speers	Provide Wellbeing	Contractor	GPwSI	Adult	07930 387770	alan.speers@nhs.net
Ali Rasekhi	Provide Wellbeing	Contractor	GPwSI	Adult	07545 565657	ali.rasekhi@nhs.net
Alison Ram	Provide Wellbeing	Contractor	PSCH	Children	07738 966002	dralisonram@gmail.com (also alisonram@nhs.net)
Alix Willmore	Provide Wellbeing	Contractor	OSALT	Children		WILLMORE, Alix (TOLLGATE CLINIC LTD) <alix.willmore@nhs.net>
Amanda Stirling	Workforce Solutions	Third Party Contract	Nurse	Adult	07905 131395	amanda.stirling6@nhs.net
Amber Congrave	Provide Wellbeing	Contractor	Nurse	Adults		
Amin Band	Provide Wellbeing	Contractor	Paed Cons	Both	07888 682233	BAND, Amin (PROVIDE) <amin.band@nhs.net>
Amy Bignell	Provide Wellbeing	Contractor	OSALT	Both	07736 272864	BIGNELL, Amy (PROVIDE) <amybignell@nhs.net>
Andrew Biggins	Provide Wellbeing	Substantive	Data/Comp	Both	07508 743951	andrew.biggins@nhs.net
Angela Harper	Provide Wellbeing	Contractor	Nurse	Adults		
Anna Jones	Provide Wellbeing	Contractor	OSALT	Both	07809409360	Anna.l.jones1251@gmail.com
Annamarie Webster	Provide Wellbeing	Contractor	NMP	Both		anna.webster11@nhs.net
Ashraf Ali	Provide Wellbeing	Contractor	GPwSI	Adult	07931 984742	ashraf.ali5@nhs.net
Bethan Mant	Provide CIC	Contractor	Med Sec	Both	07580 912211	beth.mant@nhs.net
Bridget Taylor	Workforce Solutions	Third Party Contract	Med Sec	Both	01245 546306	TAYLOR, Bridget (PROVIDE) <bridget.taylor2@nhs.net> / abbeylib@gmail.com
Carla Booty	Provide CIC	Contractor	Vas Assessment Nurse	Adult	07910 060211	BOOTY, Carla (PROVIDE) <c.booty@nhs.net>
Caroline Brock	Provide Wellbeing	Substantive	GM	Both		caroline.brock@nhs.net
Catherine Hudson	Provide Wellbeing	Contractor	OSALT	Children	07580 912920	HUDSON, Catherine (PROVIDE) <catherinehudson@nhs.net>
Catherine Smith	Provide CIC	Contractor	OT	Children		SMITH, Cath (PROVIDE) <cathsmith1@nhs.net>

Figure 4 [00:03:13] — Staff Management.xlsm › Names: Names table scrolled to columns E–G: Works with, Tel Num, Email (free-text contact data, mixed formats)

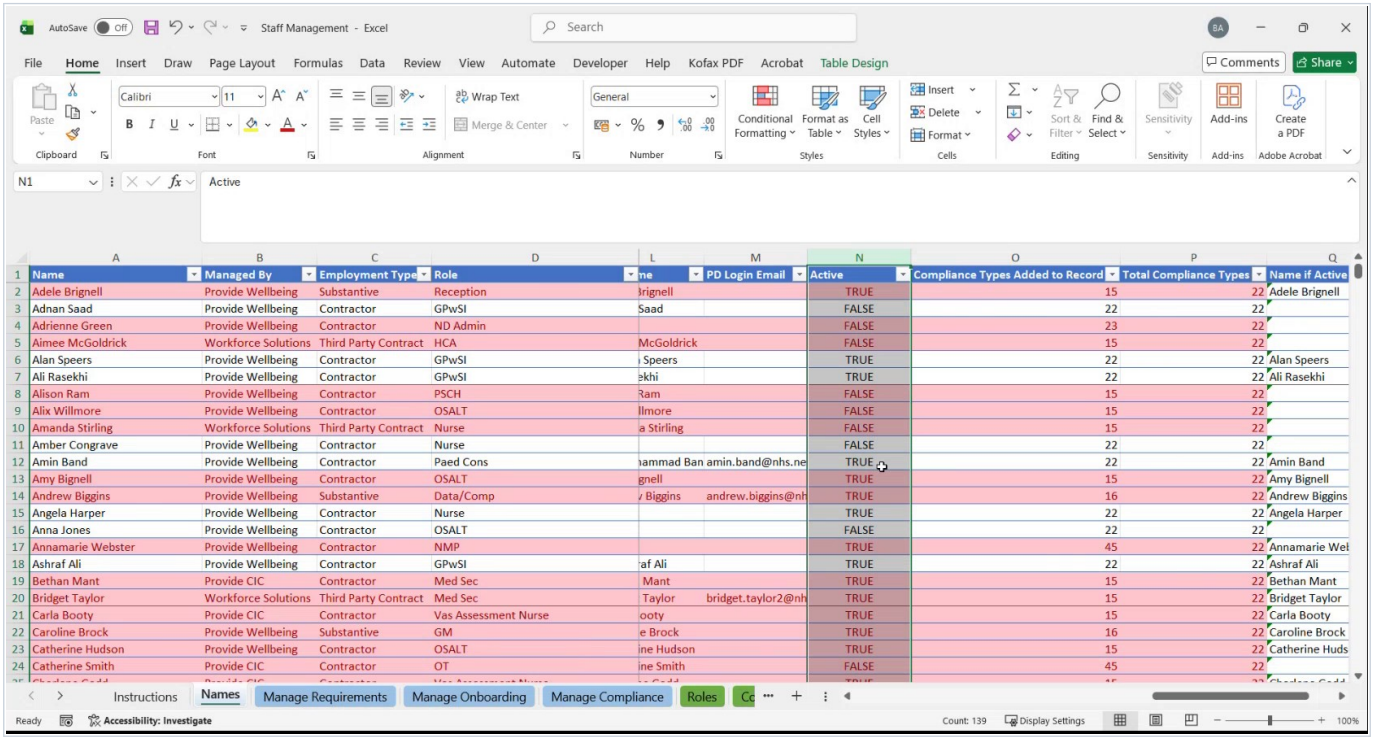


Figure 5 [00:04:23] — Staff Management.xlsm › Names: Column N Active (TRUE/FALSE) selected; O = Compliance Types Added to Record, P = Total Compliance Types, Q = Name if Active. Note O values 23 and 45 exceed P = 22

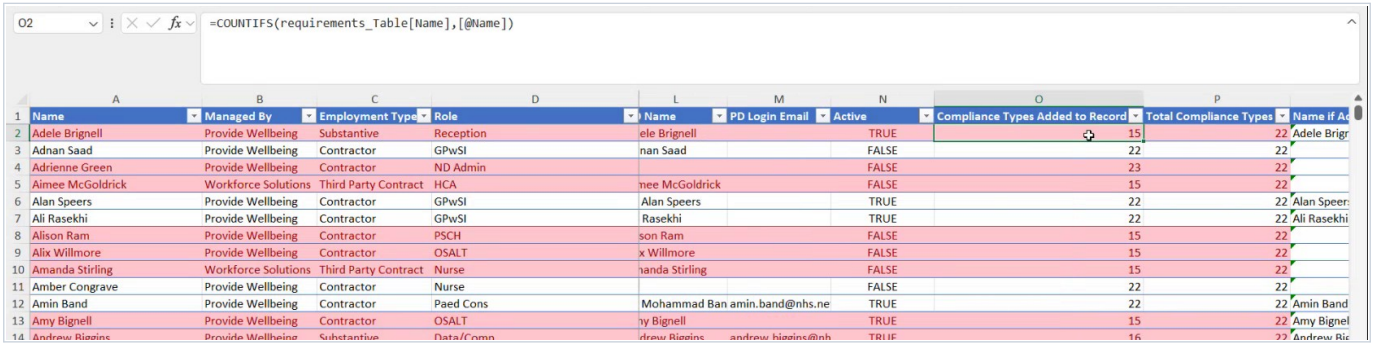


Figure 6 [00:04:53] — Staff Management.xlsm › Names: Formula bar, O2: =COUNTIFS(requirements_Table[Name],[@Name])

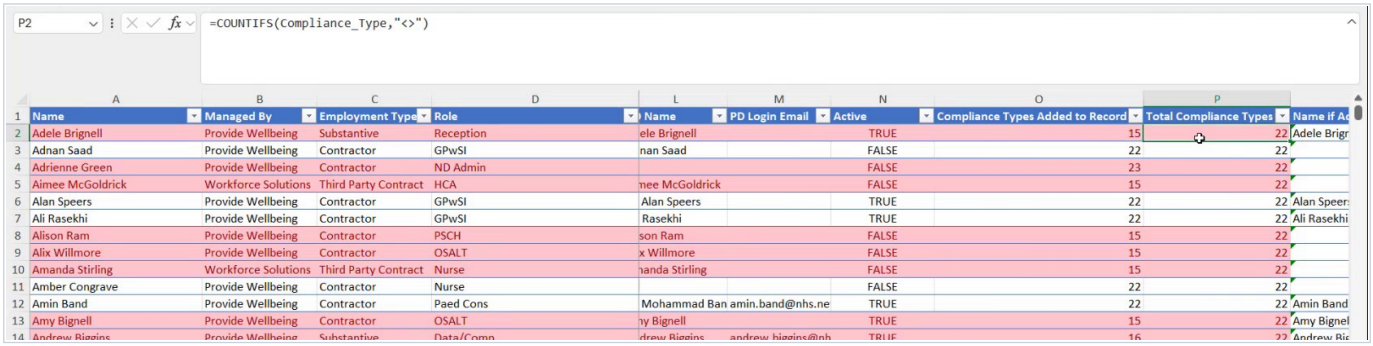
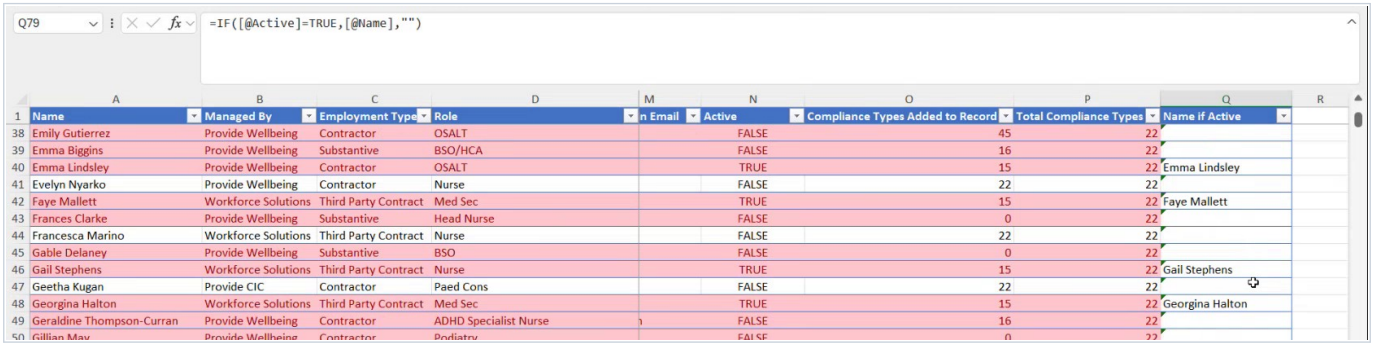


Figure 7 [00:05:03] — Staff Management.xlsm › Names: Formula bar, P2: =COUNTIFS(Compliance_Type,"<>") – counts the 22-item named list Compliance_Type

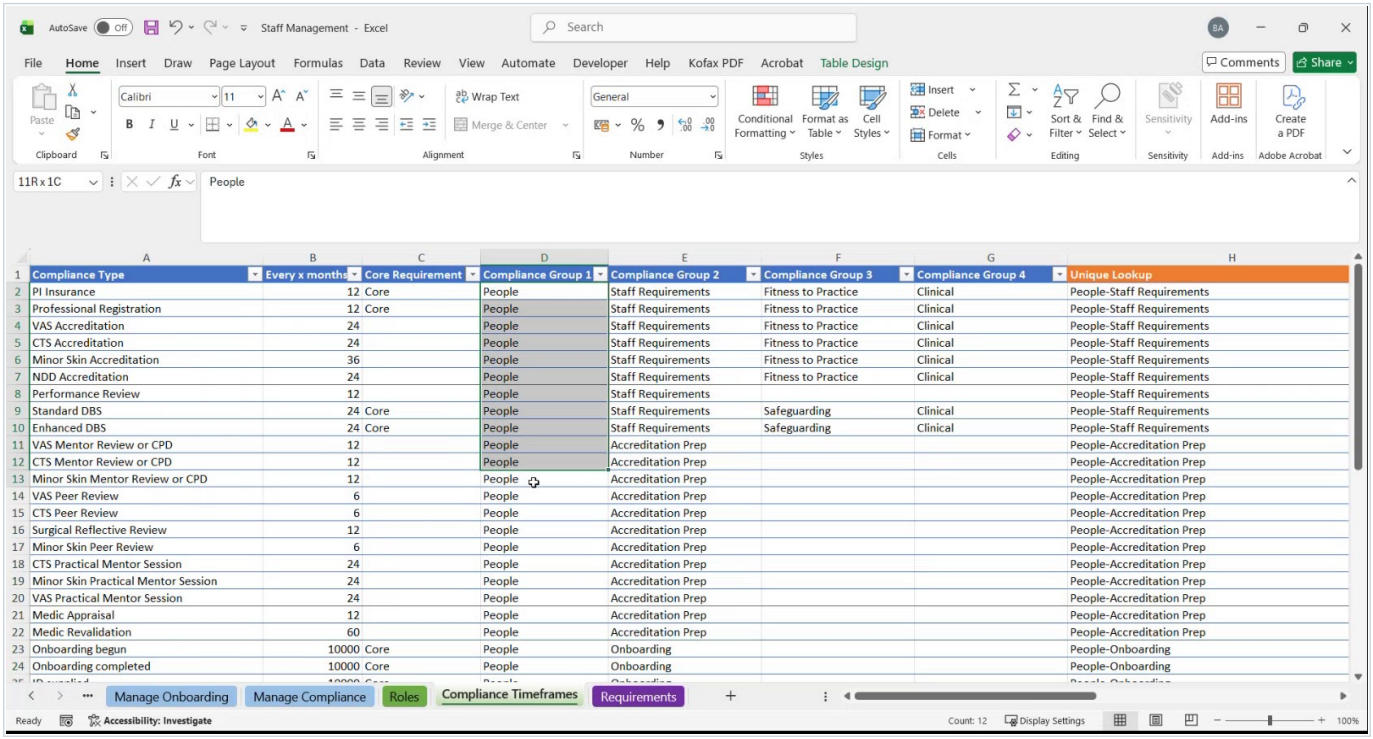


	A	B	C	D	M	N	O	P	Q	R
1	Name	Managed By	Employment Type	Role	n Email	Active	Compliance Types Added to Record	Total Compliance Types	Name if Active	
38	Emily Gutierrez	Provide Wellbeing	Contractor	OSALT		FALSE	45	22		
39	Emma Biggins	Provide Wellbeing	Substantive	BSO/HCA		FALSE	16	22		
40	Emma Lindsley	Provide Wellbeing	Contractor	OSALT		TRUE	15	22	Emma Lindsley	
41	Evelyn Nyarko	Provide Wellbeing	Contractor	Nurse		FALSE	22	22		
42	Faye Mallett	Workforce Solutions	Third Party Contract	Med Sec		TRUE	15	22	Faye Mallett	
43	Frances Clarke	Provide Wellbeing	Substantive	Head Nurse		FALSE	0	22		
44	Francesca Marino	Workforce Solutions	Third Party Contract	Nurse		FALSE	22	22		
45	Gable Delaney	Provide Wellbeing	Substantive	BSO		FALSE	0	22		
46	Gail Stephens	Workforce Solutions	Third Party Contract	Nurse		TRUE	15	22	Gail Stephens	
47	Geetha Kugan	Provide CIC	Contractor	Paed Cons		FALSE	22	22		
48	Georgina Halton	Workforce Solutions	Third Party Contract	Med Sec		TRUE	15	22	Georgina Halton	
49	Geraldine Thompson-Curran	Provide Wellbeing	Contractor	ADHD Specialist Nurse		FALSE	16	22		
50	Gillian May	Provide Wellbeing	Contractor	Podiatry		FALSE	0	22		

Figure 8 [00:05:23] — Staff Management.xlsm › Names: Formula bar, Q79: =IF([@Active]=TRUE,[@Name],"")

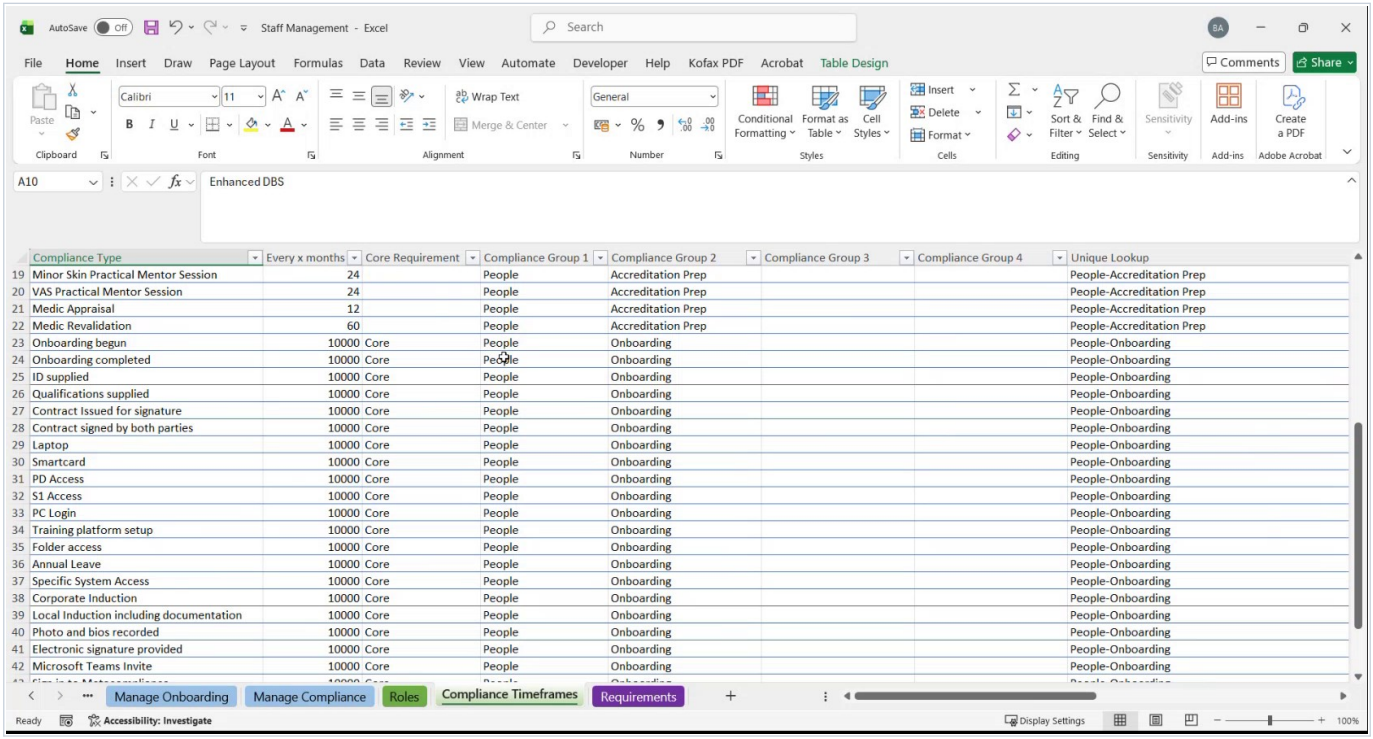
2.2 Staff Management.xlsm › Compliance Timeframes (Compliance_Timeframe_Table)

Col	Header	Type / format	Origin	Lineage, rule and evidence
A	Compliance Type	Text · key	REF	Catalogue key: 22 staff types (rows 2–22) plus onboarding items (rows 23→, “Onboarding begun”, “ID supplied”, “Laptop”, …)
B	Every x months	Integer	REF	Review frequency: 6, 12, 24, 36, 60. **10000** = “never expires” sentinel for onboarding. Drives the due date (completion + B months) and `requirements_Table[Compliance Timeframe]`
C	Core Requirement	Text (Core / blank)	REF	Catalogue-level default. Operator disputes the values (accreditation and performance review “should be core”) [Said 08:32]. The per-person value on Requirements!D overrides it
D	Compliance Group 1	Text (People / Operational / Clinical)	REF	Top-level dashboard grouping; all staff types = People
E	Compliance Group 2	Text (Staff Requirements / Accreditation Prep / Onboarding)	REF	Sub-group. “Accreditation Prep” was a proposed change [Said 08:06]
F	Compliance Group 3	Text (Fitness to Practice / Safeguarding / blank)	REF	Cross-cutting grouping (e.g. PI insurance also affects the Clinical score). Future of groups 3–4 is for Richard to decide [Said 09:17–09:54]
G	Compliance Group 4	Text (Clinical / blank)	REF	As F
H	Unique Lookup	Text	CALC	`Group 1 & "-" & Group 2`, e.g. “People-Staff Requirements” [Inferred from values; formula not shown]. Pulled into PQ tables [Said 10:06]
I	Unique Lookup 2	Text	CALC	`=IF(CONCAT([@[Compliance Group 4]],"-",[@[Compliance Group 3]])="-","",CONCAT(…))`, e.g. “Clinical-Fitness to Practice”. Figure 13 [00:10:03]
J	Missing	Boolean	CALC	`=XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),requirements_Table[Unique Lookup],requirements_Table[Unique Lookup],"")=""`: TRUE if the person *currently selected* on Manage Requirements lacks this type. UI-state-dependent; probably what *List Missing* reads [Inferred]. Figure 14 [00:10:23]
K–O	Threshold 1 … Threshold 5	Decimal 0–1	REF	Gauge colour bands: 1 / 0.9 / 0.8 / 0.7 / 0.6 (DBS rows 1 / 0.98 / 0.9 / 0.8 / 0.7). ≥0.9 green, <0.6 red, graded between [Said 10:40]. Blank for onboarding rows
P	(blank spacer)	—	—	—
Q	Compliance Type (list)	Text, 22 items	REF	Stand-alone list PI Insurance … **IR35 Compliant**. Probably the named range `Compliance_Type` used by Names!P [Inferred]. IR35 Compliant is in this list but not in table rows 2–22 [Gap G-05]
S	Onboarding (list)	Text, ~24 items	REF	Onboarding checklist items (Onboarding begun … Prescribing)



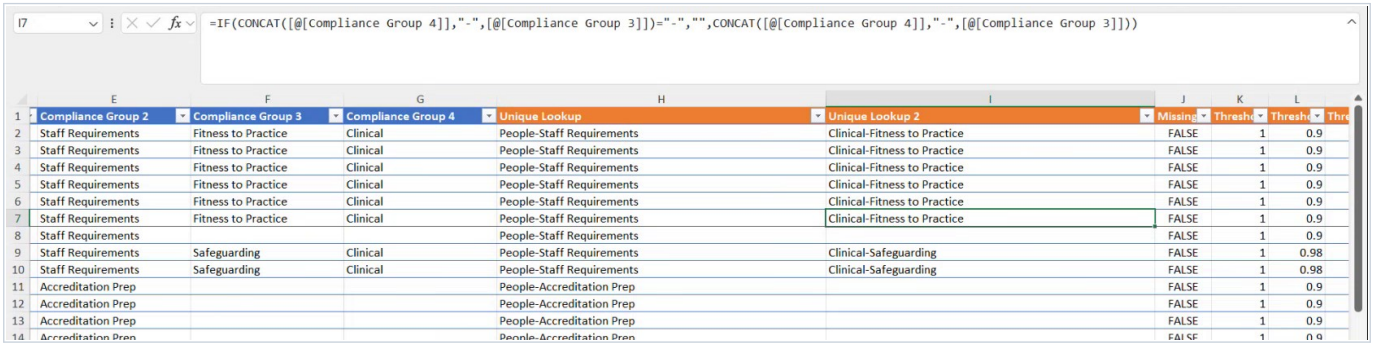
Compliance Type	Every x months	Core Requirement	Compliance Group 1	Compliance Group 2	Compliance Group 3	Compliance Group 4	Unique Lookup
PI Insurance	12	Core	People	Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements
Professional Registration	12	Core	People	Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements
VAS Accreditation	24		People	Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements
CTS Accreditation	24		People	Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements
Minor Skin Accreditation	36		People	Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements
NDD Accreditation	24		People	Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements
Performance Review	12		People	Staff Requirements			People-Staff Requirements
Standard DBS	24	Core	People	Staff Requirements	Safeguarding	Clinical	People-Staff Requirements
Enhanced DBS	24	Core	People	Staff Requirements	Safeguarding	Clinical	People-Staff Requirements
VAS Mentor Review or CPD	12		People	Accreditation Prep			People-Accreditation Prep
CTS Mentor Review or CPD	12		People	Accreditation Prep			People-Accreditation Prep
Minor Skin Mentor Review or CPD	12		People	Accreditation Prep			People-Accreditation Prep
VAS Peer Review	6		People	Accreditation Prep			People-Accreditation Prep
CTS Peer Review	6		People	Accreditation Prep			People-Accreditation Prep
Surgical Reflective Review	12		People	Accreditation Prep			People-Accreditation Prep
Minor Skin Peer Review	6		People	Accreditation Prep			People-Accreditation Prep
CTS Practical Mentor Session	24		People	Accreditation Prep			People-Accreditation Prep
Minor Skin Practical Mentor Session	24		People	Accreditation Prep			People-Accreditation Prep
VAS Practical Mentor Session	24		People	Accreditation Prep			People-Accreditation Prep
Medic Appraisal	12		People	Accreditation Prep			People-Accreditation Prep
Medic Revalidation	60		People	Accreditation Prep			People-Accreditation Prep
Onboarding begun	10000	Core	People	Onboarding			People-Onboarding
Onboarding completed	10000	Core	People	Onboarding			People-Onboarding

Figure 11 [00:07:33] — Staff Management.xlsm › Compliance Timeframes: Compliance_Timeframe_Table columns A–H: Compliance Type, Every x months, Core Requirement, Compliance Group 1–4, Unique Lookup



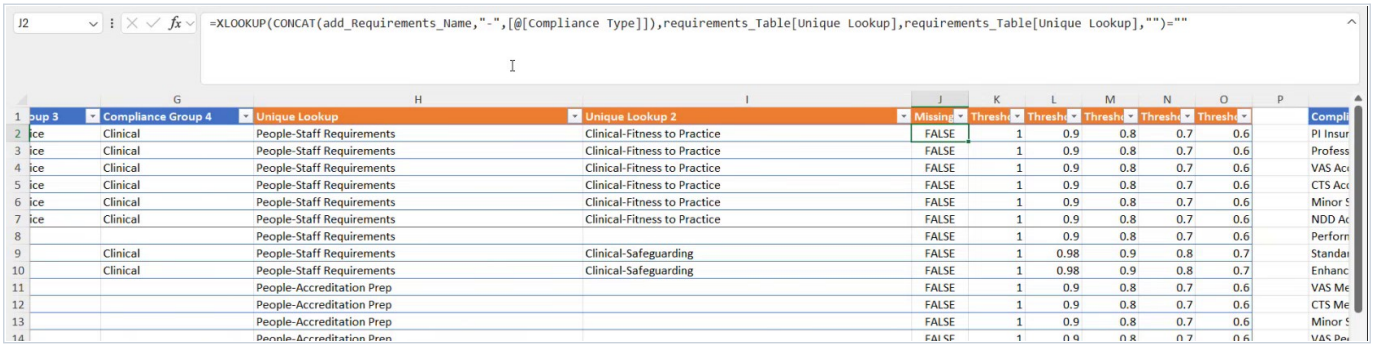
Compliance Type	Every x months	Core Requirement	Compliance Group 1	Compliance Group 2	Compliance Group 3	Compliance Group 4	Unique Lookup
Minor Skin Practical Mentor Session	24		People	Accreditation Prep			People-Accreditation Prep
VAS Practical Mentor Session	24		People	Accreditation Prep			People-Accreditation Prep
Medic Appraisal	12		People	Accreditation Prep			People-Accreditation Prep
Medic Revalidation	60		People	Accreditation Prep			People-Accreditation Prep
Onboarding begun	10000	Core	People	Onboarding			People-Onboarding
Onboarding completed	10000	Core	People	Onboarding			People-Onboarding
ID supplied	10000	Core	People	Onboarding			People-Onboarding
Qualifications supplied	10000	Core	People	Onboarding			People-Onboarding
Contract Issued for signature	10000	Core	People	Onboarding			People-Onboarding
Contract signed by both parties	10000	Core	People	Onboarding			People-Onboarding
Laptop	10000	Core	People	Onboarding			People-Onboarding
Smartcard	10000	Core	People	Onboarding			People-Onboarding
PD Access	10000	Core	People	Onboarding			People-Onboarding
S1 Access	10000	Core	People	Onboarding			People-Onboarding
PC Login	10000	Core	People	Onboarding			People-Onboarding
Training platform setup	10000	Core	People	Onboarding			People-Onboarding
Folder access	10000	Core	People	Onboarding			People-Onboarding
Annual Leave	10000	Core	People	Onboarding			People-Onboarding
Specific System Access	10000	Core	People	Onboarding			People-Onboarding
Corporate Induction	10000	Core	People	Onboarding			People-Onboarding
Local Induction including documentation	10000	Core	People	Onboarding			People-Onboarding
Photo and bios recorded	10000	Core	People	Onboarding			People-Onboarding
Electronic signature provided	10000	Core	People	Onboarding			People-Onboarding
Microsoft Teams Invite	10000	Core	People	Onboarding			People-Onboarding

Figure 12 [00:09:13] — Staff Management.xlsm › Compliance Timeframes: Rows 19–42: staff types end at row 22 (Medic Revalidation, 60 months); Onboarding types from row 23 use 10000 months as a 'never expires' sentinel



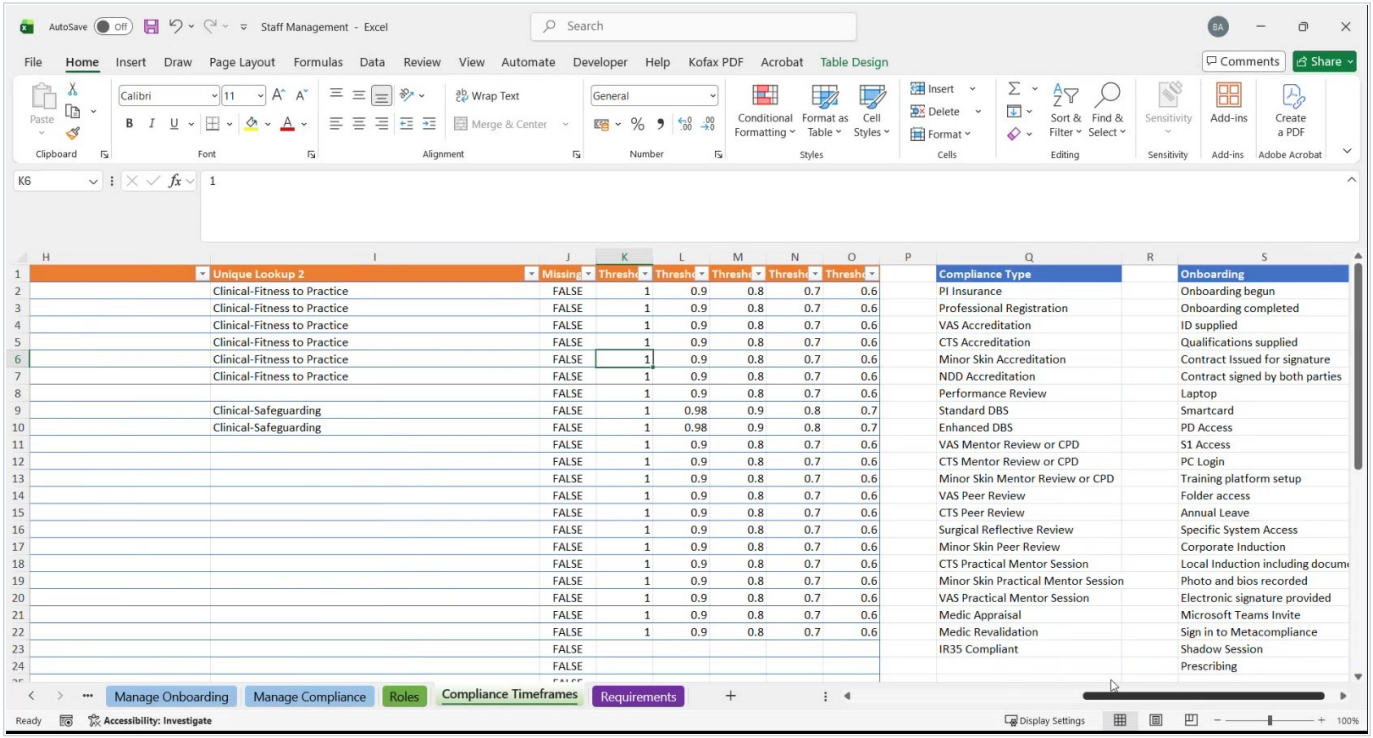
Compliance Group 2	Compliance Group 3	Compliance Group 4	Unique Lookup	Unique Lookup 2	Missing	Thresh	Thresh	Thre
Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	
Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	
Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	
Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	
Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	
Staff Requirements	Fitness to Practice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	
Staff Requirements			People-Staff Requirements		FALSE	1	0.9	
Staff Requirements	Safeguarding	Clinical	People-Staff Requirements	Clinical-Safeguarding	FALSE	1	0.98	
Staff Requirements	Safeguarding	Clinical	People-Staff Requirements	Clinical-Safeguarding	FALSE	1	0.98	
Accreditation Prep			People-Accreditation Prep		FALSE	1	0.9	
Accreditation Prep			People-Accreditation Prep		FALSE	1	0.9	
Accreditation Prep			People-Accreditation Prep		FALSE	1	0.9	

Figure 13 [00:10:03] — Staff Management.xlsm › Compliance Timeframes: Formula bar, I7 Unique Lookup 2: =IF(CONCAT([@[Compliance Group 4]],"-",[@[Compliance Group 3]])="-","",CONCAT(...))



Group 3	Compliance Group 4	Unique Lookup	Unique Lookup 2	Missing	Thresh	Thresh	Thresh	Thresh	Thresh	Compli
ice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6	PI Insur
ice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6	Profess
ice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6	VAS Acc
ice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6	CTS Acc
ice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6	Minor S
ice	Clinical	People-Staff Requirements	Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6	NDD Ac
		People-Staff Requirements		FALSE	1	0.9	0.8	0.7	0.6	Perforn
	Clinical	People-Staff Requirements	Clinical-Safeguarding	FALSE	1	0.98	0.9	0.8	0.7	Standar
	Clinical	People-Staff Requirements	Clinical-Safeguarding	FALSE	1	0.98	0.9	0.8	0.7	Enhanc
		People-Accreditation Prep		FALSE	1	0.9	0.8	0.7	0.6	VAS Me
		People-Accreditation Prep		FALSE	1	0.9	0.8	0.7	0.6	CTS Me
		People-Accreditation Prep		FALSE	1	0.9	0.8	0.7	0.6	Minor S
		People-Accreditation Prep		FALSE	1	0.9	0.8	0.7	0.6	VAS Pe

Figure 14 [00:10:23] — Staff Management.xlsm › Compliance Timeframes: Formula bar, J2 Missing: =XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),requirements_Table[Unique Lookup],...,"")="" – depends on the name currently selected on Manage Requirements



Unique Lookup 2	Missing	Thresh	Thresh	Thresh	Thresh	Thresh
Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6
Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6
Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6
Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6
Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6
Clinical-Fitness to Practice	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
Clinical-Safeguarding	FALSE	1	0.98	0.9	0.8	0.7
Clinical-Safeguarding	FALSE	1	0.98	0.9	0.8	0.7
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE	1	0.9	0.8	0.7	0.6
	FALSE					
	FALSE					

Compliance Type	Onboarding
PI Insurance	Onboarding begun
Professional Registration	Onboarding completed
VAS Accreditation	ID supplied
CTS Accreditation	Qualifications supplied
Minor Skin Accreditation	Contract Issued for signature
NDD Accreditation	Contract signed by both parties
Performance Review	Laptop
Standard DBS	Smartcard
Enhanced DBS	PD Access
VAS Mentor Review or CPD	S1 Access
CTS Mentor Review or CPD	PC Login
Minor Skin Mentor Review or CPD	Training platform setup
VAS Peer Review	Folder access
CTS Peer Review	Annual Leave
Surgical Reflective Review	Specific System Access
Minor Skin Peer Review	Corporate Induction
CTS Practical Mentor Session	Local Induction including docum
Minor Skin Practical Mentor Session	Photo and bios recorded
VAS Practical Mentor Session	Electronic signature provided
Medic Appraisal	Microsoft Teams Invite
Medic Revalidation	Sign in to Metacompliance
IR35 Compliant	Shadow Session
	Prescribing

Figure 15 [00:11:12] — Staff Management.xlsm › Compliance Timeframes: Unique Lookup 2 (I), Missing (J), Threshold 1–5 (K–O: 1/0.9/0.8/0.7/0.6; DBS rows 1/0.98/0.9/0.8/0.7) and the side lists Compliance Type (Q, 22 items incl. IR35 Compliant) and Onboarding (S)

2.3 Staff Management.xlsm › Roles

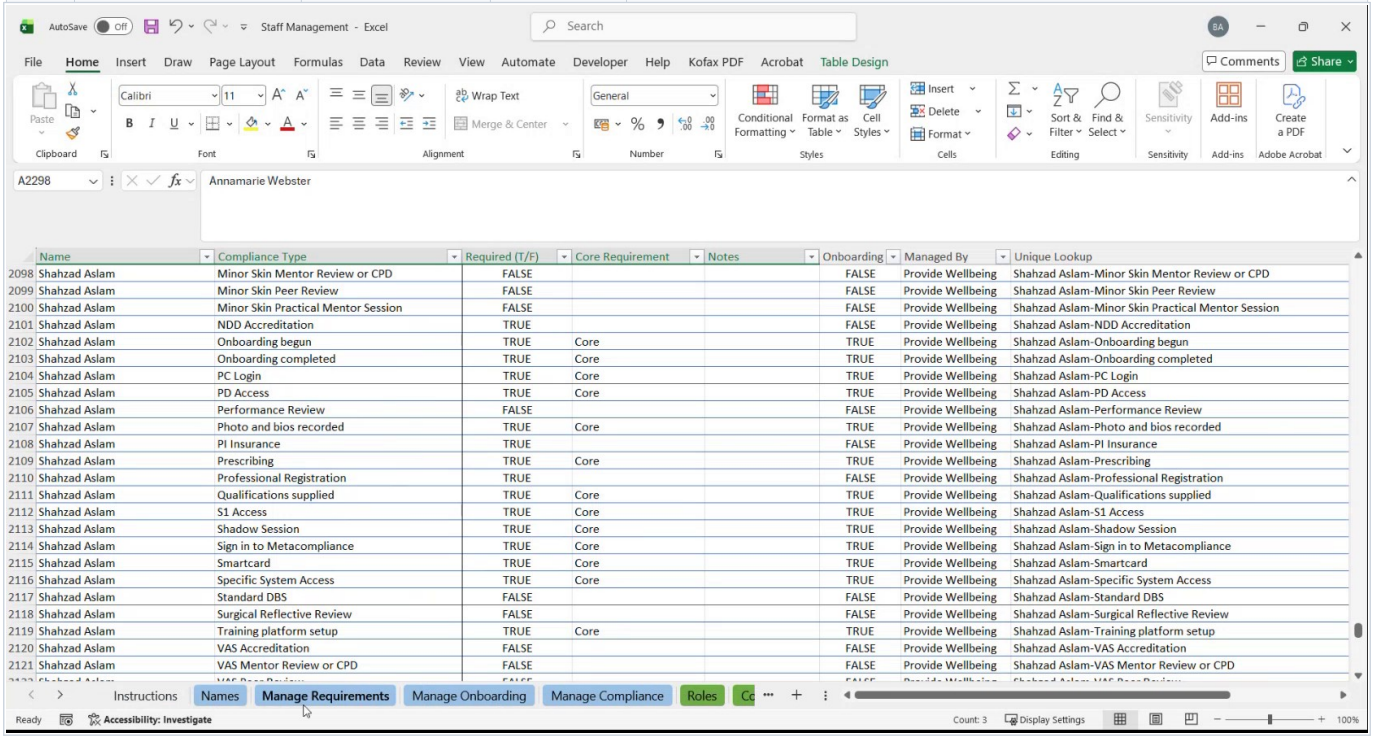
Col	Header	Type / format	Origin	Lineage, rule and evidence
A	“Reception” (header)	Text	REF	Role code list: GPwSI, HCA, PSCH, OSALT, Nurse, Paed Cons, Data/Comp, Med Sec, Vas Assessment, GM, OT, BSO, Physio, BSO/HCA, ADHD Specialist, Audio, Head Nurse, DGM, MD, OD, Podiatry. **Defect:** the header cell holds the data value “Reception”, so that role has been lost to the header [Seen]

Col	Header	Type / format	Origin	Lineage, rule and evidence
B	“Non-clinical” (header)	Text (Clinical / Non-clinical)	REF	Classification. Header is again a data value

See Figure 16 [00:12:23] in §1.4.

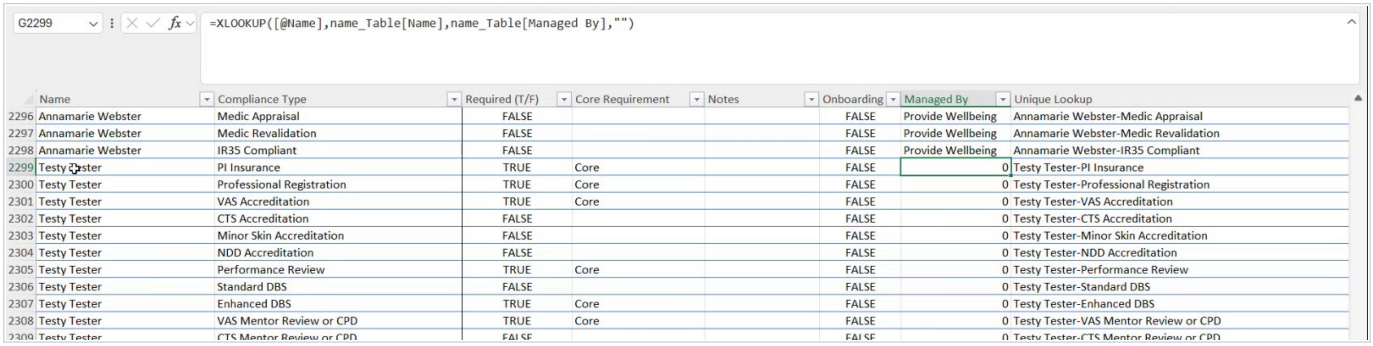
2.4 Staff Management.xlsm › Requirements (requirements_Table)

Col	Header	Type / format	Origin	Lineage, rule and evidence
A	Name	Text	VBA	From Manage Requirements D6 (`add_Requirements_Name`). Figure 24 [00:18:33]
B	Compliance Type	Text	VBA	From the grid, column C
C	Required (T/F)	Boolean	VBA (operator choice)	Whether the type applies to the person
D	Core Requirement	Text (Core / Non Core / blank)	VBA (operator choice)	Person-level core flag; overrides the catalogue default
E	Notes	Text	VBA	Grid column F; blank in demo
F	Onboarding	Boolean	VBA / default	FALSE for staff types, TRUE for onboarding items (e.g. Shahzad Aslam rows 2102–2119). Operator doubts it is used [Said 18:39]
G	Managed By	Text	CALC	`=XLOOKUP([@Name],name_Table[Name],name_Table[Managed By],"")`. Returns **0** (not blank) when the source is empty. Figure 25 [00:18:43]
H	Unique Lookup	Text · composite key	CALC	`=CONCAT([@Name],"-",[@[Compliance Type]])`, e.g. “Testy Tester-PI Insurance”. Key for Manage Requirements and CT!J. Uniqueness not enforced
I	Compliance Timeframe	Integer months	CALC	`=XLOOKUP([@[Compliance Type]],Compliance_Timeframe_Table[Compliance Type],Compliance_Timeframe_Table[Every x months],"")`. Figure 28 [00:19:33]
J	Active	Boolean	CALC	`=XLOOKUP([@Name],name_Table[Name],name_Table[Active],"")`, so deactivating a person cascades to all their rows. Figure 29 [00:19:53]



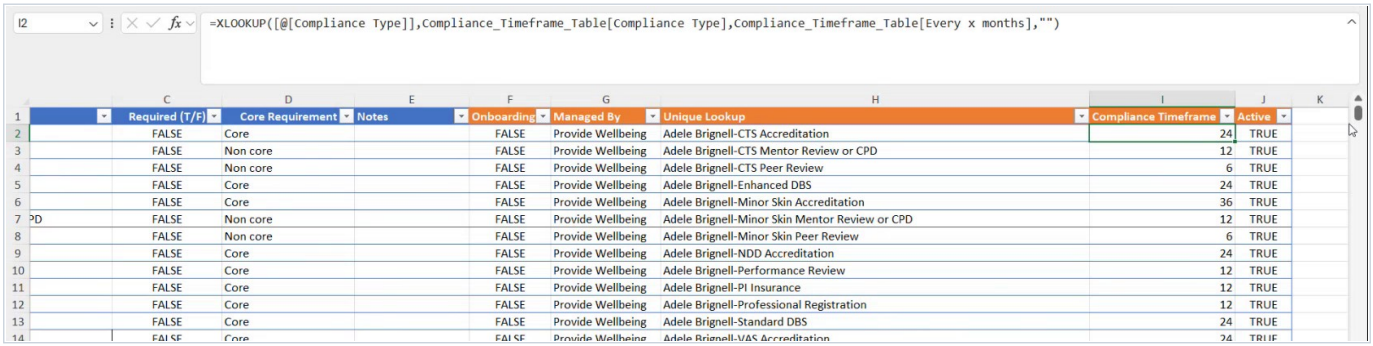
	Name	Compliance Type	Required (T/F)	Core Requirement	Notes	Onboarding	Managed By	Unique Lookup
2098	Shahzad Aslam	Minor Skin Mentor Review or CPD	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-Minor Skin Mentor Review or CPD
2099	Shahzad Aslam	Minor Skin Peer Review	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-Minor Skin Peer Review
2100	Shahzad Aslam	Minor Skin Practical Mentor Session	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-Minor Skin Practical Mentor Session
2101	Shahzad Aslam	NDD Accreditation	TRUE			FALSE	Provide Wellbeing	Shahzad Aslam-NDD Accreditation
2102	Shahzad Aslam	Onboarding begun	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Onboarding begun
2103	Shahzad Aslam	Onboarding completed	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Onboarding completed
2104	Shahzad Aslam	PC Login	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-PC Login
2105	Shahzad Aslam	PD Access	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-PD Access
2106	Shahzad Aslam	Performance Review	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-Performance Review
2107	Shahzad Aslam	Photo and bios recorded	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Photo and bios recorded
2108	Shahzad Aslam	PI Insurance	TRUE			FALSE	Provide Wellbeing	Shahzad Aslam-PI Insurance
2109	Shahzad Aslam	Prescribing	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Prescribing
2110	Shahzad Aslam	Professional Registration	TRUE			FALSE	Provide Wellbeing	Shahzad Aslam-Professional Registration
2111	Shahzad Aslam	Qualifications supplied	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Qualifications supplied
2112	Shahzad Aslam	S1 Access	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-S1 Access
2113	Shahzad Aslam	Shadow Session	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Shadow Session
2114	Shahzad Aslam	Sign in to Metacompliance	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Sign in to Metacompliance
2115	Shahzad Aslam	Smartcard	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Smartcard
2116	Shahzad Aslam	Specific System Access	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Specific System Access
2117	Shahzad Aslam	Standard DBS	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-Standard DBS
2118	Shahzad Aslam	Surgical Reflective Review	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-Surgical Reflective Review
2119	Shahzad Aslam	Training platform setup	TRUE	Core		TRUE	Provide Wellbeing	Shahzad Aslam-Training platform setup
2120	Shahzad Aslam	VAS Accreditation	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-VAS Accreditation
2121	Shahzad Aslam	VAS Mentor Review or CPD	FALSE			FALSE	Provide Wellbeing	Shahzad Aslam-VAS Mentor Review or CPD

Figure 17 [00:12:53] — Staff Management.xlsm › Requirements: requirements_Table (~2,300 rows): Name, Compliance Type, Required (T/F), Core Requirement, Notes, Onboarding, Managed By, Unique Lookup



	Name	Compliance Type	Required (T/F)	Core Requirement	Notes	Onboarding	Managed By	Unique Lookup
2296	Annamarie Webster	Medic Appraisal	FALSE			FALSE	Provide Wellbeing	Annamarie Webster-Medic Appraisal
2297	Annamarie Webster	Medic Revalidation	FALSE			FALSE	Provide Wellbeing	Annamarie Webster-Medic Revalidation
2298	Annamarie Webster	IR35 Compliant	FALSE			FALSE	Provide Wellbeing	Annamarie Webster-IR35 Compliant
2299	Testy Tester	PI Insurance	TRUE	Core		FALSE	0	Testy Tester-PI Insurance
2300	Testy Tester	Professional Registration	TRUE	Core		FALSE	0	Testy Tester-Professional Registration
2301	Testy Tester	VAS Accreditation	TRUE	Core		FALSE	0	Testy Tester-VAS Accreditation
2302	Testy Tester	CTS Accreditation	FALSE			FALSE	0	Testy Tester-CTS Accreditation
2303	Testy Tester	Minor Skin Accreditation	FALSE			FALSE	0	Testy Tester-Minor Skin Accreditation
2304	Testy Tester	NDD Accreditation	FALSE			FALSE	0	Testy Tester-NDD Accreditation
2305	Testy Tester	Performance Review	TRUE	Core		FALSE	0	Testy Tester-Performance Review
2306	Testy Tester	Standard DBS	FALSE			FALSE	0	Testy Tester-Standard DBS
2307	Testy Tester	Enhanced DBS	TRUE	Core		FALSE	0	Testy Tester-Enhanced DBS
2308	Testy Tester	VAS Mentor Review or CPD	TRUE	Core		FALSE	0	Testy Tester-VAS Mentor Review or CPD
2309	Testy Tester	CTS Mentor Review or CPD	FALSE			FALSE	0	Testy Tester-CTS Mentor Review or CPD

Figure 25 [00:18:43] — Staff Management.xlsm › Requirements: Formula bar, G2299 Managed By: =XLOOKUP([@Name],name_Table[Name],name_Table[Managed By],"")



	Required (T/F)	Core Requirement	Notes	Onboarding	Managed By	Unique Lookup	Compliance Timeframe	Active
2	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-CTS Accreditation	24	TRUE
3	FALSE	Non core		FALSE	Provide Wellbeing	Adele Brignell-CTS Mentor Review or CPD	12	TRUE
4	FALSE	Non core		FALSE	Provide Wellbeing	Adele Brignell-CTS Peer Review	6	TRUE
5	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-Enhanced DBS	24	TRUE
6	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-Minor Skin Accreditation	36	TRUE
7	FALSE	Non core		FALSE	Provide Wellbeing	Adele Brignell-Minor Skin Mentor Review or CPD	12	TRUE
8	FALSE	Non core		FALSE	Provide Wellbeing	Adele Brignell-Minor Skin Peer Review	6	TRUE
9	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-NDD Accreditation	24	TRUE
10	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-Performance Review	12	TRUE
11	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-PI Insurance	12	TRUE
12	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-Professional Registration	12	TRUE
13	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-Standard DBS	24	TRUE
14	FALSE	Core		FALSE	Provide Wellbeing	Adele Brignell-VAS Accreditation	24	TRUE

Figure 28 [00:19:33] — Staff Management.xlsm › Requirements: Formula bar, I2 Compliance Timeframe: =XLOOKUP([@[Compliance Type]],Compliance_Timeframe_Table[Compliance Type],Compliance_Timeframe_Table[Every x months],"")

2.5 Staff Management.xlsm › Manage Requirements (entry form)

Col	Header	Type / format	Origin	Lineage, rule and evidence
D6	Name (add_Requirements_Name)	Text, dropdown of names	RAW	Drives CT!J “Missing” and the Update panel. Figure 18 [00:13:23]
D8	Role	Text	CALC? [Unknown]	Showed “Surgical Clinician – NHS Only” for the previous person **and** for Testy Tester (whose Role is GPwSI), so it is stale or unrelated. Operator: “I don’t think it actually uses this” [Said 15:05]
C11:C32	Compliance Type	Text	VBA (List Missing)	Populated by List Missing (Figure 21 [00:14:28])
D11:D32	Required	Boolean dropdown	RAW	TRUE/FALSE per type (Figure 22 [00:15:13])
E11:E32	Requirement	Core / Non Core dropdown	RAW	Core flag
F11:F32	Notes	Text	RAW	Blank in demo
J6	Name (update panel)	Text	CALC	Mirrors D6 (shows Testy Tester)
I11:I…	Compliance Type (update)	Text	CALC	Full type list for the update panel
J11	Required (existing)	Boolean	CALC	`=IF(I11<>"",XLOOKUP(CONCAT($J$6,"-",I11),requirements_Table[Unique Lookup],requirements_Table[Required (T/F)],""),"")`
K11	Requirement (existing)	Text	CALC	`=IF(I11<>"",IF(XLOOKUP(CONCAT($J$6,"-",I11),…,requirements_Table[Core Requirement],"")<>0,XLOOKUP(…),""),""):` suppresses 0 from blank sources. Figure 19 [00:13:43]
L11	Notes (existing)	Text	CALC	Same pattern on [Notes] [Inferred]
M:O	Required / Core Requirement / Notes (new values)	Mixed	RAW	Operator types replacement values here for *Update Requirements*
P:Q…T	Site / Compliance Type / … / helper	Text	CALC	“Originally hidden” helper columns for the macro [Said 13:56]. T11: `=IF(I11<>"",IF(011<>"",011,IF(L11<>"",L11,"")),"")` = coalesce(new, existing). Figure 20 [00:14:03]. Header “Site” shows the form was cloned from the Estates tool

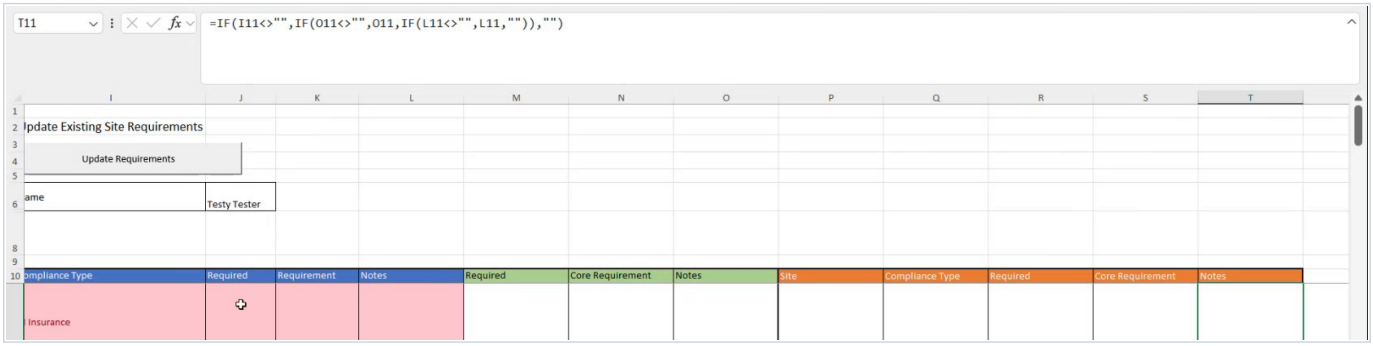


Figure 20 [00:14:03] — Staff Management.xlsm › Manage Requirements: Formula bar, T11 helper: =IF(I11<>"",IF(O11<>"",O11,IF(L11<>"",L11,"")),"") – merges new vs existing values for the update macro

2.6 Staff Management.xlsm › Manage Compliance (entry form)

Col	Header	Type / format	Origin	Lineage, rule and evidence
B	Name	Text, dropdown/autocomplete	RAW	Per row (allows several people in one batch)
C	Compliance Type	Text, dropdown	RAW	Per row
D	Completion Date	Date dd/mm/yyyy	RAW	Date the evidence was seen or the activity done (e.g. GMC check “today” = 11/08/2026)
E	Completion Status	List: Completed / Sign-off Required / Scheduled	RAW	Figure 33 [00:23:02]
F	RAG Rating	List: Green / Amber / Red (cell fill follows)	RAW	Chosen by hand, not derived from dates [Seen]
G	Notes	Text	RAW	“Certificate on file”, “Copy on file”
H–N	(hidden)	—	—	Columns H–N are hidden (letters jump G→O) [Gap G-10]
O	Time Frame (Months)	Integer	CALC	Fills as soon as C is set (12, 24, 60…), so it is a lookup on `Compliance_Timeframe_Table[Every x months]` [Inferred]. Figure 31 [00:20:33]
R5	Name (update_Co…)	Text	RAW	Selects the person whose stored records are shown. Figure 30 [00:20:13]
R9:R…	Compliance Type	Text	CALC	Rows filled for the selected person (pink until a record exists)
S	Completion Date (stored)	Date	CALC (x-wb)	XLOOKUP into the data store, key R5&"-"&R [Inferred, same pattern as V]
T	Completion Status (stored)	Text	CALC (x-wb)	As S
U	RAG Rating (stored)	Text	CALC (x-wb)	As S; conditional fill
V	Notes (stored)	Text	CALC (x-wb)	`=IF(XLOOKUP(CONCAT($R$5,"-",R9),'Compliance Data.xlsm'!compliance_Data_Store_Table[Unique Lookup 3],'Compliance Data.xlsm'!compliance_Data_Store_Table[Compliance Notes],"")<>0, XLOOKUP(…),"")`. Figure 36 [00:24:03]
W	Due Date (stored)	Date	CALC (x-wb)	Completion + Time Frame months (01/06/2026 → 01/06/2027). Whether it is stored or derived is [Unknown]
X	Completion Date (new)	Date	RAW	Amendment value (01/05/2026). Figure 37 [00:25:03]
Y	Completion Status (new)	Text	RAW	Blank = keep existing [Inferred from T-column pattern]
Z	RAG Rating (new)	Text	RAW	As Y
AA	Notes (new)	Text	RAW	As Y
AB	Due Date (new)	Date	**RAW**	**Typed by hand** (“1/5/27”) on the update path, so it is not computed: a due-date/timeframe mismatch is possible. Spoken: “the due date’s gonna have to change” [Said 24:52]
AC	Delete	Flag	RAW	Marks a stored record for deletion [Said 24:59]; value format [Unknown]

2.7 Compliance Data.xlsm › compliance_Data_Store_Table (external, partially visible)

Only the column names used in formulas, and the destination of the VBA writes, are visible. Inferred schema:

Col	Header	Type / format	Origin	Lineage, rule and evidence
?	Name	Text	VBA	From Manage Compliance B [Inferred]
?	Compliance Type	Text	VBA	From C
?	Completion Date / Status / RAG / Due Date	Date / Text / Text / Date	VBA	From D/E/F and the computed due date; overwritten in place by *Update*
?	Compliance Notes	Text	VBA	Seen in V9 formula
?	Unique Lookup 3	Text · key	CALC/VBA	Seen in V9 formula; evidently Name & "-" & Compliance Type. “Unique” fails once history accumulates (“build up a picture over time”, 25:08), and XLOOKUP returns only the *first* match [Gap G-04]
?	(Unique Lookup list with 024/025/026 prefixes)	Text	PQ? [Unknown]	Glimpsed in the taskbar preview of F2: values such as “026 Clinical-Audit”, “024 People/Workforce-Employment Requirements”. Possibly a group code + Unique Lookup [Gap G-06]. Figure 44 [00:29:38]
PQ	Requirements / Timeframes (built)	Tables	PQ	Built from the saved Requirements and Compliance Timeframes tabs of every tool [Said 11:21, 27:02]. Operator says “there we go” at 11:50 but the window shown is still Staff Management, so the PQ table was not visible to the viewer [Gap G-02]

3. Chronological Step-by-Step Operator Workflow

Twenty-two steps cover the whole 30:54 recording with no gaps. Start and end times are the utterance boundaries from the re-transcription; macro moments are pinned to the first frame showing the result.

Step	Window	Action	Tab
1	00:00:00–00:00:56	Start recording and share the correct screen	—
2	00:00:56–00:02:34	Introduce the user-input tool; clear a prior attempt	Manage Compliance
3	00:02:34–00:05:28	Tour the Names tab (master register)	Names
4	00:05:28–00:07:11	Register new person “Testy Tester”	Names
5	00:07:11–00:11:13	Review Compliance Timeframes (reference catalogue)	Compliance Timeframes
6	00:11:13–00:12:09	Explain Power Query build of timeframe tables	Compliance Timeframes
7	00:12:09–00:12:41	Glance at Roles	Roles
8	00:12:41–00:13:11	Introduce Requirements (per-person applicability)	Requirements
9	00:13:11–00:14:16	Select the new person on Manage Requirements	Manage Requirements
10	00:14:16–00:14:50	Run VBA “List Missing”	Manage Requirements
11	00:14:50–00:17:53	Declare Required / Core for each of the 22 types	Manage Requirements
12	00:17:53–00:18:29	Run VBA “Add Requirements”	Requirements
13	00:18:29–00:19:14	Fix Managed By (manual exception)	Names → Requirements
14	00:19:14–00:20:03	Inspect H–J and save the workbook	Requirements
15	00:20:03–00:23:24	Enter compliance evidence on Manage Compliance	Manage Compliance
16	00:23:24–00:24:30	Run VBA “Add New Compliance”	Manage Compliance
17	00:24:30–00:26:18	Amend an existing record via “Update Existing Compliance”	Manage Compliance
18	00:26:18–00:28:24	Explain the monthly cycle and start the data refresh	Manage Compliance / (Compliance Data)
19	00:28:24–00:29:13	Stop the refresh; deactivate the test person; save	Names
20	00:29:13–00:29:27	Close the workbook: VBA run-time error	Manage Compliance
21	00:29:27–00:29:45	Decline to save; background refresh restarts	Manage Compliance
22	00:29:45–00:30:54	Stop screen share and wrap up	—

Step 1 [00:00:00 – 00:00:56]: Start recording and share the correct screen

Action performed	Operator on camera, then switches off the camera and starts a screen share; first attempt shares the wrong screen and is re-done (00:33–00:48).
Source → target	—
State mutation (before → after)	No data change.

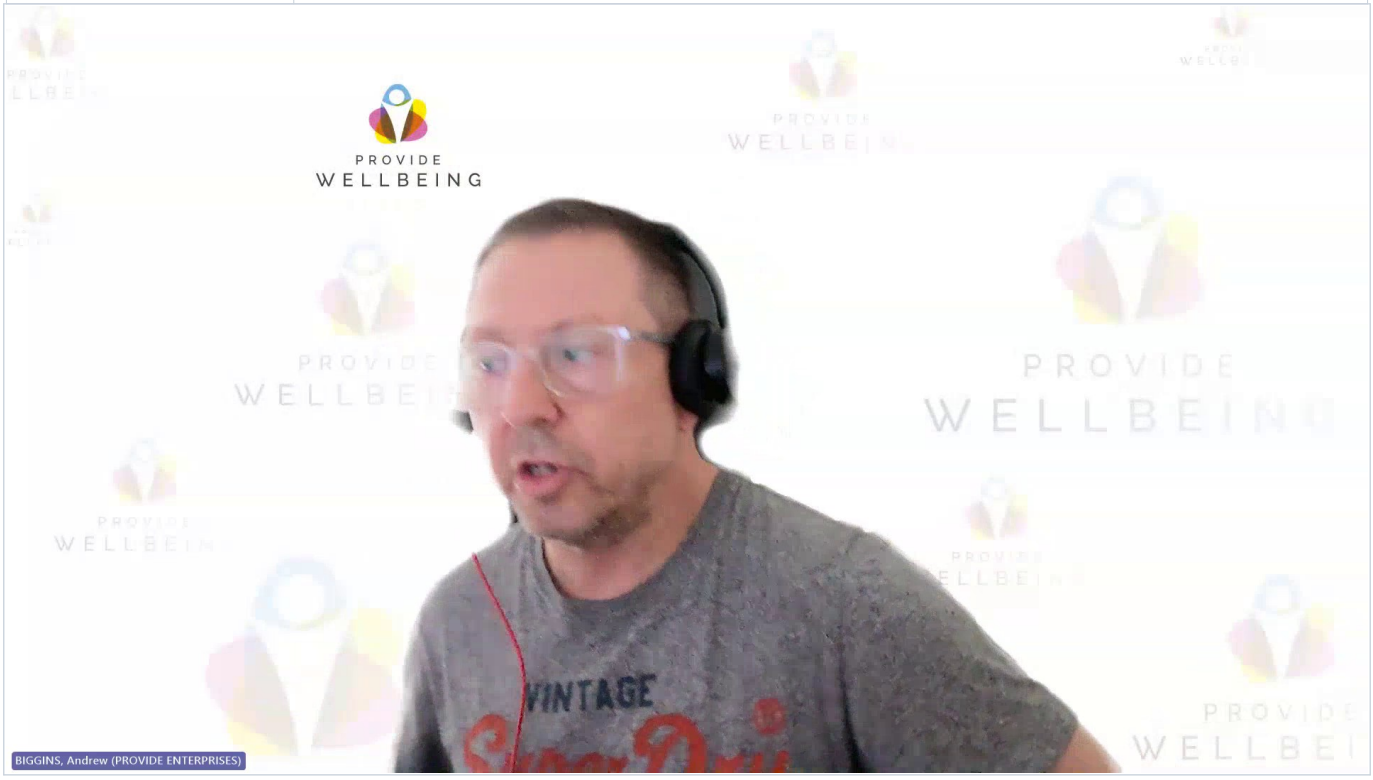


Figure 1 [00:00:05] — Teams recording › (camera): Recording starts; operator on camera before screen share (no workbook visible yet)

Step 2 [00:00:56 – 00:02:34]: Introduce the user-input tool; clear a prior attempt

Action performed	Workbook Staff Management.xlsm is open on *Manage Compliance* with a previous person (Emma Lindsley: Enhanced DBS 26/02/2026, Performance Review 08/05/2026) loaded. Operator says he "want[s] to get rid of something I did on my abortive attempt" (01:50–02:03); no visible change follows.
Source → target	—
State mutation (before → after)	Possibly a removal of test data from the earlier, unrecorded attempt [Gap G-03].

Shown above: Figure 2 [00:00:53] — First shared screen. Workbook open on Manage Compliance with a previous person (Emma Lindsley) loaded; full ta…

Step 3 [00:02:34 – 00:05:28]: Tour the Names tab (master register)

Action performed	Opens *Names*; scrolls right across A→Q; selects N (Active) and points at O, P, Q, which displays their formulas.
Source → target	Read-only.
State mutation (before → after)	None. Calculated O/P/Q explained. Pink banding = Active FALSE rows [Inferred].

Shown above: Figure 3 [00:02:58] — Names table (name_Table), columns A–D and I–N: Name, Managed By, Employment Type, Role, WTE, Start Date, Systm…

Shown above: Figure 4 [00:03:13] — Names table scrolled to columns E–G: Works with, Tel Num, Email (free-text contact data, mixed formats)

Shown above: Figure 5 [00:04:23] — Column N Active (TRUE/FALSE) selected; O = Compliance Types Added to Record, P = Total Compliance Types, Q = N…

Shown above: Figure 6 [00:04:53] — Formula bar, O2: =COUNTIFS(requirements_Table[Name],[@Name])

Shown above: Figure 7 [00:05:03] — Formula bar, P2: =COUNTIFS(Compliance_Type,"<>") – counts the 22-item named list Compliance_Type

Shown above: Figure 8 [00:05:23] — Formula bar, Q79: =IF([@Active]=TRUE,[@Name],"")

Step 4 [00:05:28 – 00:07:11]: Register new person “Testy Tester”

Action performed	Types into A140 (first blank row of `name_Table`, so the table auto-extends), then C = Contractor, D = GPwSI, J = 1 Aug 2026, N = TRUE. Leaves B (Managed By) blank on purpose (“freelance... not managed by anybody”).
Source → target	Manual keyboard entry → `name_Table` row 140.
State mutation (before → after)	BEFORE: row 140 empty. AFTER: A–D, J, N set; O = **0**, P = **22**, Q = “Testy Tester” (calculated columns fill automatically).

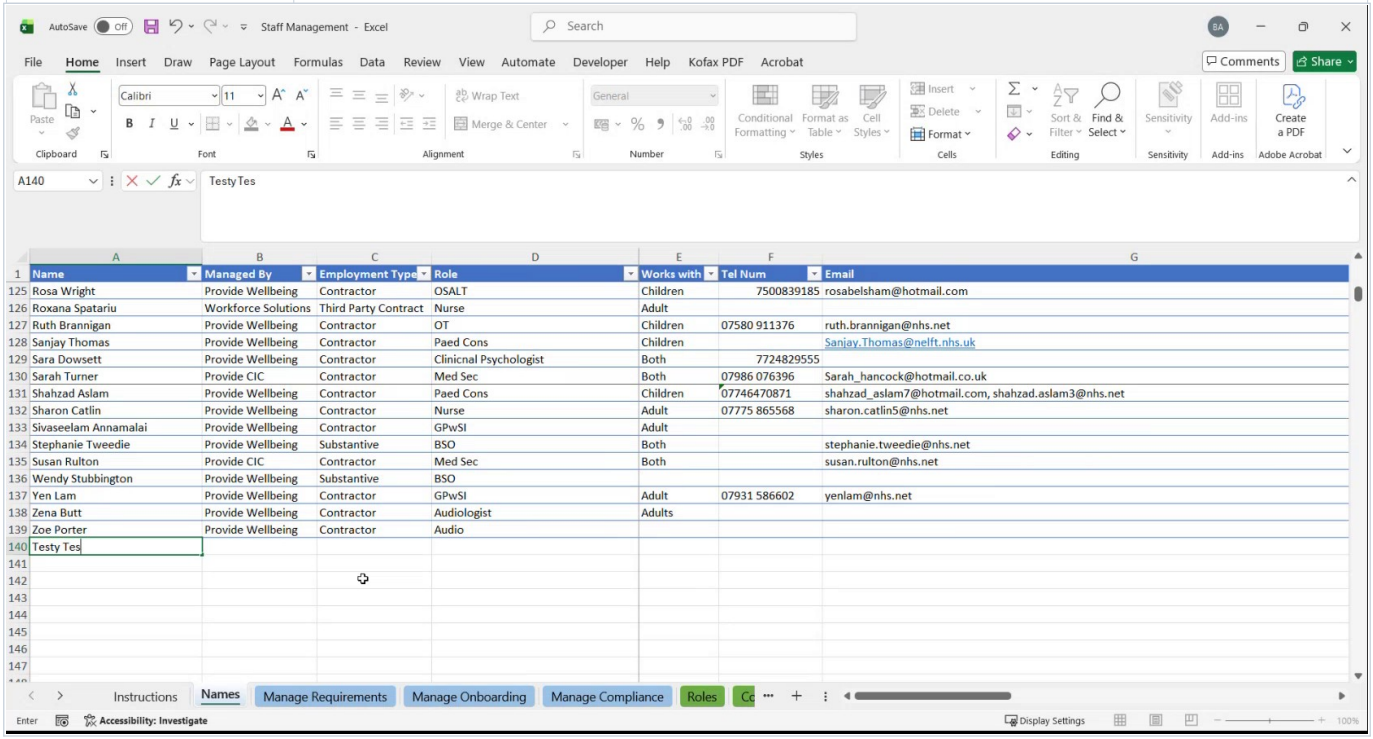
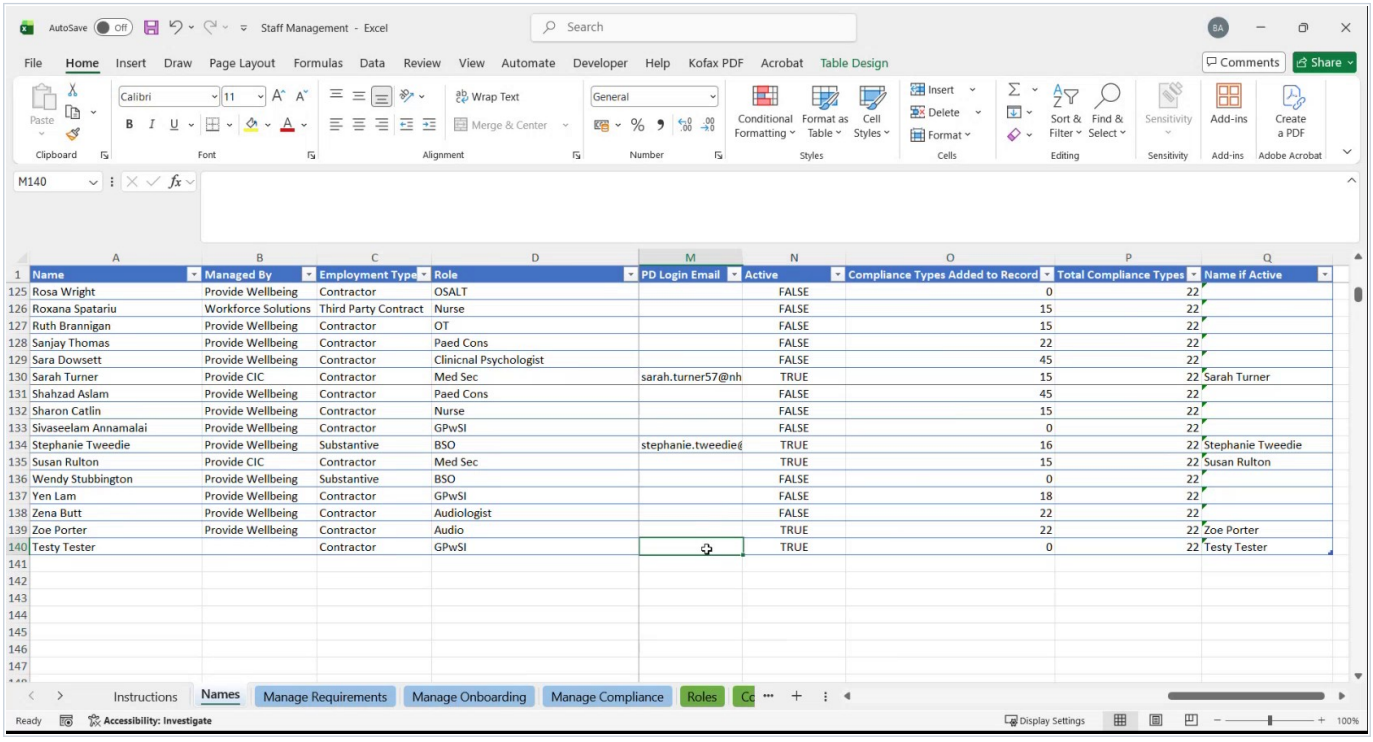


Figure 9 [00:05:43] — Staff Management.xlsm › Names: BEFORE: operator types new name "Testy T..." into A140, the first empty row of name_Table



	Name	Managed By	Employment Type	Role	PD Login Email	Active	Compliance Types Added to Record	Total Compliance Types	Name if Active
125	Rosa Wright	Provide Wellbeing	Contractor	OSALT		FALSE	0	22	
126	Roxana Spatariu	Workforce Solutions	Third Party Contract	Nurse		FALSE	15	22	
127	Ruth Brannigan	Provide Wellbeing	Contractor	OT		FALSE	15	22	
128	Sanjay Thomas	Provide Wellbeing	Contractor	Paed Cons		FALSE	22	22	
129	Sara Dowsett	Provide Wellbeing	Contractor	Clinicnal Psychologist		FALSE	45	22	
130	Sarah Turner	Provide CIC	Contractor	Med Sec	sarah.turner57@nh	TRUE	15	22	Sarah Turner
131	Shahzad Aslam	Provide Wellbeing	Contractor	Paed Cons		FALSE	45	22	
132	Sharon Catlin	Provide Wellbeing	Contractor	Nurse		FALSE	15	22	
133	Sivaseelam Annamalai	Provide Wellbeing	Contractor	GPwSI		FALSE	0	22	
134	Stephanie Tweedie	Provide Wellbeing	Substantive	BSO	stephanie.tweedie@	TRUE	16	22	Stephanie Tweedie
135	Susan Rulton	Provide CIC	Contractor	Med Sec		TRUE	15	22	Susan Rulton
136	Wendy Stubbington	Provide Wellbeing	Substantive	BSO		FALSE	0	22	
137	Yen Lam	Provide Wellbeing	Contractor	GPwSI		FALSE	18	22	
138	Zena Butt	Provide Wellbeing	Contractor	Audiologist		FALSE	22	22	
139	Zoe Porter	Provide Wellbeing	Contractor	Audio		TRUE	22	22	Zoe Porter
140	Testy Tester		Contractor	GPwSI		TRUE	0	22	Testy Tester

Figure 10 [00:07:03] — Staff Management.xlsm › Names: AFTER: row 140 Testy Tester – Contractor, GPwSI, Active TRUE; calculated O = 0 (no requirements yet), P = 22, Q = Testy Tester. Managed By (B140) left blank

Step 5 [00:07:11 – 00:11:13]: Review Compliance Timeframes (reference catalogue)

Action performed	Switches to *Compliance Timeframes*; selects Group 1 (People), Group 2 (Accreditation Prep block E11:E22), Group 3/4, scrolls to rows 23–42 (onboarding, 10000 months), opens formulas I7 and J2, then selects K–O (thresholds).
Source → target	Read-only.
State mutation (before → after)	None. Operator voices disagreement with the Core flags and the grouping design (see §4.3).

Shown above: Figure 11 [00:07:33] — Compliance_Timeframe_Table columns A–H: Compliance Type, Every x months, Core Requirement, Compliance Group 1–…

Shown above: Figure 12 [00:09:13] — Rows 19–42: staff types end at row 22 (Medic Revalidation, 60 months); Onboarding types from row 23 use 10000 …

Shown above: Figure 13 [00:10:03] — Formula bar, I7 Unique Lookup 2: =IF(CONCAT([@[Compliance Group 4]],"-",[@[Compliance Group 3]])="-","",CONCAT…

Shown above: Figure 14 [00:10:23] — Formula bar, J2 Missing: =XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),requirements_Table[Un…

Shown above: Figure 15 [00:11:12] — Unique Lookup 2 (I), Missing (J), Threshold 1–5 (K–O: 1/0.9/0.8/0.7/0.6; DBS rows 1/0.98/0.9/0.8/0.7) and the …

Step 6 [00:11:13 – 00:12:09]: Explain Power Query build of timeframe tables

Action performed	States that on the first refresh the data store builds compliance-timeframe tables from every tool by Power Query, and that the staff-management compliance table is there. Screen stays on Compliance Timeframes.
Source → target	PQ: Staff Management!Compliance_Timeframe_Table → Compliance Data model.
State mutation (before → after)	None on screen.

Step 7 [00:12:09 – 00:12:41]: Glance at Roles

Action performed	Clicks the *Roles* tab; “I’m not sure this is actually used at all… I don’t think it’s essential”.

Source → target	—
State mutation (before → after)	None.

Shown above: Figure 16 [00:12:23] — Roles table: Role → Clinical/Non-clinical. Header row reads 'Reception \| Non-clinical' – a data row has been p…

Step 8 [00:12:41 – 00:13:11]: Introduce Requirements (per-person applicability)

Action performed	Opens *Requirements* (~2,300 rows; Shahzad Aslam's block visible, onboarding rows TRUE/Core).
Source → target	Read-only.
State mutation (before → after)	None.

Shown above: Figure 17 [00:12:53] — requirements_Table (~2,300 rows): Name, Compliance Type, Required (T/F), Core Requirement, Notes, Onboarding, …

Step 9 [00:13:11 – 00:14:16]: Select the new person on Manage Requirements

Action performed	Opens *Manage Requirements*; the D6 dropdown now offers Testy Tester (proving name_Table drives it); selects Testy Tester; inspects the Update panel and helper formulas K11 and T11.
Source → target	`add_Requirements_Name` ← "Testy Tester".
State mutation (before → after)	BEFORE: D6 = Annamarie Webster, panel J6 = Zoe Porter (stale). AFTER: D6 = J6 = Testy Tester; Update panel pink with blank Required/Requirement, so **no existing requirements**.

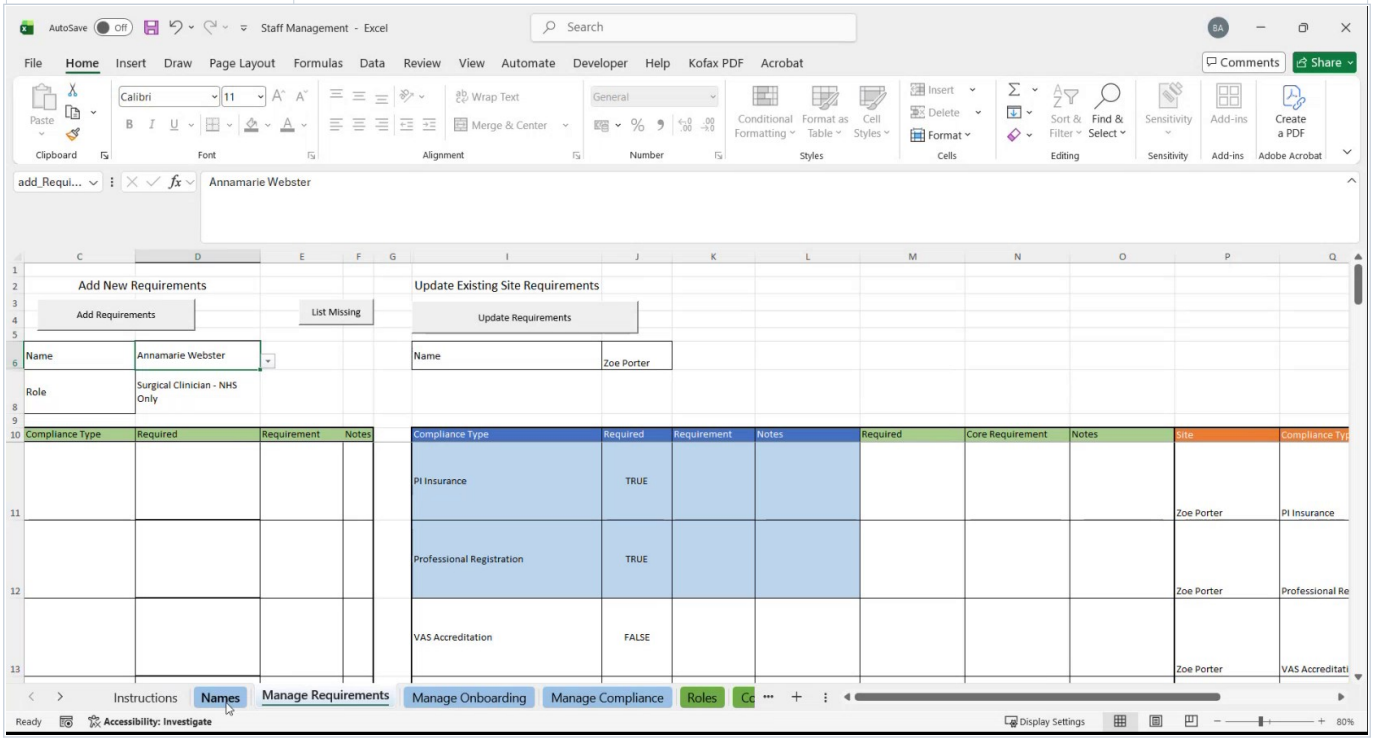


Figure 18 [00:13:23] — Staff Management.xlsm › Manage Requirements: Manage Requirements opened: name selector add_Requirements_Name (D6) still holds the previous person (Annamarie Webster); buttons Add Requirements, List Missing, Update Requirements; right-hand panel shows Zoe Porter's existing requirements

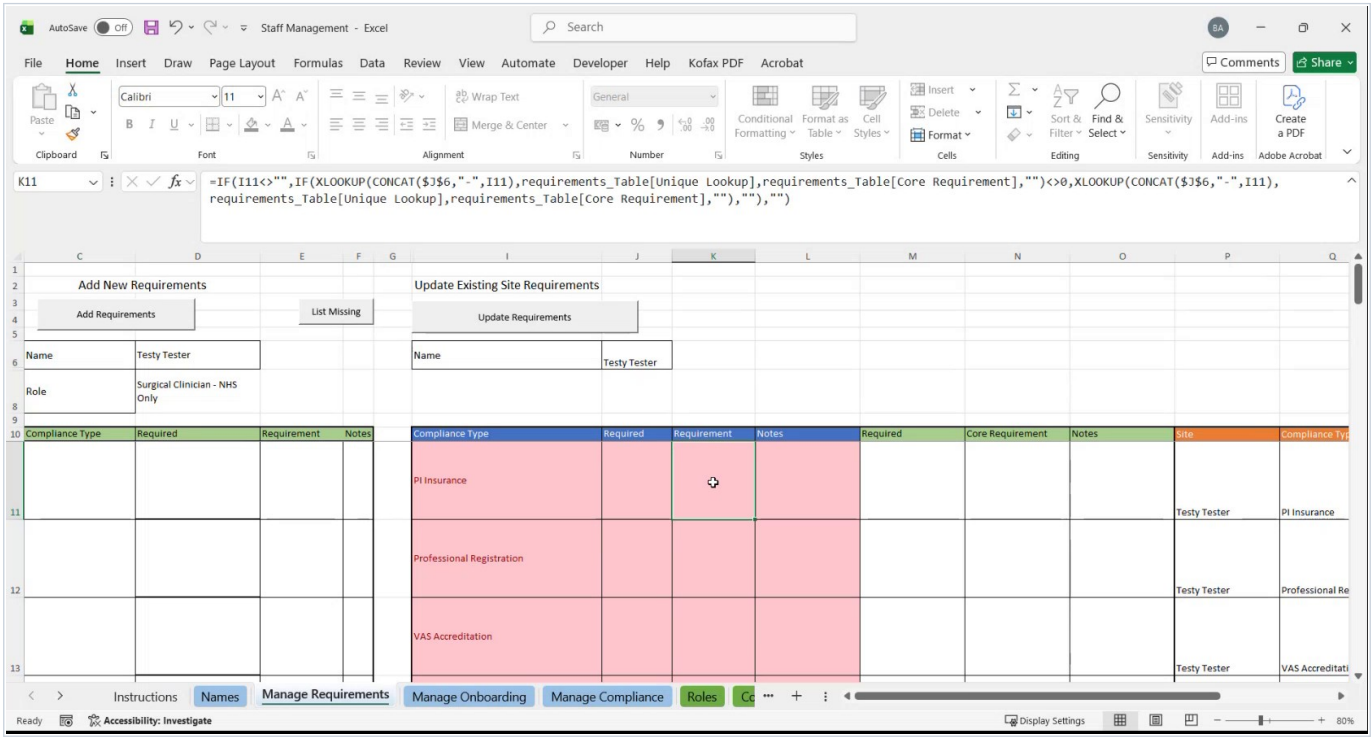


Figure 19 [00:13:43] — Staff Management.xlsm › Manage Requirements: BEFORE: Testy Tester selected; 'Update Existing Site Requirements' panel (I–L) lists every compliance type with blank Required – nothing set up yet. Formula bar K11: IF(I11<>"",IF(XLOOKUP(CONCAT(J6,"-",I11),requirements_Table[Unique Lookup],requirements_Table[Core Requirement],"")<>0,...)). Role (D8) still shows 'Surgical Clinician – NHS Only' from the previous person

Shown above: Figure 20 [00:14:03] — Formula bar, T11 helper: =IF(I11<>"",IF(O11<>"",O11,IF(L11<>"",L11,"")),"") – merges new vs existing values fo…

Step 10 [00:14:16 – 00:14:50]: Run VBA “List Missing”

Action performed	Clicks *List Missing*.
Source → target	VBA reads the types with no requirement row for D6 (the CT!J logic) → writes them into C11:C32.
State mutation (before → after)	BEFORE: grid C11:C32 empty. AFTER: 22 compliance types listed (all of them, as expected for a new person).

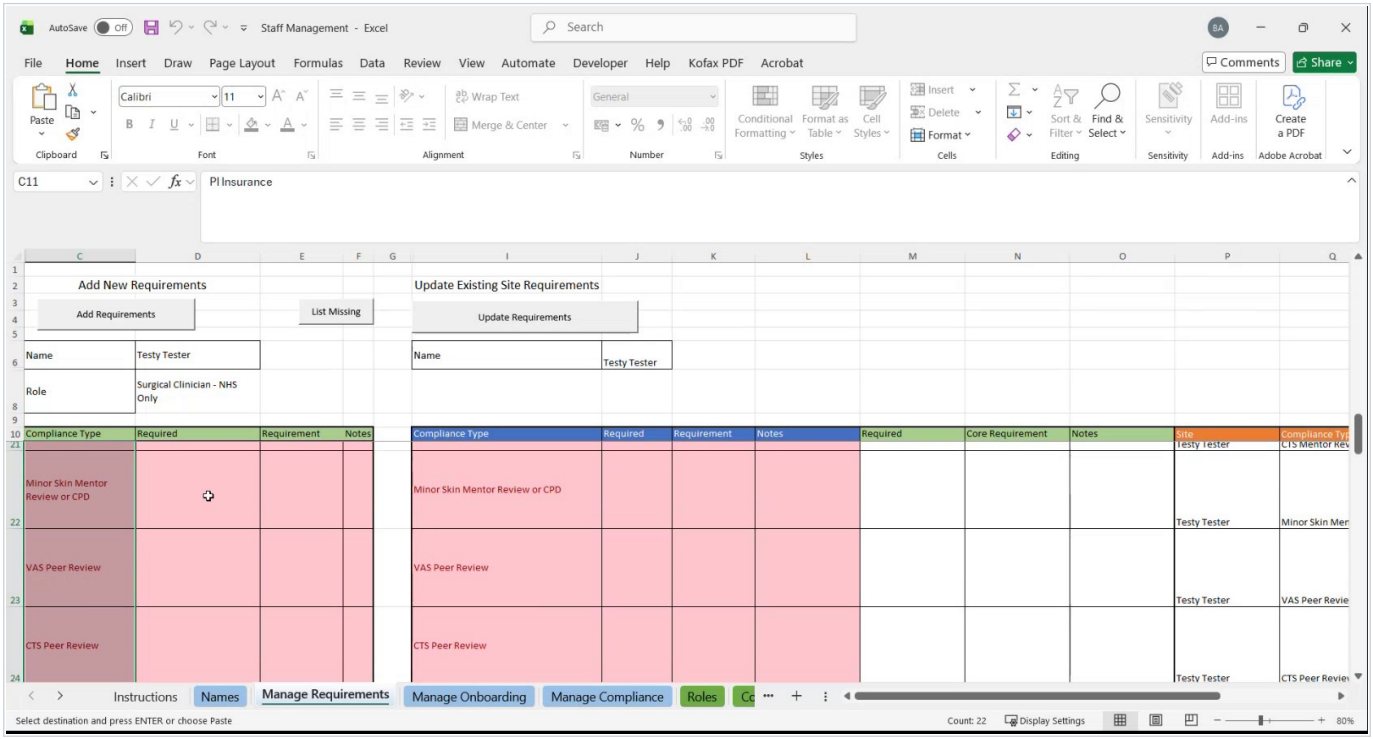


Figure 21 [00:14:28] — Staff Management.xlsm › Manage Requirements: AFTER 'List Missing': the 22 missing compliance types now fill the Add New Requirements grid (C11:C32); status bar shows a copy marquee ('Select destination…', Count: 22) – see Gap G-07

Step 11 [00:14:50 – 00:17:53]: Declare Required / Core for each of the 22 types

Action performed	Row by row selects Required (TRUE/FALSE) and Requirement (Core/Non Core) from dropdowns, partly by copy-and-paste (status bar “Select destination and press ENTER or choose Paste”).
Source → target	Manual → D11:E32.
State mutation (before → after)	See the table under this step: what the operator said against what was actually entered (two mismatches).

Spoken intent against values recorded (Requirements rows 2299–2320, Figure 24 [00:18:33]):

Compliance type	Spoken (timestamp)	Entered: Required / Core (Requirements row)	Match?
PI Insurance	Needed, core (15:10)	TRUE / Core (2299)	✓
Professional Registration	Core (15:17)	TRUE / Core (2300)	✓
VAS Accreditation	Core, every 2 years (15:26)	TRUE / Core (2301)	✓
CTS / Minor Skin / NDD Accreditation	Not doing (15:35–15:41)	FALSE / – (2302–2304)	✓
Performance Review	Core (15:46)	TRUE / Core (2305)	✓
Standard DBS	Not needed, has enhanced (15:46)	FALSE / – (2306)	✓
Enhanced DBS	Core (15:46–16:06)	TRUE / Core (2307)	✓
VAS Mentor Review or CPD	**Needed but not core** (16:12)	TRUE / **Core** (2308)	✗
CTS / Minor Skin Mentor Review	Not needed (16:25)	FALSE (2309–2310)	✓
VAS Peer Review	Recommended, not core (16:42–16:53)	TRUE / Non Core (2311)	✓
CTS Peer Review	Not core / not required (16:55)	FALSE (2312)	✓
Surgical Reflective Review	**Required but not core** (17:03)	TRUE / **Core** (2313)	✗
Minor Skin Peer / CTS & MS Practical Mentor	Not required (17:08)	FALSE (2314–2316)	✓
VAS Practical Mentor Session	Required, non-core (17:19)	TRUE / Non Core (2317)	✓
Medic Appraisal	Required, core (17:24)	TRUE / Core (2318)	✓
Medic Revalidation	Core (17:33)	TRUE / Core (2319)	✓
IR35 Compliant	Core (trades as a company) (17:33–17:47)	TRUE / Core (2320)	✓

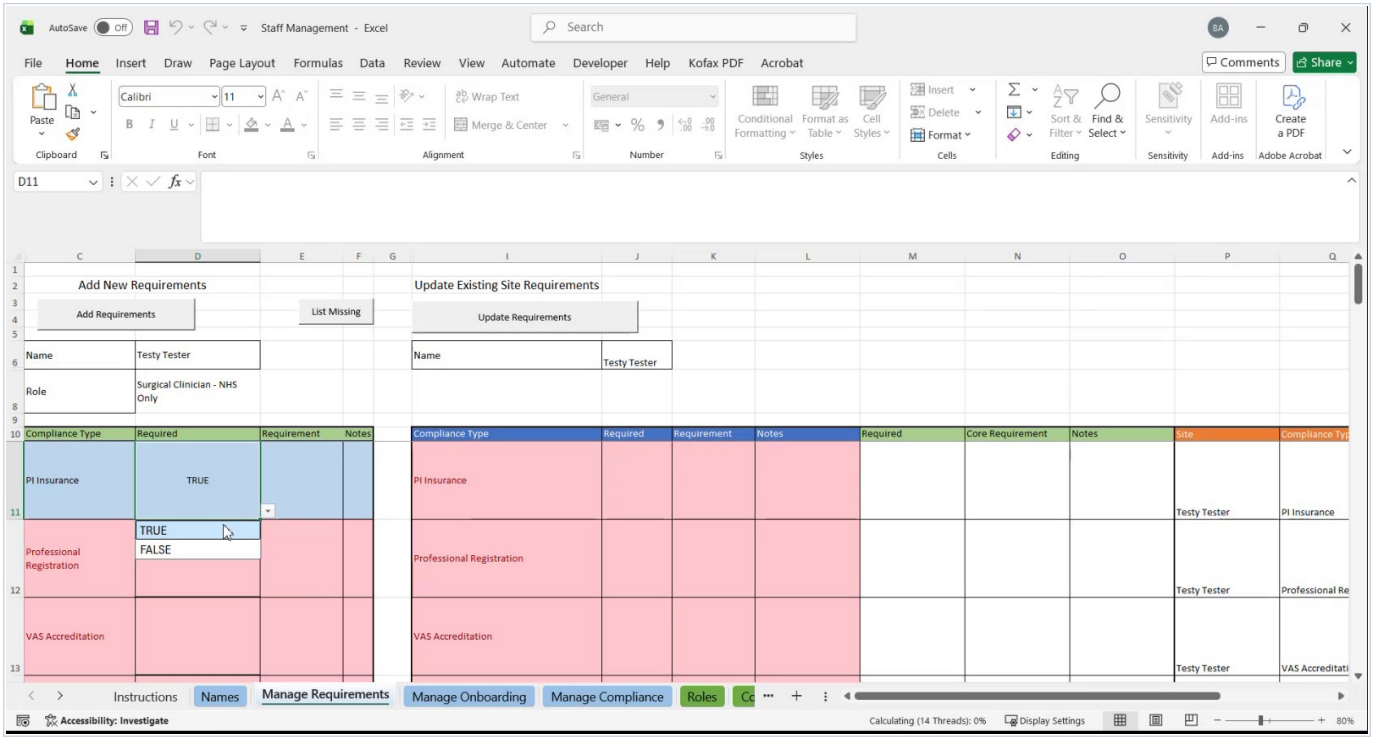


Figure 22 [00:15:13] — Staff Management.xlsm › Manage Requirements: Required (D11) data-validation dropdown open with TRUE / FALSE for PI Insurance (Requirement column E offers Core / Non Core)

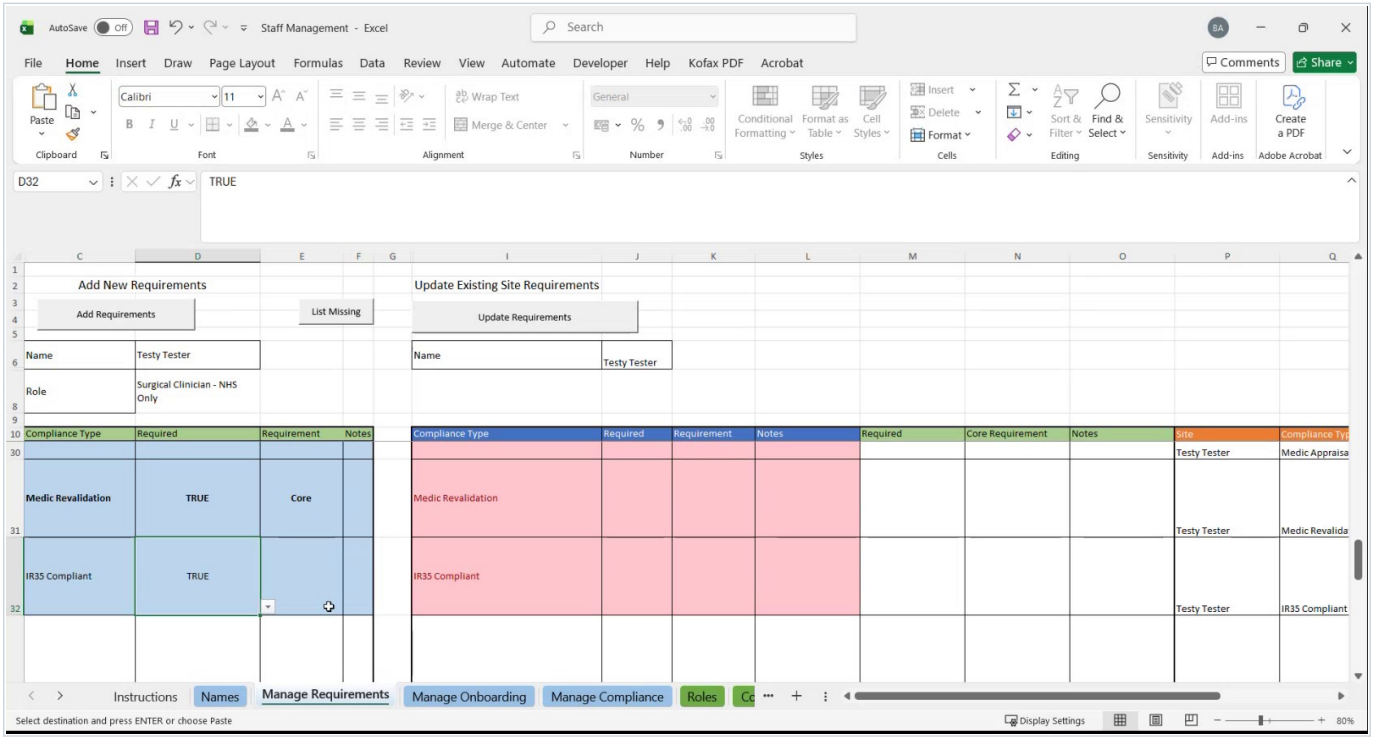


Figure 23 [00:17:43] — Staff Management.xlsm › Manage Requirements: Grid complete down to IR35 Compliant = TRUE; status bar 'Select destination and press ENTER or choose Paste' shows values being copy-pasted between rows

Step 12 [00:17:53 – 00:18:29]: Run VBA “Add Requirements”

Action performed	Clicks *Add Requirements*; waits about 15 s.
Source → target	VBA: grid C/D/E/F + D6 → appends 22 rows to `requirements_Table` (rows 2299–2320). Values written as values; G–J recalculate.
State mutation (before → after)	BEFORE: last row 2298 (Annamarie Webster-IR35 Compliant). AFTER: rows 2299–2320 Testy Tester; G (Managed By) = **0** for all; I = 12/12/24/24/36/24/12/24/24/12/12/12/6/6/12/6/24/24/24/12/60/12; J = TRUE.

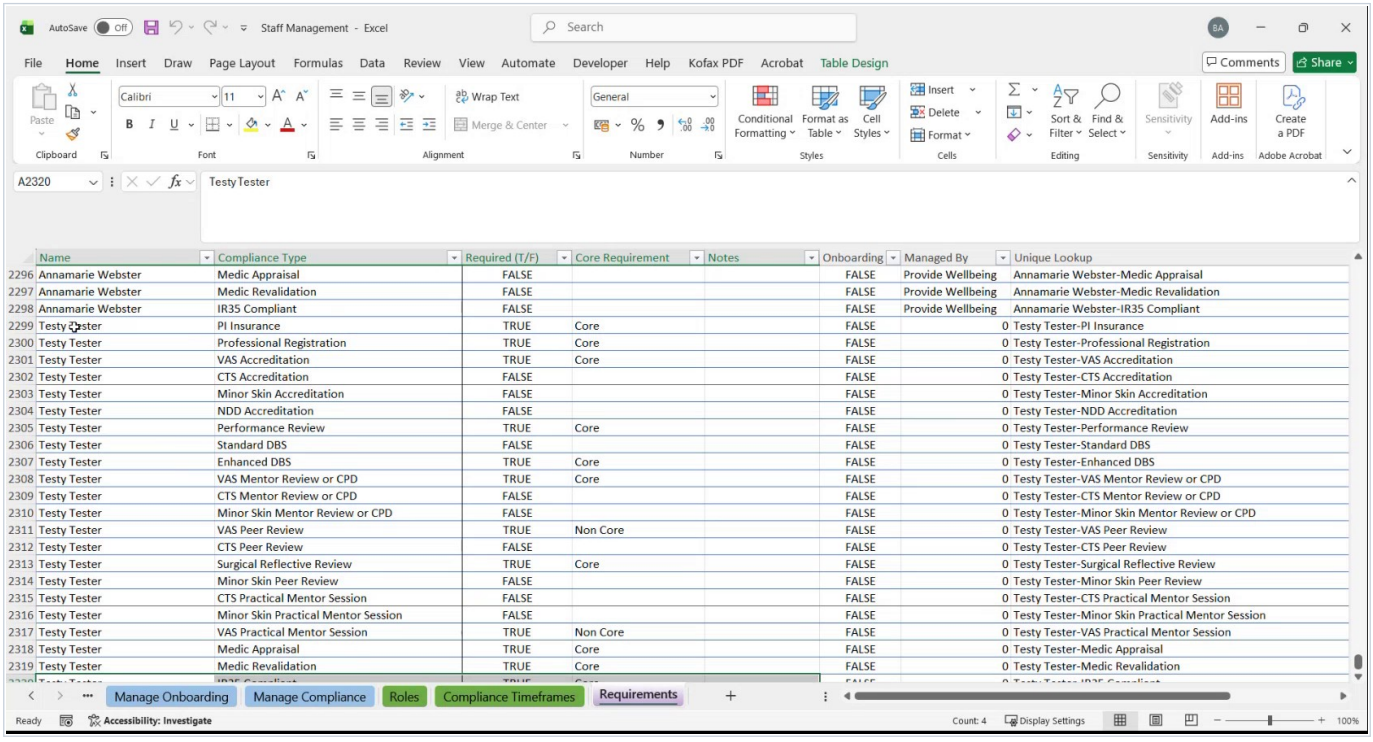


Figure 24 [00:18:33] — Staff Management.xlsm › Requirements: AFTER 'Add Requirements': 22 Testy Tester rows appended (2299–2320). Managed By returns 0 because Names!B140 is blank

Shown above: Figure 25 [00:18:43] — Formula bar, G2299 Managed By: =XLOOKUP([@Name],name_Table[Name],name_Table[Managed By],"")

Step 13 [00:18:29 – 00:19:14]: Fix Managed By (manual exception)

Action performed	Notices Managed By = 0, decides “it will probably actually be Provide Wellbeing for everybody”, goes to *Names*!B140 and selects Provide Wellbeing.
Source → target	Manual → name_Table[Managed By] → recalc → requirements_Table[Managed By].
State mutation (before → after)	BEFORE: Requirements!G2299:G2320 = 0. AFTER: “Provide Wellbeing”. Names!O140 goes 0 → 22.

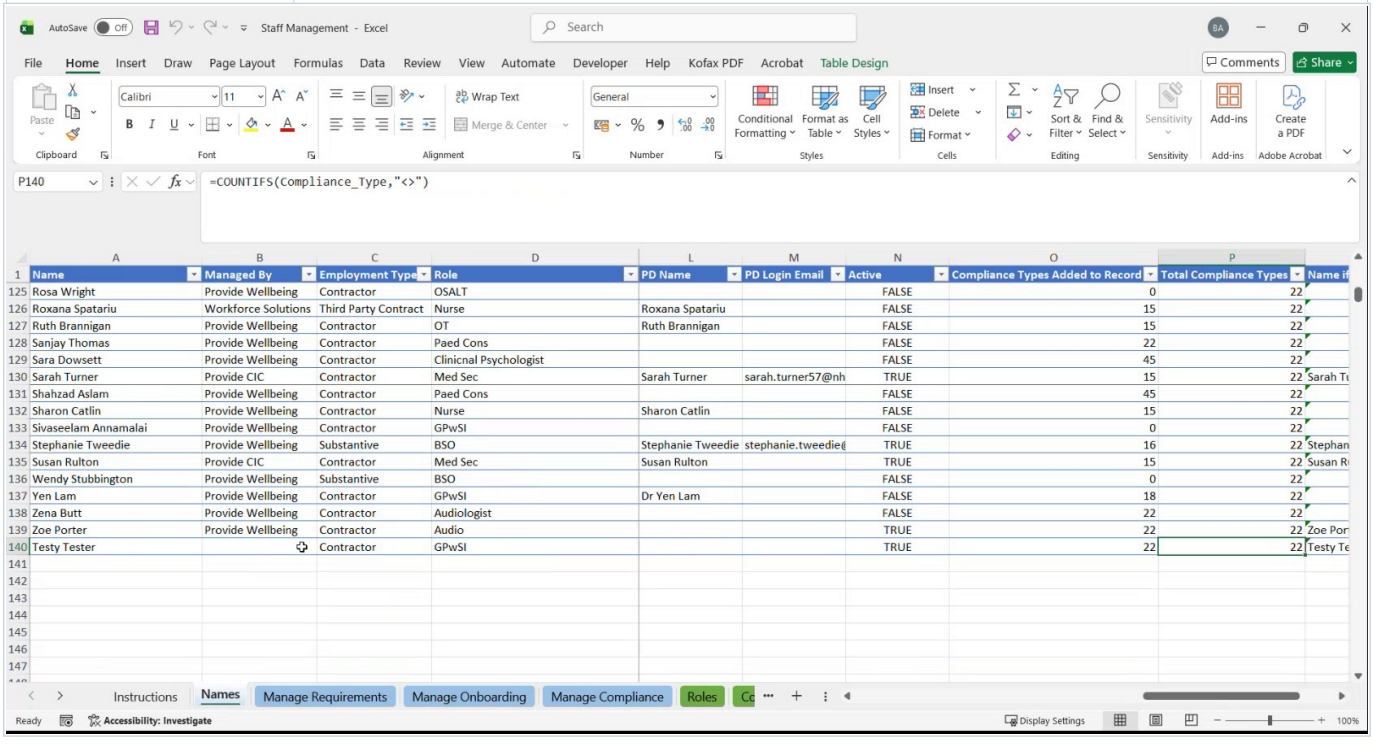


Figure 26 [00:19:03] — Staff Management.xlsm › Names: Manual fix: Managed By dropdown on Names!B140 set to 'Provide Wellbeing'

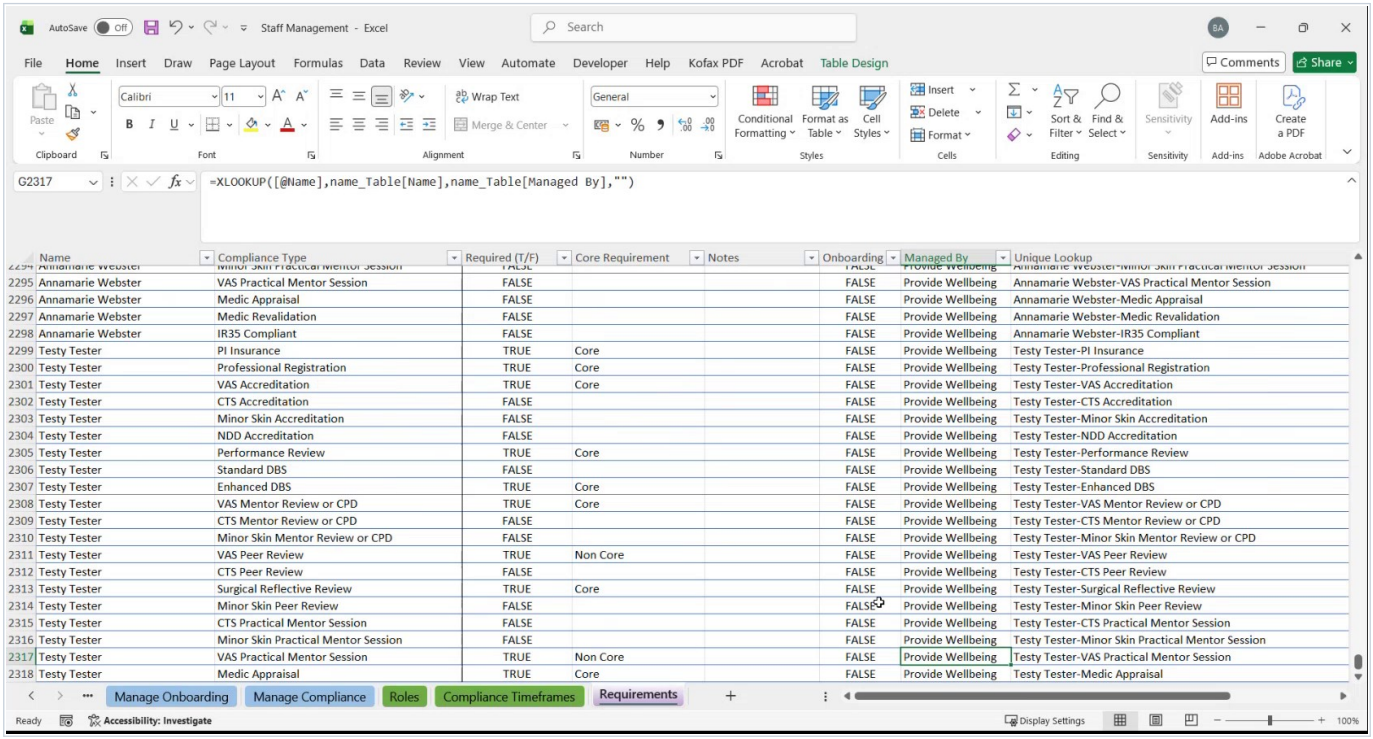


Figure 27 [00:19:13] — Staff Management.xlsm › Requirements: Managed By now resolves to Provide Wellbeing for the Testy Tester rows

Step 14 [00:19:14 – 00:20:03]: Inspect H-J and save the workbook

Action performed	Explains Unique Lookup, Compliance Timeframe and Active; saves (Ctrl+S), because the data store can only build from a *saved* tool.
Source → target	File save → Z:\...\Staff Management.xlsm.
State mutation (before → after)	Persisted state of name_Table and requirements_Table.

Shown above: Figure 28 [00:19:33] — Formula bar, I2 Compliance Timeframe: =XLOOKUP([@[Compliance Type]],Compliance_Timeframe_Table[Compliance Type…

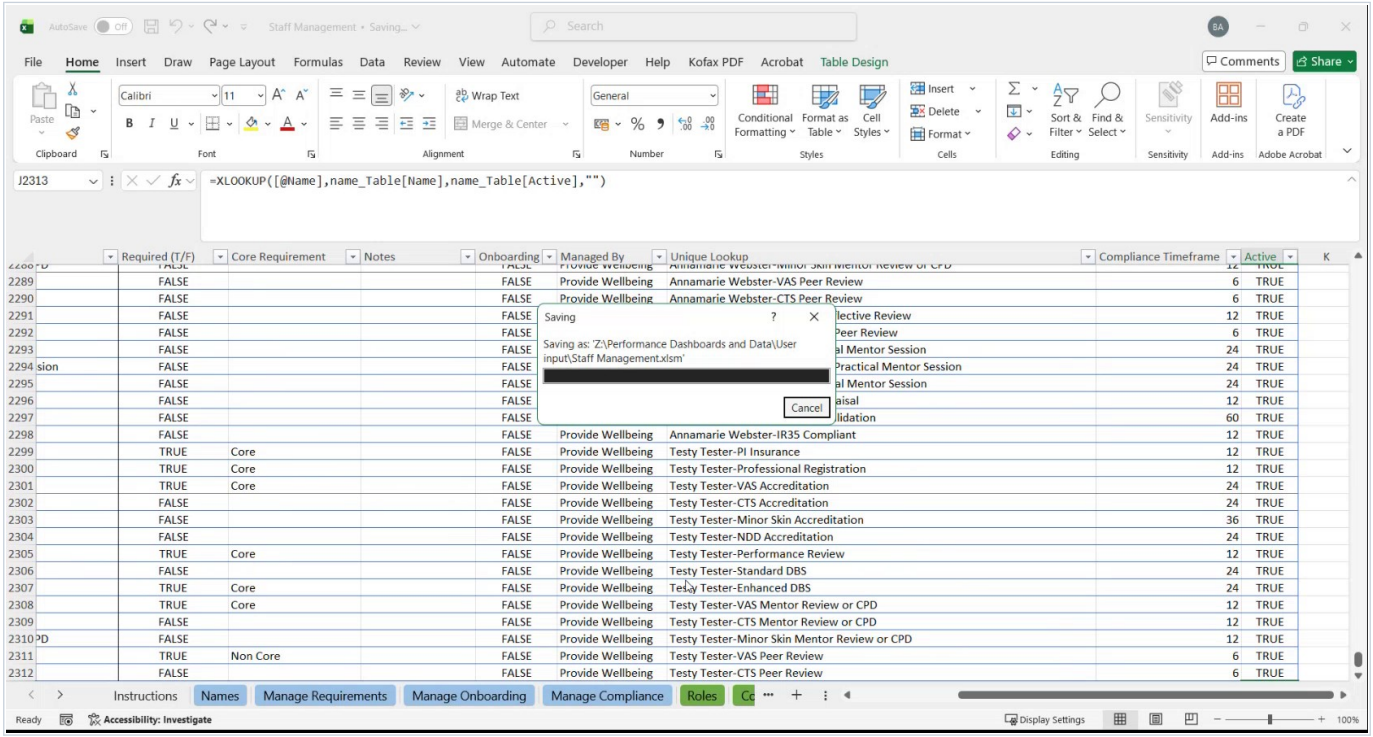


Figure 29 [00:19:53] — Staff Management.xlsm › Requirements: Save in progress ('Saving as Z:\Performance Dashboards and Data\User input\Staff Management.xlsm'); J Active = XLOOKUP into name_Table[Active]

Step 15 [00:20:03 – 00:23:24]: Enter compliance evidence on Manage Compliance

Action performed	Opens *Manage Compliance*; types “TEST” in R5 (autocompletes to Testy Tester); fills the Add grid one row per evidence item (Name, Type, Date, Status, RAG, Notes). Leaves non-applicable types out. A stray “test” entry in row 16 is overtyped. Warns that the VBA “doesn’t like it if you miss a row”.
Source → target	Manual → Add grid B9:G16; O fills automatically.
State mutation (before → after)	Rows entered (Completion Date → Time Frame): PI Insurance 01/06/2026 (12) Completed/Green “Certificate on file”; Professional Registration 11/08/2026 (12) “Copy on file”; VAS Accreditation 01/08/2026 (24); Performance Review 01/08/2026 (12); Enhanced DBS 15/07/2026 (24); Medic Appraisal 01/06/2026 (12); Medic Revalidation 01/03/2025 (60); IR35 Compliant 15/07/2026 (12). All Completed / Green.

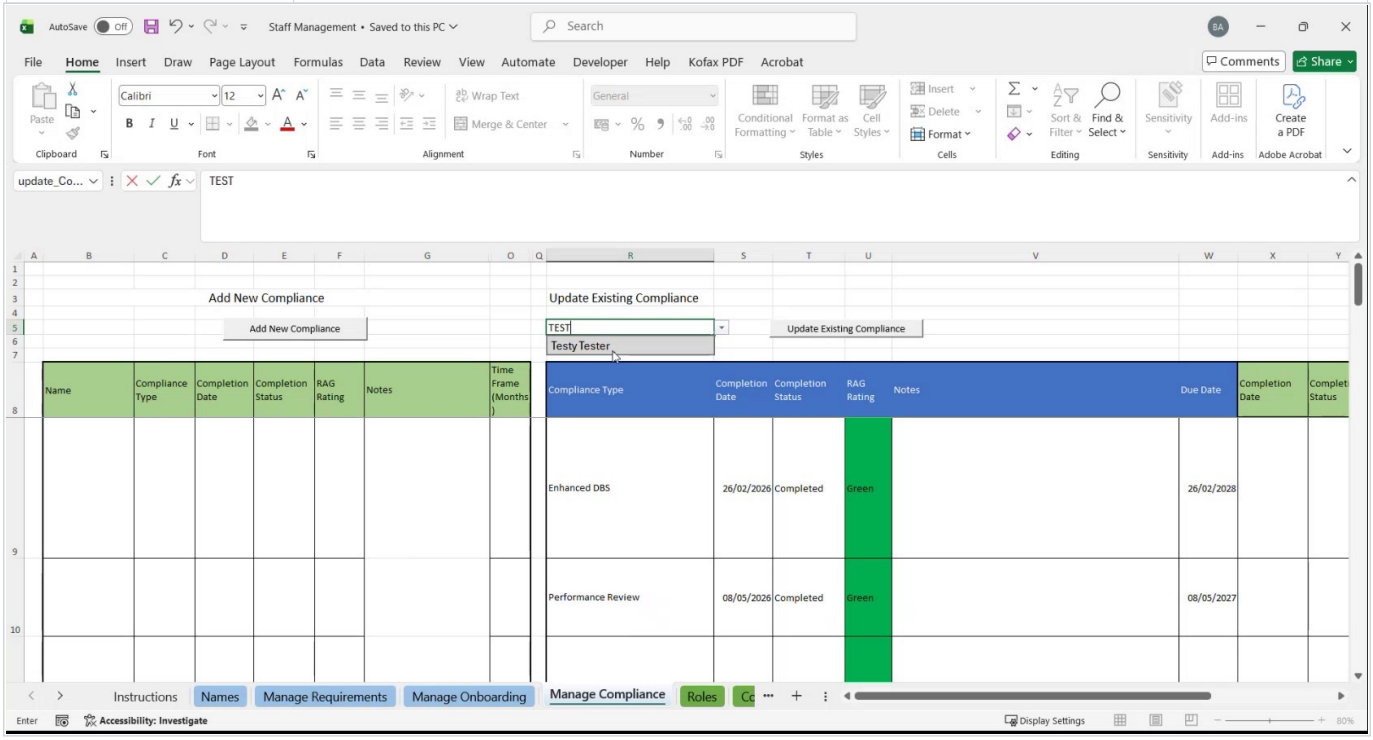


Figure 30 [00:20:13] — Staff Management.xlsm › Manage Compliance: Operator types 'TEST' into the Update Existing Compliance name cell (update_Co..., R5); autocomplete to Testy Tester

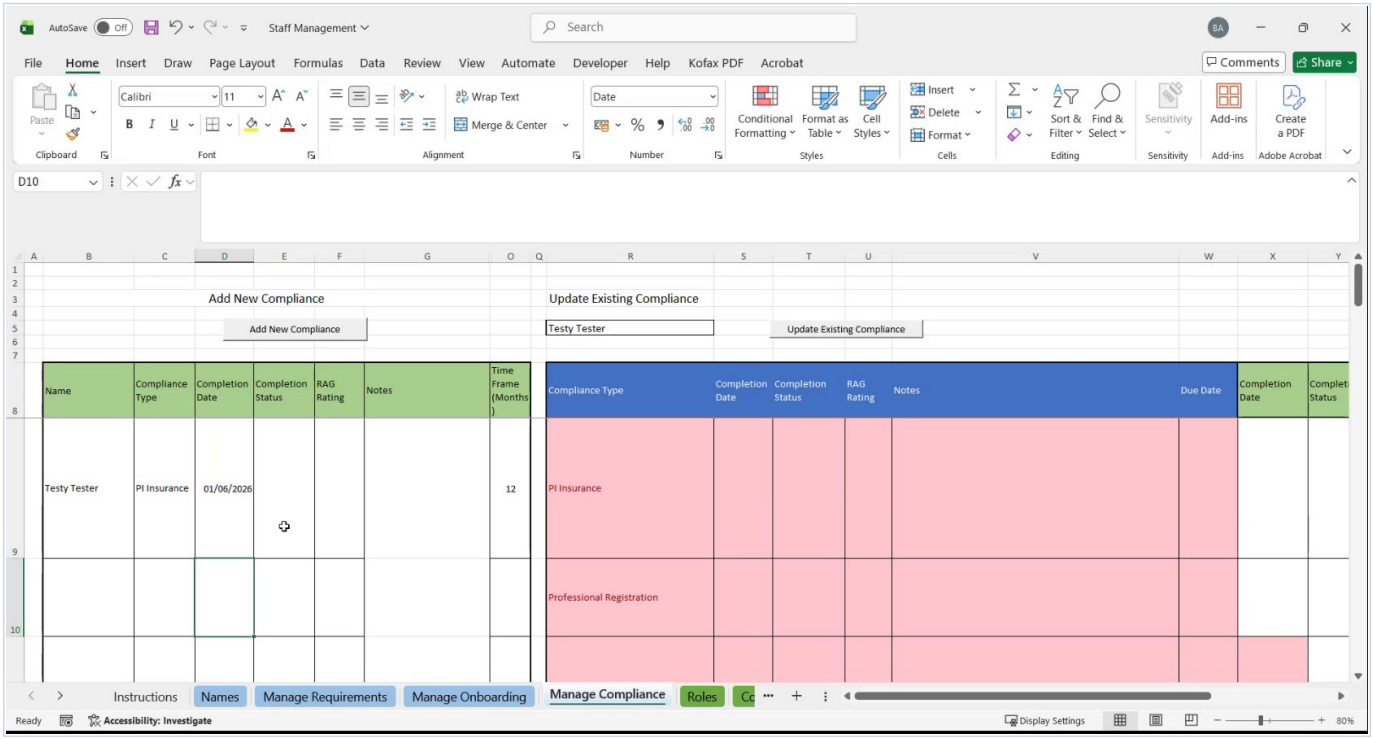


Figure 31 [00:20:33] — Staff Management.xlsm › Manage Compliance: Add New Compliance grid, row 9 PI Insurance 01/06/2026; Time Frame (O) auto-fills 12

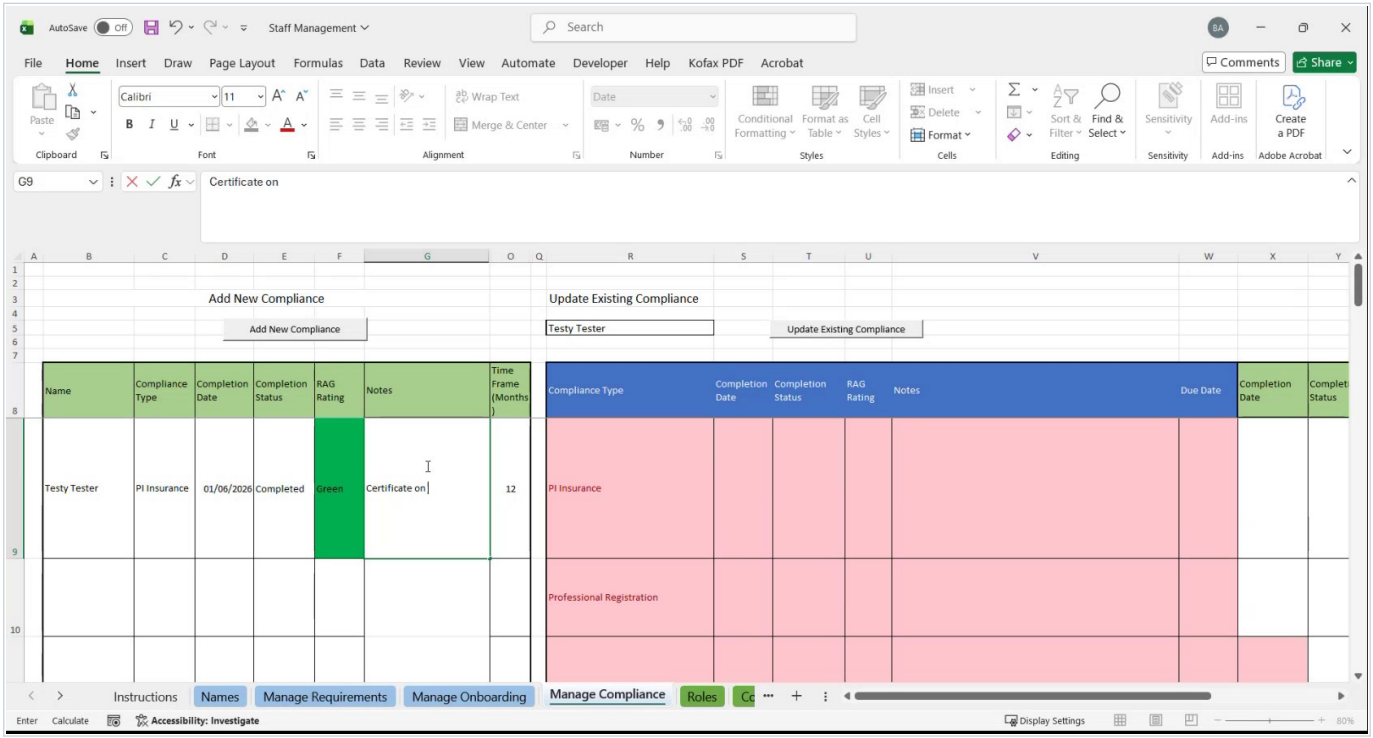


Figure 32 [00:20:43] — Staff Management.xlsm › Manage Compliance: Row 9 complete: Completed, RAG Green, Notes 'Certificate'

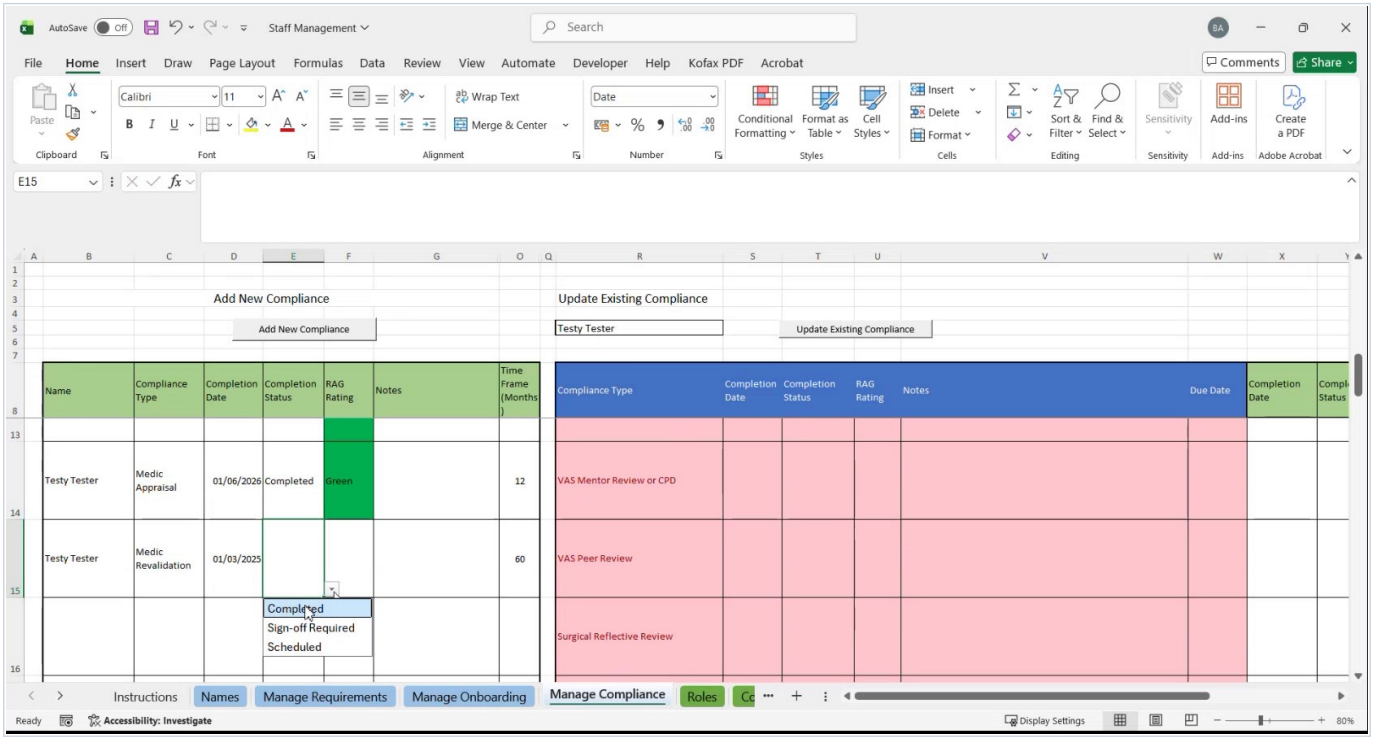


Figure 33 [00:23:02] — Staff Management.xlsm › Manage Compliance: Completion Status dropdown values: Completed / Sign-off Required / Scheduled (row 15 Medic Revalidation 01/03/2025, 60 months)

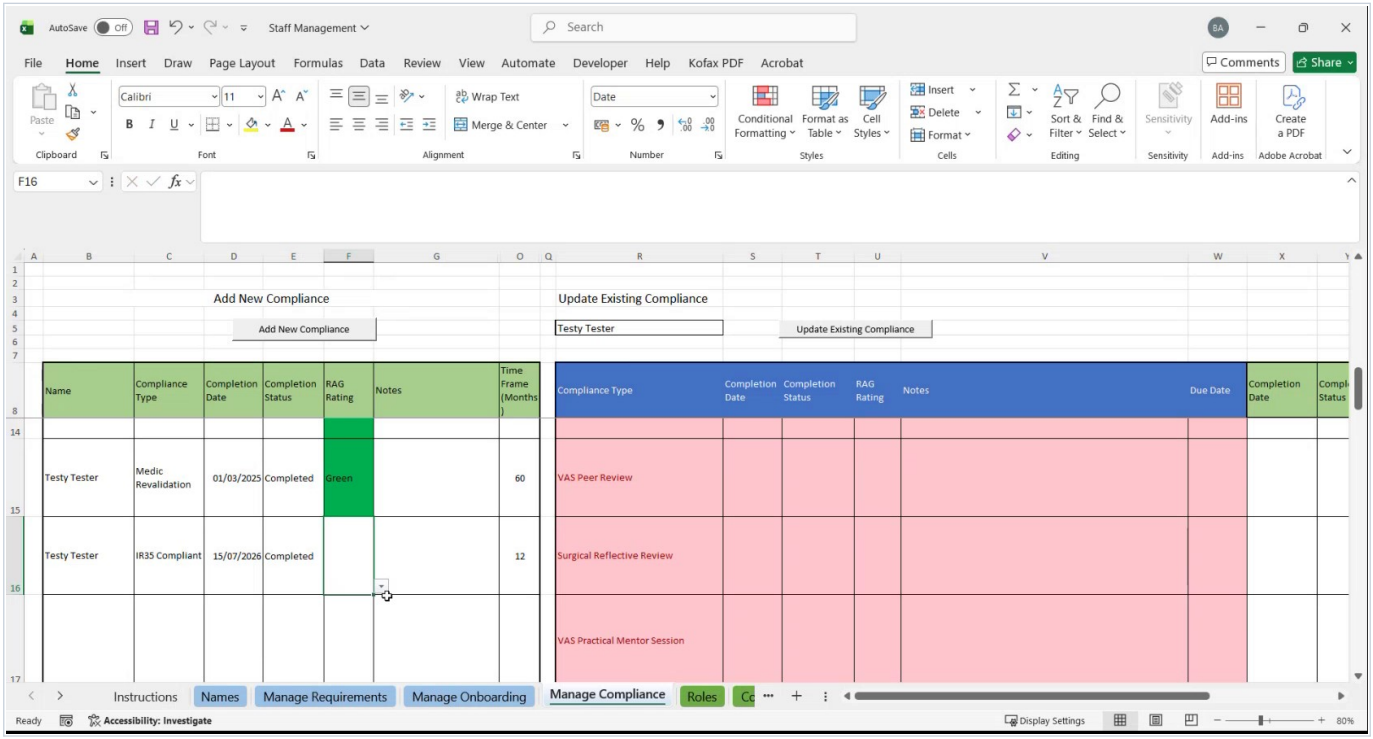


Figure 34 [00:23:23] — Staff Management.xlsm › Manage Compliance: BEFORE 'Add New Compliance': last rows of the entry grid; blue 'existing' panel still empty for Testy Tester

Step 16 [00:23:24 – 00:24:30]: Run VBA “Add New Compliance”

Action performed	Clicks *Add New Compliance*; the grid clears; the Existing panel fills from the external store.
Source → target	VBA: grid rows → append to `'Compliance Data.xlsm'!compliance_Data_Store_Table`; R:W re-evaluate by XLOOKUP on Unique Lookup 3.
State mutation (before → after)	BEFORE: S:W blank and pink for Testy Tester. AFTER: e.g. PI Insurance 01/06/2026 · Completed · Green · “Certificate on file” · Due **01/06/2027**; Professional Registration 11/08/2026 → Due 11/08/2027.

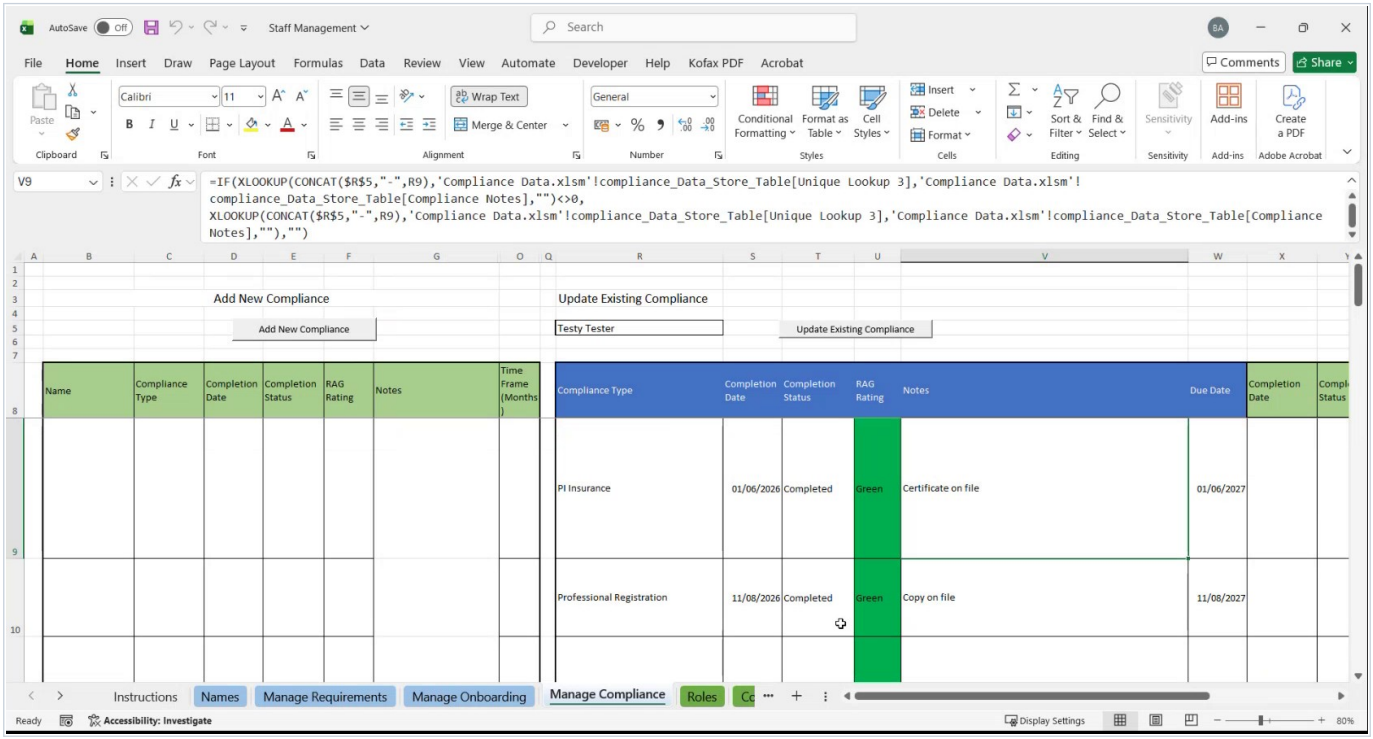


Figure 35 [00:23:53] — Staff Management.xlsm › Manage Compliance: AFTER 'Add New Compliance': entry grid cleared; blue panel now shows PI Insurance 01/06/2026 → Due 01/06/2027, Professional Registration 11/08/2026 → 11/08/2027 read back from the data store

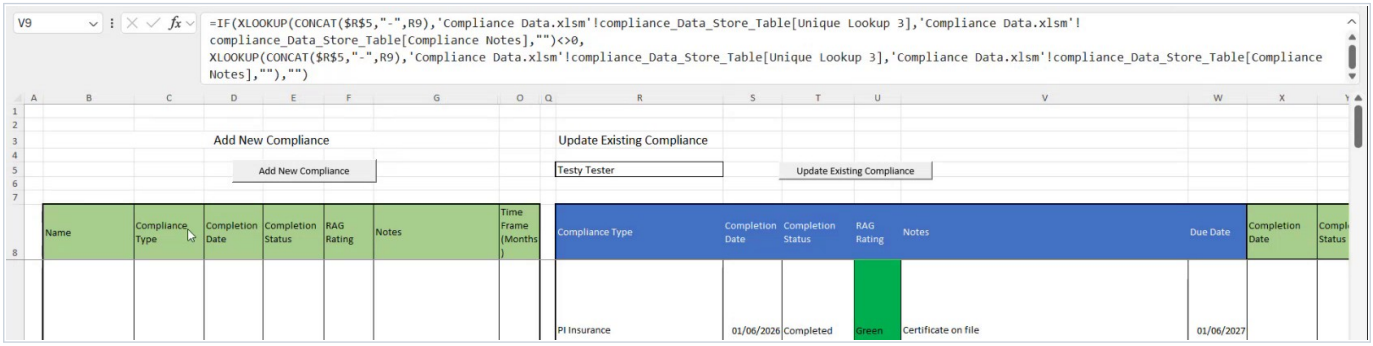


Figure 36 [00:24:03] — Staff Management.xlsm › Manage Compliance: Formula bar, V9 Notes: XLOOKUP(CONCAT(R5,"-",R9),'Compliance Data.xlsm'!compliance_Data_Store_Table[Unique Lookup 3], ...[Compliance Notes]) – cross-workbook link

Step 17 [00:24:30 – 00:26:18]: Amend an existing record via “Update Existing Compliance”

Action performed	Scenario: PI insurance actually ran from 1 May, not 1 June. Types X9 = 01/05/2026 and AB9 = 01/05/2027 (due date by hand), then clicks *Update Existing Compliance* (about 25:47). Mentions the Delete column AC.
Source → target	VBA: X:AC for the row → overwrite the matching store record in place.
State mutation (before → after)	BEFORE: S9 01/06/2026, W9 01/06/2027. AFTER: S9 **01/05/2026**, W9 **01/05/2027**; X:AC cleared. The previous value is not kept (no audit trail).

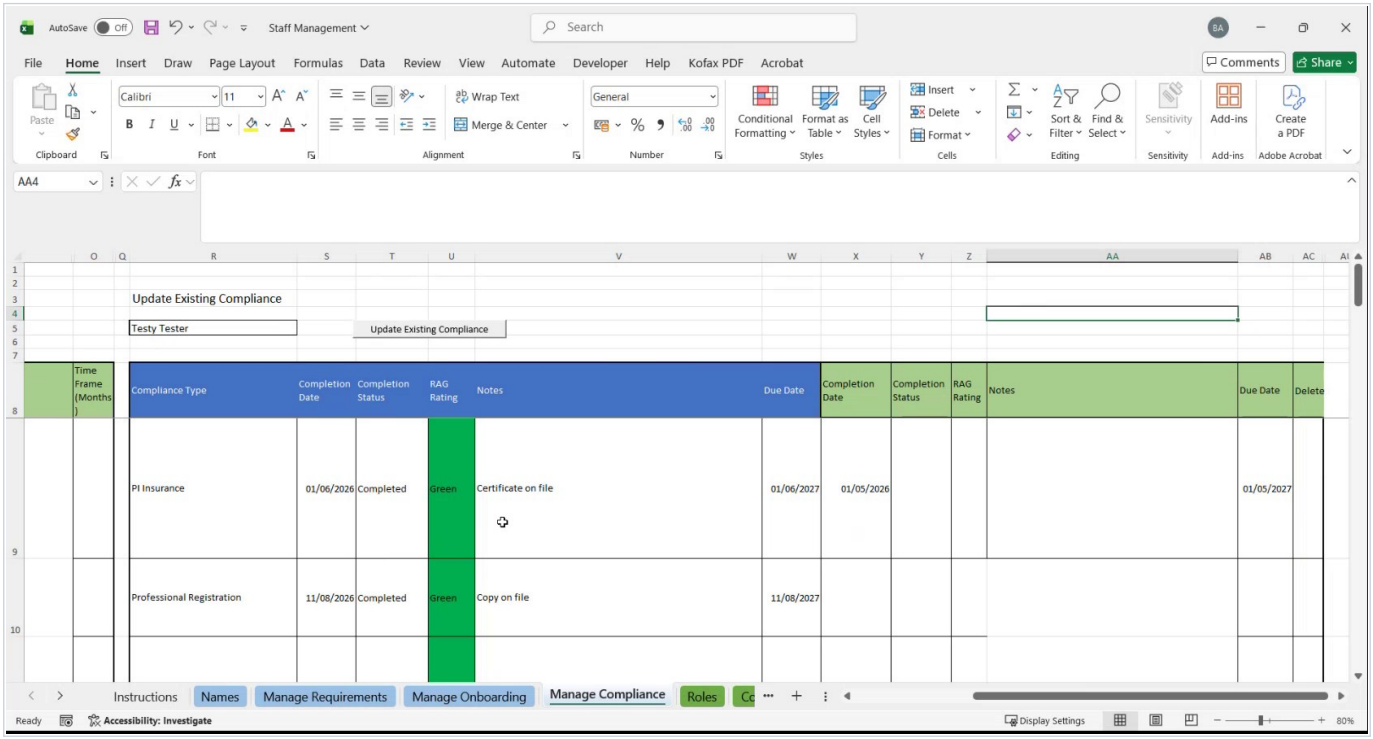


Figure 37 [00:25:03] — Staff Management.xlsm › Manage Compliance: Amendment entered in the green update panel: Completion Date X9 = 01/05/2026 and Due Date AB9 = 01/05/2027 typed by hand (Delete flag column AC)

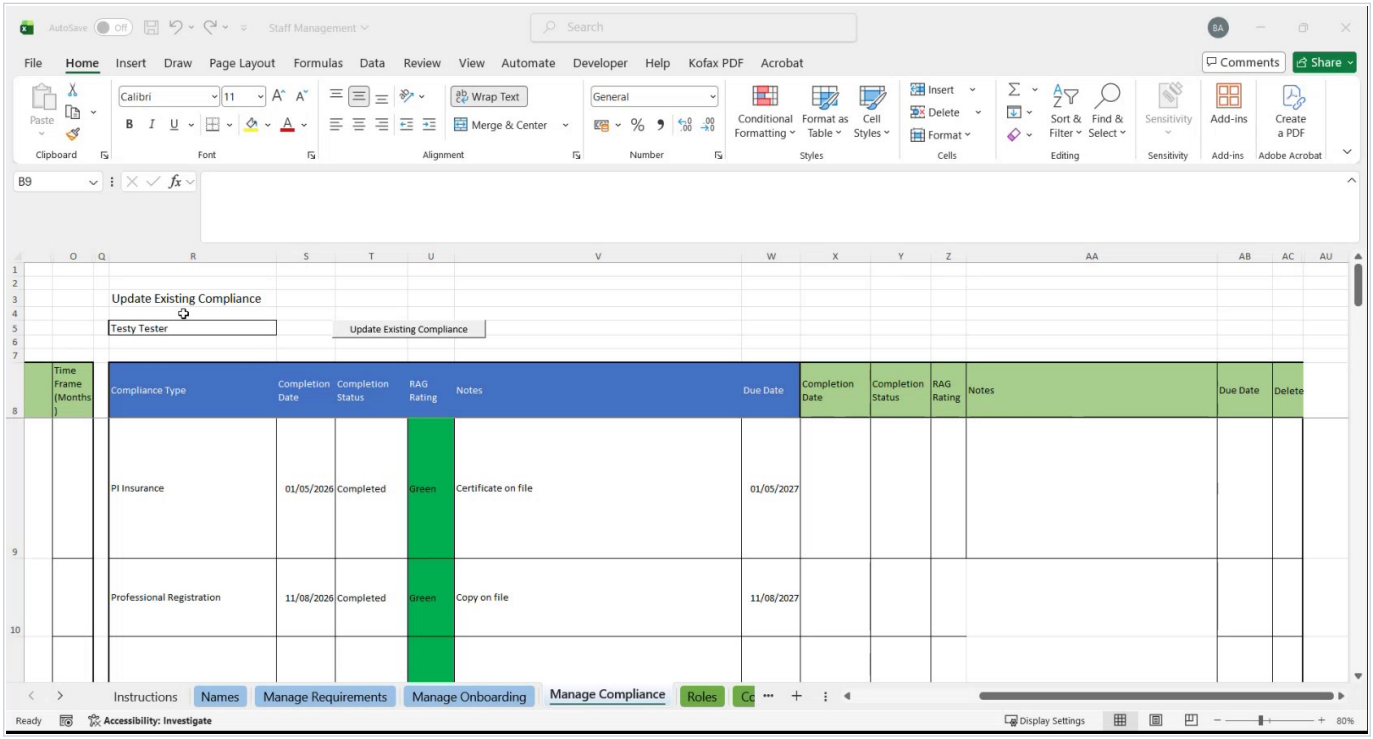


Figure 38 [00:26:03] — Staff Management.xlsm › Manage Compliance: AFTER 'Update Existing Compliance': PI Insurance now 01/05/2026, due 01/05/2027 – record overwritten in place

Step 18 [00:26:18 – 00:28:24]: Explain the monthly cycle and start the data refresh

Action performed	Describes the monthly loop (update what you can, chase staff, input replies) and the publish sequence **save data store → Refresh → save → Refresh → save**, plus updating the “last updated” tab by hand because its VBA is broken. Triggers a refresh in Compliance Data.xlsm (status: “Calculating (14 Threads)” 26:20, “Loading .Net Framework” 27:20, “Running background query” 27:27–28:35).
Source → target	PQ: all saved tools → Compliance Data model tables.

State mutation (before → after)	Not visible: the PQ output tables are in the other window.

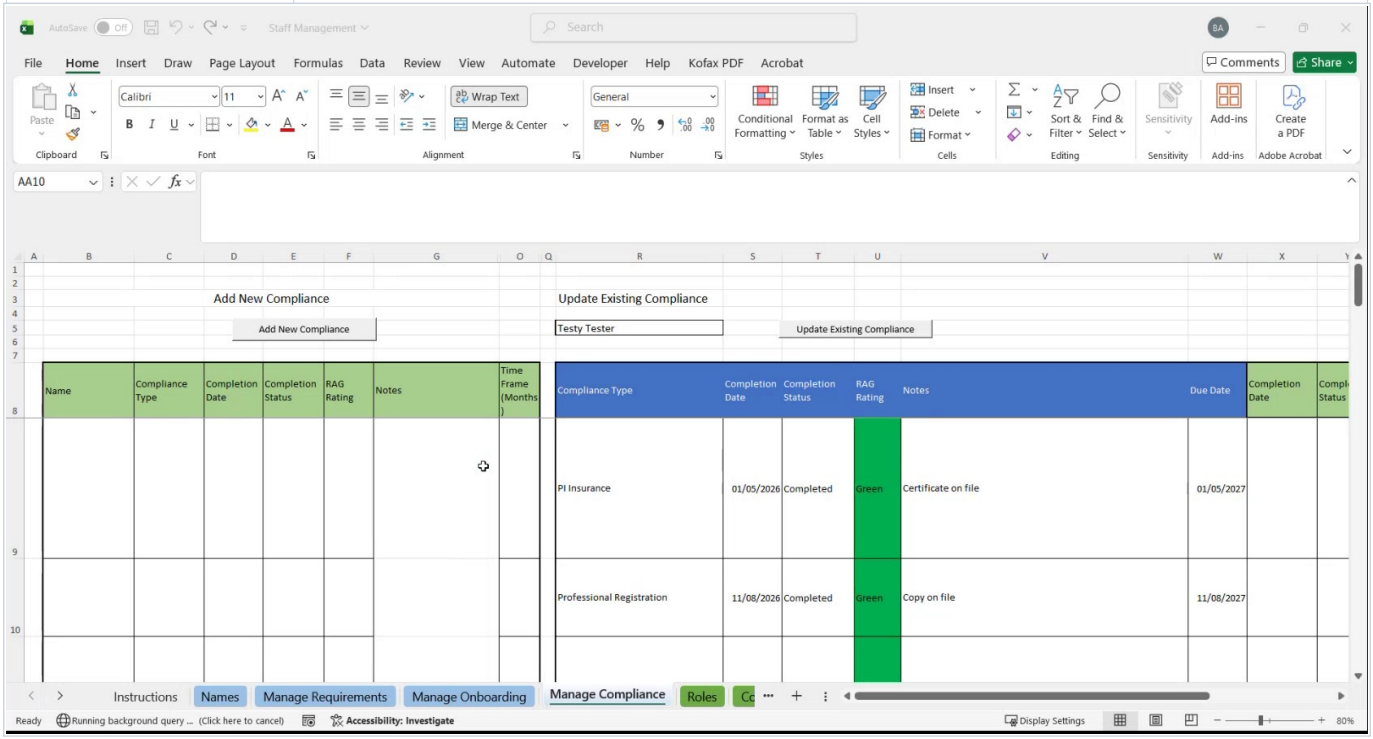


Figure 39 [00:27:33] — Staff Management.xlsm › Manage Compliance: Data refresh: status bar 'Running background query ... (Click here to cancel)' – Power Query refresh; Excel window is inactive (refresh driven from another window)

Step 19 [00:28:24 – 00:29:13]: Stop the refresh; deactivate the test person; save

Action performed	“I’m going to stop that... I don’t particularly want to add Testy Tester in any way.” Cancels the query, sets Names!N140 = FALSE, saves.
Source → target	Manual → name_Table[Active] → requirements_Table[Active].
State mutation (before → after)	BEFORE: N140 = TRUE. AFTER: N140 = FALSE (person stays in the store, flagged inactive). The 22 requirement rows and 8 store records are **not** deleted.

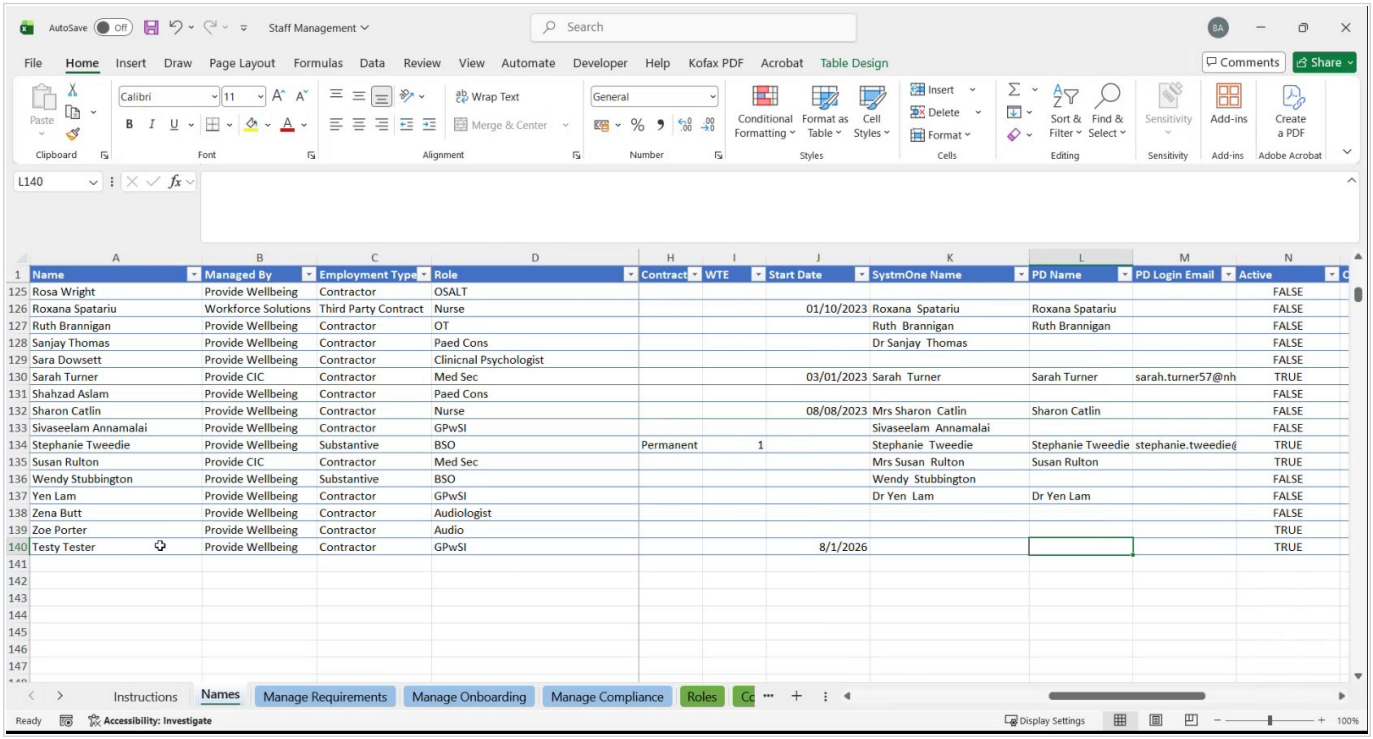


Figure 40 [00:28:53] — Staff Management.xlsm › Names: BEFORE deactivation: Testy Tester Active = TRUE; Start Date shows '8/1/2026' (different format to other rows' dd/mm/yyyy); column H 'Contract'

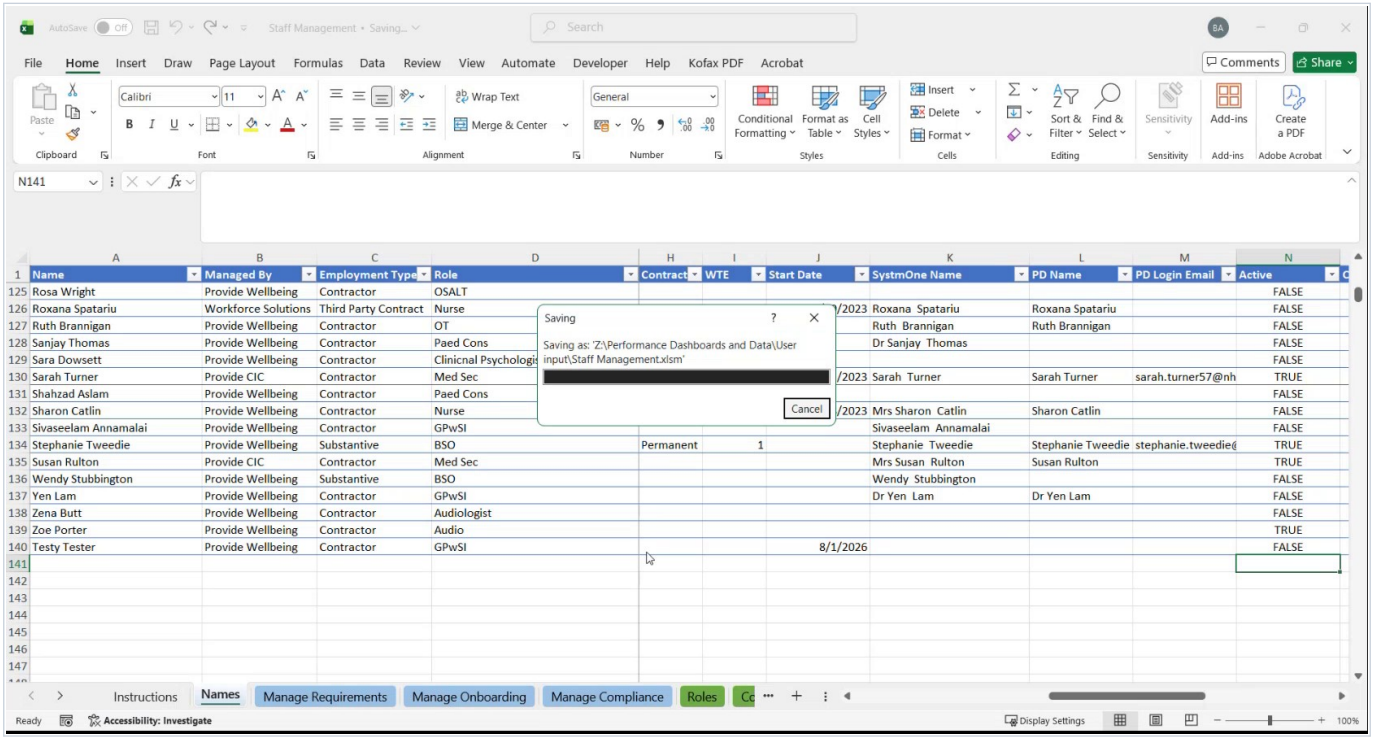


Figure 41 [00:29:03] — Staff Management.xlsm › Names: AFTER: Active = FALSE; workbook saving

Step 20 [00:29:13 – 00:29:27]: Close the workbook: VBA run-time error

Action performed	Closes Staff Management; the close macro (“do you want to update the last updated date?”) fails.
Source → target	—
State mutation (before → after)	**Run-time error ‘13’: Type mismatch** dialog (Continue disabled; End/Debug/Help). Operator dismisses it.

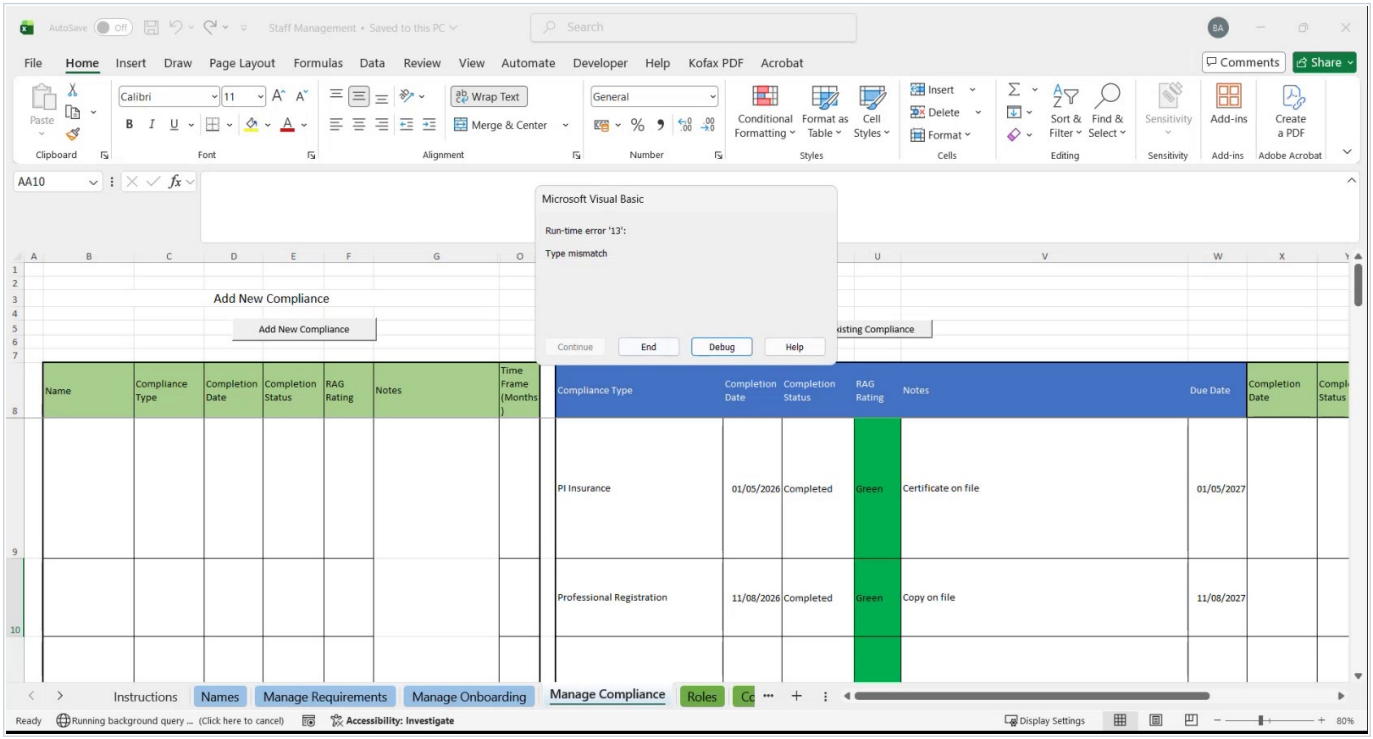


Figure 42 [00:29:23] — Staff Management.xlsm › Manage Compliance: On close: 'Microsoft Visual Basic – Run-time error 13: Type mismatch' from the last-updated prompt macro

Step 21 [00:29:27 – 00:29:45]: Decline to save; background refresh restarts

Action performed	“Save your changes to this file?” → operator chooses **Don’t Save**, explaining that saving now can corrupt the lookups if Compliance Data closes first, and it re-triggers a data update. Taskbar hover (29:38) shows a second workbook “Comp…” open.
Source → target	—
State mutation (before → after)	Workbook closed without saving the post-error state; background query running again in Compliance Data.

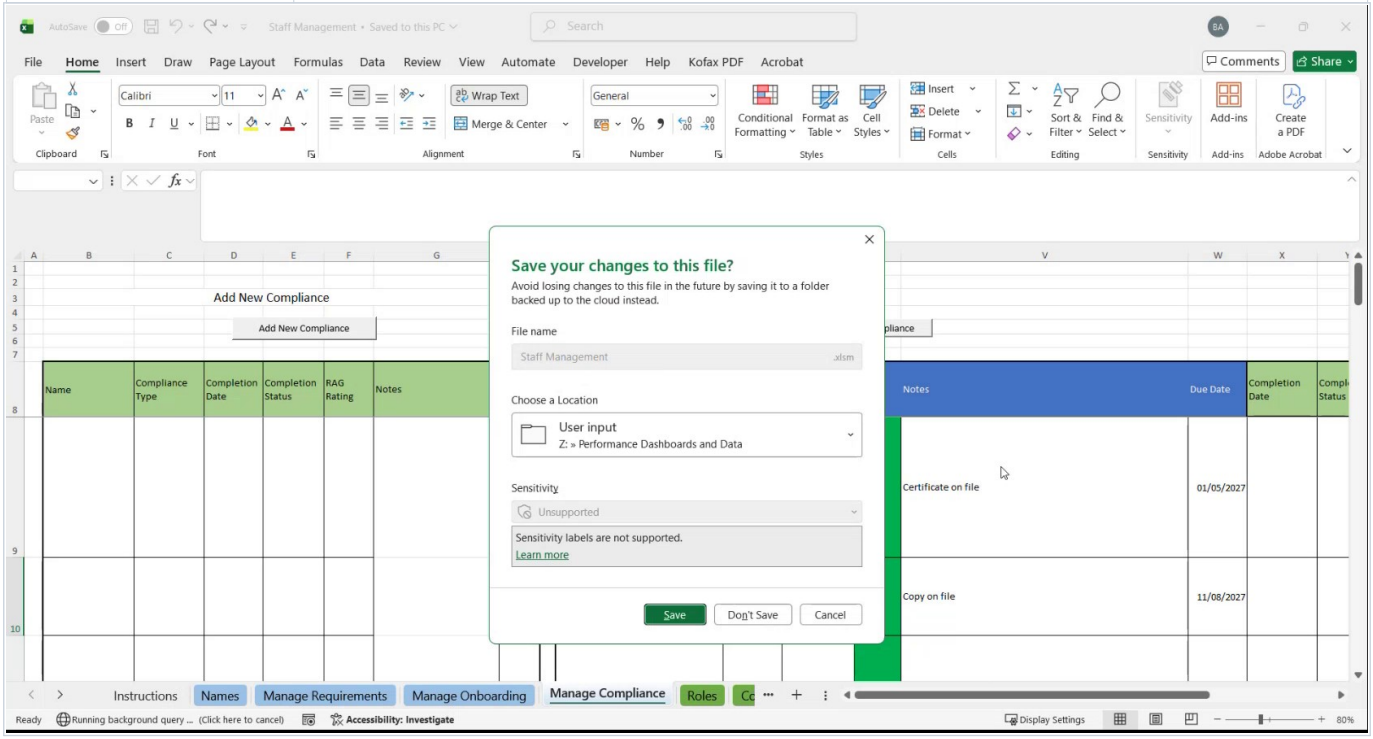


Figure 43 [00:29:33] — Staff Management.xlsm › Manage Compliance: 'Save your changes to this file?' – location Z: › Performance Dashboards and Data › User input; operator chooses Don't Save

Shown above: Figure 44 [00:29:38] — Brief hover over the Excel taskbar group: a second workbook ('Comp…' = Compliance Data.xlsm) is open, showing …

Step 22 [00:29:45 – 00:30:54]: Stop screen share and wrap up

Action performed	Camera tile only. Operator says he will send the video plus a list of compliance areas that use this method, and record the others.
Source → target	—
State mutation (before → after)	—

4. Data Quality Checks, Manual Exceptions & Edge Cases

4.1 Implicit business rules and keys

ID	Rule	Where enforced today	Evidence
BR-01	A person is identified by **Name** alone; no ID	Every XLOOKUP on name_Table[Name]	Figure 25 [00:18:43]
BR-02	Requirement identity = `Name-ComplianceType`; at most one per pair (intended)	requirements_Table[Unique Lookup]; List Missing excludes existing pairs. **Not** enforced against duplicates	Figure 21 [00:14:28]
BR-03	Compliance-record identity = `Name-ComplianceType` (Unique Lookup 3)	Cross-workbook XLOOKUP (first match)	Figure 36 [00:24:03]
BR-04	Due Date = Completion Date + Every x months (EDATE)	Add path: automatic. **Update path: typed by hand**	Figure 35 [00:23:53], Figure 37 [00:25:03]
BR-05	Only Active = TRUE people appear on the dashboard; deactivation keeps history	name_Table[Active] → requirements_Table[Active] → dashboard filter	Figure 41 [00:29:03]
BR-06	Gauge colour: ≥T2 green … <T5 red (0.9/0.8/0.7/0.6; DBS 0.98/0.9/0.8/0.7)	Thresholds on Compliance Timeframes; applied in the dashboard	Figure 15 [00:11:12]
BR-07	Onboarding items never expire (10000 months)	Sentinel value in Every x months	Figure 12 [00:09:13]
BR-08	Core flag is per person and overrides the catalogue default	requirements_Table[Core Requirement]	Figure 24 [00:18:33]
BR-09	The data store builds only from **saved** tool files	Power Query reads files on disk	Said 19:41–19:58
BR-10	Add-compliance batches must be contiguous (no blank rows)	VBA loop stops at the first blank [Inferred]	Said 20:45
BR-11	Evidence is cumulative (“build up a picture over time”) but corrections overwrite	Add appends; Update overwrites	Said 25:08, 25:52

4.2 Manual exceptions and workarounds observed

ID	Exception	How the operator handled it	Evidence
EX-01	Managed By blank → Requirements!G shows 0	Back-filled the master record (Names!B140 = Provide Wellbeing); recalculation fixed 22 rows	Figure 26 [00:19:03] → Figure 27 [00:19:13]
EX-02	Wrong completion date stored (PI from 1 May, not 1 June)	Update panel X/AB with the due date re-typed by hand, then *Update Existing Compliance*	Figure 37 [00:25:03] → Figure 38 [00:26:03]
EX-03	“Last updated” VBA does not run (error 13)	Update the Last Updated tab by hand (“not a big deal”)	Figure 42 [00:29:23]; Said 27:57–29:22
EX-04	Close-time save risks corrupting lookups	Choose **Don’t Save** at close; save only at controlled points	Figure 43 [00:29:33]
EX-05	Slow Power Query over VPN	Accept the wait; stop the query when not needed	Figure 39 [00:27:33]
EX-06	Test person must not reach the dashboard	Deactivate (Active = FALSE) rather than delete; rows remain in both workbooks	Figure 41 [00:29:03]
EX-07	Non-applicable compliance types	Set Required = FALSE at requirement stage; leave them out of compliance entry	Figure 23 [00:17:43]

4.3 Spoken vs. visual discrepancies

ID	Spoken	On screen / in data	Impact
SV-01	VAS Mentor Review or CPD: “needed, but it’s not core” (16:12)	Requirements!D2308 = **Core**	Inflates the core denominator; dashboard core % wrong for this person
SV-02	Surgical Reflective Review: “required, but it’s not core” (17:03)	Requirements!D2313 = **Core**	As SV-01. Probably a copy-paste carry-over (status bar shows paste mode)
SV-03	PI insurance “goes through to the 31st of May next year” (20:28)	Due date shown 01/06/2027 (EDATE +12)	One-day boundary ambiguity: is the due date the expiry day or the day after?
SV-04	Testy Tester is “not actually managed by anybody” (05:45)	Managed By later set to Provide Wellbeing to clear a 0	Managed By semantics unclear (contracting entity vs line manager)
SV-05	Person is a vasectomist / GPwSI (05:55)	Manage Requirements!D8 Role = “Surgical Clinician – NHS Only” (left over from previous person)	The form shows misleading context; field possibly unused

ID	Spoken	On screen / in data	Impact
SV-06	“There is the compliance table... it’s pulled all these in” (11:50)	Screen still on Staff Management!Compliance Timeframes	PQ output never shown; schema unknown
SV-07	“We would save the data store first... do one refresh, save, refresh, save” (26:49–27:43)	No visible save of Compliance Data.xlsm; one refresh started then cancelled	Real publication sequence not demonstrated
SV-08	“There are 22 compliance areas” (06:56)	Names!O shows 23 and 45 for some people against P = 22	“Added / Total” ratio is not a completeness measure
SV-09	Missing: “I think it’s saying is it in the requirements or not” (10:25)	Formula depends on `add_Requirements_Name` (one selected person), not a per-row person	Column is only meaningful for the person on the form at that moment

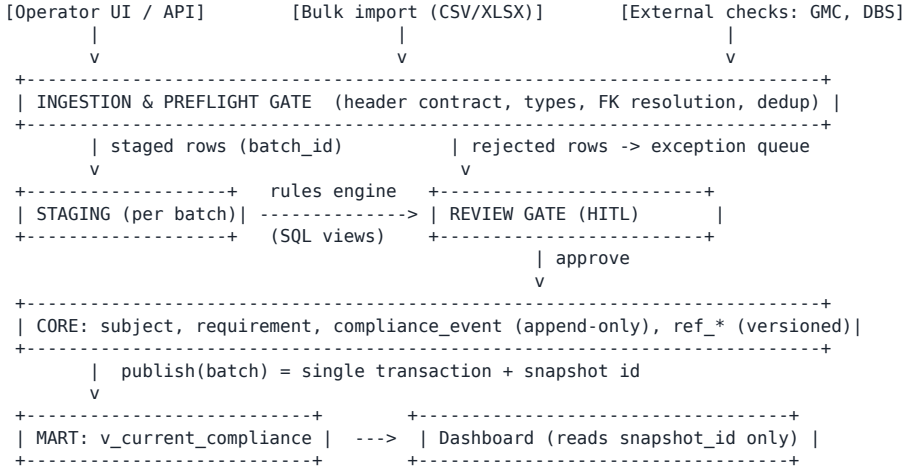
4.4 Hidden failure modes in the current process

ID	Failure mode	Mechanism	Severity
FM-01	**Key drift**	String keys (Name, Name-Type). A rename, trailing space or a duplicate name (two “John Smith”) breaks or merges records silently; XLOOKUP falls back to ""/0 with no error	High
FM-02	**First-match lookups on a history table**	Store keeps multiple records per Name-Type, but XLOOKUP returns the first (oldest) match by default. The UI may show stale evidence, and *Update* may overwrite the wrong row	High
FM-03	**Destructive updates**	Update overwrites in place; Delete flag removes. No before-image, user stamp or timestamp, so there is no audit trail for compliance evidence	High
FM-04	**Workbook coupling and close-order corruption**	Manage Compliance formulas reference an open external workbook. Closing in the wrong order or saving at close corrupts the lookups [Said 29:29–29:43]	High
FM-05	**Unsaved-state publication**	PQ reads files on disk. Forgetting to save a tool before refresh publishes stale data, and nothing warns about it	Medium
FM-06	**Non-atomic publish**	Save → refresh → save → refresh can stop half-way (cancelled query at 28:24, VBA error at 29:17), leaving the dashboard partially updated	High
FM-07	**Silent null coercion**	XLOOKUP of a blank cell returns 0 (Managed By = 0) and is passed downstream as a value	Medium
FM-08	**Batch truncation**	Add VBA stops at a blank row, so items after a gap are silently not written [Said 20:45]	Medium
FM-09	**Manual due dates on update**	AB typed by hand can disagree with Every x months; no recalculation	Medium
FM-10	**Manual RAG**	RAG is chosen, not derived from due date/status, so a lapsed item can stay Green	Medium
FM-11	**Metric denominator mismatch**	Names!O counts onboarding rows while P counts only 22 staff types (23/22, 45/22)	Low
FM-12	**Reference-data hygiene**	Roles header row contains data (“Reception” lost); “Adult”/“Adults”; Start Date formats differ; IR35 in list Q but not in the table rows seen	Low
FM-13	**UI-state-dependent formulas**	CT!J depends on whichever name is currently selected; PQ snapshots taken while a name is selected embed that state	Medium
FM-14	**Broken/undebuggable VBA**	Error 13 on close; the last-updated macro could not be fixed by an internal expert [Said 28:09]	Medium
FM-15	**Unbounded table growth and row order**	Requirements is appended unsorted (~2,320 rows); new names are appended after Z; nothing archives inactive people	Low
FM-16	**Test data left in production**	Demo person’s 22 requirement rows and 8 store records were left (inactive); the Requirements save at 19:53 was committed	Medium

5. Target Modern System Architecture Blueprint

Design goals, from the failure modes in §4.4: (1) identity by surrogate key, never by display string; (2) compliance evidence as an append-only event ledger with derived “current” state; (3) reference data versioned and effective-dated; (4) every rule deterministic and testable as SQL; (5) publication atomic and reversible; (6) human judgement captured as explicit, audited decisions. The same pattern must serve sibling domains (Estates sites, Medical Devices, Incidents), so the model is **subject-generic**.

5.1 Logical flow



5.2 Ingestion & preflight gate

- **No copy-paste.** Data enters through (a) the web form (replacing Manage Requirements / Manage Compliance), (b) a bulk upload (CSV/XLSX template) for migrations and sibling tools, or (c) connector jobs for registers (GMC, DBS update service) where licensing allows.
- **Header contract:** each upload type has a versioned JSON-Schema/Frictionless descriptor (column names, order-insensitive, required set, types). Unknown or missing columns fail the whole file before any row is staged.
- **Cell-level validation:** dates parsed in an explicit locale (fixes the 8/1/2026 ambiguity), enums checked against ref tables (Completion Status, RAG, Role, Managed By), booleans strict, text trimmed and NFC-normalised, and every row checked to resolve to an existing subject and compliance type.
- **Staging:** rows land in `stg_*` tables tagged with `batch_id`, source file hash, uploader and line number; they are never edited in place.
- **Idempotency:** re-uploading the same file hash for the same scope is a no-op. A natural-key hash per row detects duplicates inside and across batches.

5.3 Relational data model & identity resolution

Table	Purpose / key columns	Replaces
subject_type	staff | site | device | … (one engine for all user-input tools)	One workbook per domain
subject	subject_id (PK, identity), subject_type_id, display_name, managed_by_org_id (FK), employment_type, role_id (FK), start_date, active_from/active_to (replaces Active boolean), created_by/at	Names!A:N
subject_alias	subject_id, system (SystmOne | PD | Email | Payroll), alias_value, UNIQUE(system, alias_value)	Names!K:M, free-text emails
organisation	org_id, name (Provide Wellbeing, Provide CIC, Workforce Solutions)	Managed By text
ref_role	role_id, code, label, is_clinical, valid_from/to	Roles tab (fixes header defect)
ref_compliance_type	compliance_type_id, code, label, group1..n via **ref_compliance_group** (many-to-many with weight), is_onboarding, never_expires (replaces 10000)	Compliance Timeframes A, D:I, Q, S
ref_compliance_policy (versioned)	policy_id, compliance_type_id, frequency_months, default_core, threshold_1..5, **version, valid_from, valid_to, approved_by**. Immutable once published; changes create a new version	Compliance Timeframes B, C, K:O
requirement	requirement_id (PK), subject_id, compliance_type_id, is_required, is_core, notes, valid_from/to, UNIQUE(subject_id, compliance_type_id, valid_from)	requirements_Table

Table	Purpose / key columns	Replaces
compliance_event (append-only)	event_id (PK), requirement_id (FK), event_type (RECORDED | CORRECTED | VOIDED), completed_on, status_id, rag_override, notes, evidence_uri, **supersedes_event_id**, batch_id, entered_by, entered_at	compliance_Data_Store_Table + Update/Delete VBA
evidence_file	file_id, event_id, storage_uri, sha256, uploaded_by	"Certificate on file"
ingest_batch / publication	batch_id, status (STAGED → VALIDATED → APPROVED → PUBLISHED | REJECTED), snapshot_id, approver, timestamps	Save/refresh choreography
audit_log	row-level CDC (table, pk, before, after, actor, ts) via triggers or logical replication	— (none today)

Identity resolution: every inbound row resolves to `subject_id` by (1) explicit ID, (2) exact alias match, (3) normalised-name match with uniqueness check. Ambiguous or unmatched rows go to the exception queue, never to fuzzy auto-merge. Renames update `display_name` only, so no history breaks (fixes FM-01).

5.4 Transformation & business-rules engine (Excel/VBA → SQL)

Legacy artefact	Deterministic replacement
Names!O COUNTIFS / P COUNTIFS	`v_subject_coverage`: COUNT(requirements WHERE NOT is_onboarding) / COUNT(active staff policy types), on the same domain (fixes FM-11)
Names!Q Name if Active	Filter `WHERE now() BETWEEN active_from AND coalesce(active_to,'infinity')`
CT!H/I Unique Lookup 1/2 (CONCAT)	Join to `ref_compliance_group`; no string keys
CT!J Missing (UI-state XLOOKUP)	`v_missing_requirements`: subject × policy types LEFT JOIN requirement WHERE requirement IS NULL (for all subjects; parameter-free)
VBA List Missing	API `GET /subjects/{id}/missing-requirements` backed by the view above
VBA Add / Update Requirements	`POST/PATCH /requirements` → staged; on approval, close the old row (valid_to) and insert the new one (SCD-2)
Requirements!G/I/J XLOOKUPs	Joins at query time (organisation, current policy version, subject activity); nothing duplicated
Manage Compliance O (Time Frame) and W/AB (Due Date)	`due_on = completed_on + make_interval(months => policy.frequency_months)` on the policy version effective at completed_on; never typed (fixes FM-09). Boundary rule (SV-03) parameterised: due_on or due_on − 1 day
Cross-workbook XLOOKUP (first match)	`v_current_compliance`: ROW_NUMBER() OVER (PARTITION BY requirement_id ORDER BY completed_on DESC, entered_at DESC) = 1, excluding superseded/voided events (fixes FM-02)
Update Existing Compliance (overwrite)	Insert a CORRECTED event with supersedes_event_id; the original is kept (fixes FM-03)
Delete flag (AC)	Insert a VOIDED event with a mandatory reason
Manual RAG	Derived: Green if due_on > today + amber_window; Amber if within the window; Red if overdue or status ≠ Completed. An optional rag_override needs a reason (fixes FM-10)
Gauge thresholds K:O	Dashboard reads `ref_compliance_policy.threshold_n` of the published version
Power Query merge of all tools	One database; the mart view is filtered by subject_type
Change detection (history)	LAG(completed_on) OVER (PARTITION BY requirement_id ORDER BY completed_on) gives the renewal interval and a lateness KPI

Example: the current-state view the dashboard reads.

```
CREATE VIEW mart.v_current_compliance AS
WITH live AS (
  SELECT e.*, ROW_NUMBER() OVER (PARTITION BY e.requirement_id
                                 ORDER BY e.completed_on DESC, e.entered_at DESC) AS rn
  FROM core.compliance_event e
  WHERE e.event_type <> 'VOIDED'
    AND NOT EXISTS (SELECT 1 FROM core.compliance_event s WHERE s.supersedes_event_id = e.event_id))
SELECT r.requirement_id, r.subject_id, r.compliance_type_id, r.is_core,
       l.completed_on, l.status_id,
       (l.completed_on + make_interval(months => p.frequency_months))::date AS due_on,
       core.derive_rag(l.status_id, l.completed_on, p.frequency_months, l.rag_override) AS rag
FROM core.requirement r
JOIN core.subject s ON s.subject_id = r.subject_id AND s.active_to IS NULL
LEFT JOIN live l ON l.requirement_id = r.requirement_id AND l.rn = 1
JOIN ref.compliance_policy p ON p.compliance_type_id = r.compliance_type_id
 AND coalesce(l.completed_on, current_date) BETWEEN p.valid_from AND coalesce(p.valid_to, 'infinity')
WHERE r.is_required AND r.valid_to IS NULL;
```

5.5 Exception handling & human-in-the-loop controls

- **Exception queue** per batch: unresolved subject, unknown compliance type, date out of range (in the future, or before the subject's start date), missing Managed By (EX-01 becomes a blocking validation, not a silent 0), duplicate event, Core flag differing from the policy default (SV-01/02 surfaced for confirmation).
- **Requirement wizard** replaces List Missing: pre-selects types from *role templates* (e.g. GPwSI-Vasectomy ⇒ PI, GMC, VAS Accreditation, Enhanced DBS, Appraisal, Revalidation, IR35 if trading as a company). The

operator adjusts, and each deviation from the template is logged with a reason.

- **Four-eyes review gate:** batches touching core requirements, policy versions or deletions need approval by a second user (role: Compliance Approver). The reviewer sees a diff (before → after) per subject, the same view as the BEFORE/AFTER figures in §3.
- **Overrides** (RAG override, policy exception, “not applicable”) are first-class records with reason, actor, expiry date and a link to evidence.
- **Sandbox mode** for training and demos (the Testy Tester scenario): a separate schema or a `is_test` tenant, so demo data never reaches production (fixes FM-16).
- **Last-updated** becomes `publication.published_at`, shown automatically on the dashboard (fixes EX-03 / FM-14).

5.6 Atomic publication & auditability

- **Publish = one database transaction**: move APPROVED staged rows into core, write audit_log, create `snapshot_id`, set `publication.status = PUBLISHED`. Any failure rolls back everything, so the save → refresh → save sequence goes away (fixes FM-05/06).
- **Snapshot reads:** the dashboard queries only the latest published snapshot (materialised view refreshed *inside* the publish transaction, or a `snapshot_id` pointer swap), so users never see a half-built state.
- **Rollback:** a publication can be reverted by a compensating batch (void events, close requirement versions) or by re-pointing to the previous snapshot. Both are audited.
- **Row-level lineage:** every core row carries batch_id → source (form/session or file hash + line) → actor → approval, and the evidence file sha256 is stored with it.
- **Concurrency:** optimistic locking (row version) on requirement and subject edits; publications serialised per subject_type with an advisory lock.
- **Retention and security:** RBAC (Operator, Approver, Viewer, Admin), personal data limited to what is needed (emails/phones in a restricted table), and an audit log retained according to NHS records-management rules (to be confirmed).

5.7 Migration from the legacy workbooks

- Extract name_Table, requirements_Table, Compliance_Timeframe_Table and compliance_Data_Store_Table (read-only) → load into staging → run the same preflight; expect FM-01/02/11/12 issues to surface as exceptions (duplicate Name-Type keys, Managed By = 0, onboarding rows).
- Seed ref_compliance_policy v1 from the current Compliance Timeframes values, then let the business owner approve v2 with the corrected Core flags (SV/§6).
- Run in parallel for one monthly cycle: compare v_current_compliance against the Excel-built dashboard per subject and type; sign off before switch-over.

6. Gap & Ambiguity Register

ID	Timestamp	Gap / ambiguity	Why it matters	Ask of business owner
G-01	whole .srt	The supplied .srt is from a different meeting (Incidents/Datix, 13:59:33). Its Datix export, category-grouping lookup and #REF error are not in this video	That workflow (raw-export ingestion, a different shape) is unspecified	Provide the matching .mp4 for 135933 (and its .txt) for a separate spec
G-02	00:11:43–00:11:54	PQ-built “staff management compliance table” referred to but not visible (other window)	Schema of PQ outputs and the dashboard grain unknown	Share Compliance Data.xlsm (queries + tables)
G-03	00:01:50–00:02:03	“Get rid of something I did on my abortive attempt”: no visible change	Possible undocumented clean-up step	Confirm what was removed and how
G-04	00:24:03, 00:25:08	Store keeps history, yet lookups use first-match XLOOKUP on Unique Lookup 3. Which record does Update overwrite?	Correctness of “current” state	Share VBA modules; confirm whether history is kept or only the latest row
G-05	00:09:13, 00:11:12	IR35 Compliant appears in list Q and gets a 12-month timeframe, but is not in table rows 2–22 seen	Catalogue completeness	Confirm the full Compliance_Timeframe_Table (rows 43+)
G-06	00:29:38	Compliance Data shows a Unique Lookup list with 024/025/026 prefixes (0.3 s glimpse)	Unknown coding scheme	Explain the prefixes (period? area code? query ID?)
G-07	00:14:28, 00:17:43	Copy marquee and “Select destination…” during requirement entry; two Core values disagree with speech	Data-entry error vs intended values	Confirm the Core flag for VAS Mentor/CPD and Surgical Reflective Review
G-08	00:08:32–00:09:54	Core flags and Compliance Group 3/4 disputed (“Matt decided… Richard needs to make a call”); onboarding scope undecided	Drives policy v2 and dashboard grouping	Richard to approve the Core list, groups and onboarding scope
G-09	00:06:47, 00:28:53	Start Date typed as 1 Aug shows “8/1/2026” while others are dd/mm/yyyy	Locale / text-date risk at migration	Confirm the column’s cell type and format
G-10	00:20:13	Manage Compliance columns H–N hidden; Manage Requirements helper M:T “originally hidden”	Possible extra fields or logic	Unhide and share
G-11	00:13:23–00:15:05	Role (D8) source and use unknown; stale value	May be dead UI or broken lookup	Confirm whether Role drives anything
G-12	tab bar	*Instructions* and *Manage Onboarding* never opened	Onboarding workflow unspecified	Record the onboarding flow
G-13	00:20:28, 00:24:52	Expiry semantics: “to 31 May” against due 01/06; the update path needs the due date typed	Off-by-one and consistency	Define due date = expiry day or day after
G-14	00:26:49–00:27:43	Publish sequence (save store, refresh ×2, save ×2, Last Updated tab) described, not performed	Needed to replicate the outputs exactly	Walk through one full monthly publish on video
G-15	00:27:57–00:29:22	VBA ‘last updated’ macro: error 13 on close; code not shown	Root cause unknown	Share the VBA project (all modules)
G-16	00:04:27, 00:10:40	Dashboard product, gauge formula and the meaning of Threshold 1 (=1) not shown	Visual spec	Share the dashboard file / measure definitions
G-17	00:03:56–00:04:03	Whether Managed By is required (“I don’t think it is, but it might be”)	Validation rule (EX-01)	Confirm mandatory fields per subject type
G-18	00:29:29–00:29:50	The corruption mechanism if Compliance Data closes first is described only vaguely	Explains the Don’t Save practice	Describe the observed corruption symptoms
G-19	00:00:00–00:30:54	Audio quality: Whisper mis-hears names (e.g. “Dheeraj”); the TurboScribe .txt ends at ~30:00	Small wording uncertainty only	None. Wording cross-checked against both transcripts

7. What Is Required to Build the Modern System

This section lists everything needed to build the replacement described in §5: business inputs, functional and non-functional requirements, technology, data migration, testing, people and phasing. **Every item gives the reason it is required**, traced to the evidence in this report (figure, timestamp, failure mode FM-xx, exception EX-xx, discrepancy SV-xx or gap G-xx). An item with no evidence-backed reason is left out on purpose. Priority: **M** = must have for go-live, **S** = should have, **C** = could have later.

7.1 Prerequisites from the business (before any build starts)

These are inputs only the business can supply. Without them the build would copy unknown behaviour or encode unconfirmed rules.

ID	What is required	Why it is required
P-01	**Export of the full VBA project** from Staff Management.xlsm (all modules: List Missing, Add/Update Requirements, Add New / Update Existing Compliance, close/last-updated)	The macros hold the write logic, and none of it was visible. Without the code we cannot tell which store row *Update* overwrites (G-04), why a blank row stops a batch (FM-08), or what causes run-time error 13 (Figure 42 [00:29:23], G-15).
P-02	**Copy of Compliance Data.xlsm** with its Power Query definitions (M code), table list and the dashboard file	The data store and PQ outputs were never on screen (SV-06, G-02, G-06). Their schema sets the migration source and what the dashboard expects, and they are needed for the parity test in 7.6.
P-03	**Full, approved compliance-type catalogue** (all rows of Compliance_Timeframe_Table incl. IR35 and onboarding items)	IR35 Compliant appears in a list but not in the table rows seen (G-05). Every requirement, due date and gauge depends on this catalogue, so it must be complete and signed off.
P-04	**Policy decisions from the business owner (Richard)**: Core vs non-core per type; whether Compliance Groups 3/4 stay; whether onboarding is in scope	The operator disputes the current Core flags and says the grouping and onboarding scope are “for Richard to make a call on” (00:08:32–00:09:54, G-08). Building on unapproved rules would bake in disputed numbers.
P-05	**Due-date rule**: is the due date the expiry day or the day after? Must a corrected record recompute it?	Spoken “to 31 May” vs the shown 01/06/2027 (SV-03), and the due date is typed by hand on update (Figure 37 [00:25:03], FM-09). The rule decides every RAG status.
P-06	**RAG rule** (amber window, how Scheduled / Sign-off Required count)	RAG is picked by hand today (Figure 32 [00:20:43], FM-10). Deriving it automatically needs an agreed rule.
P-07	**Mandatory-field list** per subject type (e.g. is Managed By required?)	A blank Managed By silently became 0 downstream (Figure 24 [00:18:33], EX-01), and the operator is unsure whether it is essential (G-17).
P-08	**Role templates**: which compliance types apply to each role (GPwSI-Vasectomy, Nurse, HCA, …)	Today the operator decides 22 types from memory for each new person (00:14:50–00:17:53), and two Core values were entered against what was said (SV-01/02). Templates make this repeatable.
P-09	**Named owners and approvers** (data owner, compliance approver, system admin)	The four-eyes review gate (§5.5) and the publication sign-off need named people. Today one operator does everything with no second check.
P-10	**Information-governance sign-off**: DPIA, lawful basis, retention periods for DBS/indemnity/appraisal evidence	The system holds personal staff data (phones, emails, DBS dates, professional registration). An NHS-facing provider must do a DPIA and set retention before storing this centrally.
P-11	**Recordings of the remaining tools** (Incidents/Datix, Estates, Medical Devices, Onboarding)	The supplied .srt is an Incidents workflow (G-01); Instructions and Manage Onboarding were never opened (G-12). The generic model (subject_type) must be checked against these before its schema is frozen.

7.2 Functional requirements

ID	Requirement (priority)	Why it is required
FR-01	**Subject register with a system ID**; names are display-only and can be renamed (M)	Every lookup today is on the name text (BR-01). A typo, rename or a duplicate name breaks or merges records silently (FM-01).
FR-02	**Alias table** for SystmOne / PD / email identities, unique per system (S)	Names!K:M hold the same person under other system names (Figure 3 [00:02:58]). Imports and connectors need a dependable way to match a person.
FR-03	**Effective-dated activity** (active from/to) instead of a TRUE/FALSE flag (M)	Active is the main dashboard filter (00:04:16) and was flipped by hand to hide a test person (Figure 41 [00:29:03]). With dates, history stays correct for past reporting periods.

ID	Requirement (priority)	Why it is required
FR-04	**Requirement set-up wizard** pre-filled from role templates, with a reason logged for each deviation (M)	Replaces List Missing and a manual 22-row grid (Figure 21 [00:14:28], Figure 23 [00:17:43]) that produced two wrong Core flags (SV-01/02).
FR-05	**"Missing requirements" view** for all subjects, independent of what is on screen (M)	The current Missing column only works for whichever name is selected on the form (Figure 14 [00:10:23], SV-09, FM-13).
FR-06	**Compliance evidence entry**: date, status, notes and an **evidence file upload** (M)	Evidence today is a note like "Certificate on file" with no link to the document (Figure 32 [00:20:43]). Audits need the actual file and its hash.
FR-07	**Automatic due date** from the policy version in force on the completion date (M)	The due date is typed by hand on the update path (Figure 37 [00:25:03]) and can drift from the frequency (FM-09).
FR-08	**Automatic RAG** with a justified, time-limited override (M)	RAG is selected by hand (Figure 33 [00:23:02]), so a lapsed item can stay Green (FM-10).
FR-09	**Corrections and voids as new events**, keeping the original (M)	*Update Existing Compliance* overwrites the record (Figure 38 [00:26:03]), and the Delete flag removes it, leaving no audit trail (FM-03).
FR-10	**Current-state view** that always picks the latest valid event (M)	Lookups return the *first* match on a table that keeps history (Figure 36 [00:24:03], FM-02), so stale evidence can show.
FR-11	**Validation that blocks bad input** (missing mandatory fields, unknown types, future or impossible dates, duplicates) with clear messages (M)	Blanks become 0 (FM-07), batches stop at a blank row with no warning (FM-08), and date formats are mixed (G-09).
FR-12	**Bulk import** (CSV/XLSX template) through the same validation (S)	Needed for migration (§7.5) and for sibling tools and future exports such as Datix (G-01). It also removes copy-paste as a data path.
FR-13	**Review/approval gate** with a before/after diff per subject (M)	One operator changes requirements and evidence and publishes with no second check; errors like SV-01/02 reach the dashboard unseen.
FR-14	**One-click atomic publish and rollback**, with a recorded publish time (M)	Today's save → refresh → save → refresh can stop half-way (cancelled at 00:28:24, error at 00:29:17; FM-05/06). "Last updated" is filled in by hand because its macro fails (EX-03).
FR-15	**Due/overdue work queue and reminders** to staff and managers (S)	The monthly cycle is to chase clinicians and staff by hand for anything not yet updated (00:26:18–00:26:38).
FR-16	**Training/sandbox mode** separate from production (S)	The demo left 22 test requirement rows and 8 store records in the live files, hidden only by Active = FALSE (FM-16).
FR-17	**Reference-data admin** for compliance types, policy versions, thresholds, roles and organisations, with approval (M)	These lists are hand-edited cells today; the Roles header holds a data value (Figure 16 [00:12:23], FM-12), and thresholds were set by one person (00:11:03).
FR-18	**Generic subject types** (staff, site, device, …) on one engine (S)	The same tool is cloned per domain: a "Site" header on the staff form gives this away (Figure 20 [00:14:03]). The operator confirms sites and devices follow the same pattern (00:03:22).
FR-19	**Correct coverage metrics** (requirements set up vs applicable types, same scope) (S)	The current count shows 23/22 and 45/22 because it mixes onboarding rows with a staff-only total (Figure 5 [00:04:23], FM-11).
FR-20	**Dashboard feed**: published views and policy thresholds exposed to the BI tool (M)	The dashboard reads Active, the requirement and compliance tables, and Threshold 1–5 for its gauges (00:10:40, Figure 15 [00:11:12]).

7.3 Non-functional requirements

ID	Requirement	Why it is required
NFR-01	**Database-enforced integrity**: primary/foreign keys, unique constraints, check constraints	Excel cannot stop duplicate keys or orphan rows (FM-01, FM-02). The database must refuse bad data, not just a form.
NFR-02	**Row-level audit log** (who, when, before/after) retained per policy	There is currently no record of who changed compliance evidence or when (FM-03). Regulators and internal audit will ask.
NFR-03	**Single sign-on and role-based access** (Operator, Approver, Viewer, Admin)	The files sit on a shared drive (Z:\…) open to anyone with drive access. The four-eyes gate (FR-13) needs real user identities.
NFR-04	**Data protection**: encryption in transit and at rest, contact details in a restricted table, UK GDPR / NHS DSP Toolkit alignment	Staff personal data and DBS details are held (Figure 4 [00:03:13]). See P-10.
NFR-05	**Performance**: publish in seconds; UI responses under 1 s for one subject	Refresh over the VPN is slow ("five or six seconds in the office", much longer remotely: 00:27:19), and the operator cancels it.
NFR-06	**No file-locking or open-order dependency**	Lookups "can corrupt" if Compliance Data closes first, so saving at close is avoided (00:29:29, FM-04).

ID	Requirement	Why it is required
NFR-07	**Reliability**: daily backups, tested restore; target RPO ≤ 24 h, RTO ≤ 4 h (to confirm)	Today the only copy is the workbook on Z:. A corrupt save, which the operator already guards against, loses data.
NFR-08	**Maintainability**: logic as version-controlled SQL and code with automated tests	The VBA is broken and an internal expert could not debug it (00:28:09, FM-14).
NFR-09	**Observability**: job logs, publish history, failed-validation counts, alerts	Failures today are silent (FM-05/07/08) or show up as a modal error dialog (Figure 42 [00:29:23]).
NFR-10	**Accessibility** (WCAG 2.2 AA) for the entry UI	Public-sector web services are expected to meet it. The Excel status bar already flags “Accessibility: Investigate” on these sheets.

7.4 Technology components

A recommended default is given for each component; any equivalent works if it meets the reason. Defaults favour the Microsoft 365 estate shown in the recording (Teams, Excel, Power Query).

Component	Recommended default	Why it is required
Relational database	PostgreSQL 16 (or Azure SQL)	Needs keys and constraints (NFR-01), window functions such as ROW_NUMBER and LAG for current state and history (FR-10), and transactions for atomic publish (FR-14).
Application API	A typed backend service (e.g. NestJS/TypeScript or ASP.NET Core)	One place for the business rules (BR-01…11), validation (FR-11) and the review gate (FR-13), instead of logic spread across cells and VBA.
Web user interface	React single-page app (or Power Apps for a low-code route)	Replaces the Manage Requirements / Manage Compliance forms with validated forms, the wizard (FR-04) and the review diff (FR-13).
Identity provider	Microsoft Entra ID single sign-on	The organisation already works in Microsoft 365. It gives real user identities for audit and approvals (NFR-02/03) with no new passwords.
Evidence file storage	Azure Blob Storage or SharePoint library, private, with SHA-256 per file	Needed to hold the certificates and copies now noted only as “on file” (FR-06).
Ingestion/validation library	Frictionless/JSON-Schema contracts + import service	Header and cell checks for bulk imports (FR-12) so nothing enters by copy-paste.
Job scheduler	Platform scheduler (e.g. Azure Functions timer / cron worker)	For reminders (FR-15), optional register checks (GMC, DBS update service) and nightly integrity reports (NFR-09).
BI / dashboard	Existing dashboard tool (likely Power BI, to confirm, G-16) on published views	Keeps users on the dashboard they know while removing Power Query file merges (FR-20).
Source control and CI/CD	Git repository (this one) + pipeline with tests and migrations	Makes rule changes reviewable and testable (NFR-08), unlike VBA that nobody can debug.
Infrastructure as code	Bicep/Terraform	Repeatable dev, test and prod environments; the sandbox (FR-16) becomes a normal environment.
Monitoring	Application Insights / OpenTelemetry	Supports NFR-09 (logs, alerts on failed publishes).

7.5 Data migration requirements

ID	What is required	Why it is required
DM-01	Read-only extracts of name_Table (~140 rows), requirements_Table (~2,320), Compliance_Timeframe_Table, Roles, compliance_Data_Store_Table	These are the only records of who must comply with what, and of the evidence held (Figure 17 [00:12:53]).
DM-02	Cleansing rules: trim and normalise names; map Managed By 0 → NULL or a value; fix the Roles header; standardise Start Date; split multi-address emails	Known defects seen on screen (FM-07, FM-12, G-09) would otherwise move into the new database.
DM-03	Duplicate-key report (Name-Type pairs with more than one row) for business resolution	Counts of 23 and 45 per person (FM-11) and first-match lookups (FM-02) suggest duplicates and history that someone must review.
DM-04	Policy v1 = today’s catalogue values; v2 = approved values from P-04	Keeps past reporting reproducible while the corrected rules take effect from an agreed date.
DM-05	Exclude or flag test records (e.g. “Testy Tester”)	The demo left test data in production (FM-16).
DM-06	Freeze window and final delta load	The Excel tool keeps changing during the parallel run, so the last changes must be carried over.

7.6 Testing and acceptance requirements

ID	What is required	Why it is required
T-01	Unit tests for every business rule BR-01…BR-11 and every SQL view	The rules are reverse-engineered from a video; tests pin them down and catch regressions.
T-02	Golden-case tests from this recording (Testy Tester: 22 requirements, 8 events, PI correction to 01/05/2026 → due 01/05/2027)	This is the one fully observed end-to-end case (Figure 24 [00:18:33], Figure 38 [00:26:03]), so it is ideal as an acceptance script.
T-03	Parity reconciliation for one monthly cycle: new mart vs Excel-built dashboard, per subject and type; differences explained	Shows the new system gives the same answers, or that each difference is a known defect fixed (§4.4).
T-04	Negative tests: blank mandatory fields, duplicate names, bad dates, interrupted publish	These are the exact failure modes found in this workflow (FM-05/06/07/08).
T-05	User acceptance by the operator and the approver	They own the monthly process and the policy decisions (P-04, P-09).
T-06	Security and IG testing (access control, penetration test as policy requires)	Personal and DBS data are involved (NFR-03/04, P-10).

7.7 People and roles

Role	Responsibility	Why it is required
Business owner (Richard, per recording)	Approves policy, Core flags, grouping and scope	Named by the operator as the decision-maker for open policy questions (00:09:54).
Compliance SME (current operator)	Explains the process, writes test cases, runs UAT	The only person shown who knows the monthly routine and its workarounds.
Product owner / business analyst	Owns the backlog; closes the §6 gaps	19 open gaps must be closed before the schema is final.
Data engineer	Schema, migration, views, publish transaction	Most of the work is data modelling and migration (§5.3, §7.5).
Backend and frontend developers	API, rules, UI forms, wizard, review screen	Replace the VBA and form logic (FR-04…FR-14).
QA / test analyst	Automated tests, parity reconciliation	Needed for T-01…T-04.
DevOps / platform engineer	Environments, CI/CD, backups, monitoring	Needed for NFR-07…NFR-09.
Information-governance lead / DPO	DPIA, retention, access policy	Needed for P-10 and NFR-04.

7.8 Delivery phases and exit criteria

Sizes are relative (S/M/L), not time estimates; they can only be firmed up once P-01 and P-02 are received.

Phase	Scope	Exit criteria	Size
0. Discovery	Collect P-01…P-11; close the §6 gaps; confirm the dashboard tool	Signed policy v2; VBA and PQ logic documented; DPIA started	S
1. Foundation	Database schema, reference-data admin, identity/SSO, environments, CI/CD	Reference data loaded; access roles working; empty schema deployed to test	M
2. Core workflow	Subject register, requirement wizard, evidence entry with files, derived due date/RAG, corrections	Golden case T-02 passes end to end	L
3. Controls and publish	Validation, review gate, atomic publish/rollback, audit log, dashboard views	Interrupted-publish test passes; dashboard reads published views	M
4. Migration and parity	Cleansing, load, one parallel monthly cycle	T-03 differences all explained; business sign-off	M
5. Go-live and extend	Retire Excel tool (read-only archive); add reminders; onboard next subject type	Excel no longer used for Staff Management; next domain scoped	S–M

7.9 Traceability: each observed problem and the requirement that removes it

Observed problem	Evidence	Removed by
FM-01 Key drift on name strings	BR-01, Figure 25 [00:18:43]	FR-01, FR-02, NFR-01
FM-02 First-match lookup on history	Figure 36 [00:24:03]	FR-10
FM-03 Destructive updates, no audit	Figure 38 [00:26:03]	FR-09, NFR-02
FM-04 Workbook open/close coupling	00:29:29, Figure 43 [00:29:33]	NFR-06 (single database)

Observed problem	Evidence	Removed by
FM-05/06 Unsaved or half-finished publish	00:27:35, 00:28:24, Figure 42 [00:29:23]	FR-14
FM-07 Blank → 0 coercion (Managed By)	Figure 24 [00:18:33]	FR-11, P-07
FM-08 Batch stops at a blank row	00:20:45	FR-11, FR-12
FM-09 Hand-typed due dates	Figure 37 [00:25:03]	FR-07, P-05
FM-10 Hand-picked RAG	Figure 33 [00:23:02]	FR-08, P-06
FM-11 Wrong coverage denominator	Figure 5 [00:04:23]	FR-19
FM-12 Reference-data hygiene	Figure 16 [00:12:23]	FR-17, DM-02
FM-13 Screen-dependent formula (Missing)	Figure 14 [00:10:23]	FR-05
FM-14 VBA nobody can debug	Figure 42 [00:29:23], 00:28:09	NFR-08, P-01
FM-15 Unbounded, unsorted tables	Figure 24 [00:18:33]	NFR-01 (indexed tables)
FM-16 Test data in production	Figure 41 [00:29:03]	FR-16, DM-05
SV-01/02 Core flags differ from speech	Figure 24 [00:18:33]	FR-04, FR-13, P-04, P-08
EX-03 Last-updated filled in by hand	Figure 42 [00:29:23]	FR-14
Manual monthly chasing	00:26:18	FR-15

Appendix A. Figure Index

Fig	Time	File › Tab	Shows
1	00:00:05	Teams recording › (camera)	Recording starts; operator on camera before screen share (no workbook visible yet)
2	00:00:53	Staff Management.xlsm › Manage Compliance	First shared screen. Workbook open on Manage Compliance with a previous person (Emma Lindsley) loaded; full tab bar visible: Instructions, Names, Mana…
3	00:02:58	Staff Management.xlsm › Names	Names table (name_Table), columns A–D and I–N: Name, Managed By, Employment Type, Role, WTE, Start Date, SystmOne Name, PD Name, PD Login Email, Activ…
4	00:03:13	Staff Management.xlsm › Names	Names table scrolled to columns E–G: Works with, Tel Num, Email (free-text contact data, mixed formats)
5	00:04:23	Staff Management.xlsm › Names	Column N Active (TRUE/FALSE) selected; O = Compliance Types Added to Record, P = Total Compliance Types, Q = Name if Active. Note O values 23 and 45 e…
6	00:04:53	Staff Management.xlsm › Names	Formula bar, O2: =COUNTIFS(requirements_Table[Name],[@Name])
7	00:05:03	Staff Management.xlsm › Names	Formula bar, P2: =COUNTIFS(Compliance_Type,"<>") – counts the 22-item named list Compliance_Type
8	00:05:23	Staff Management.xlsm › Names	Formula bar, Q79: =IF([@Active]=TRUE,[@Name],"")
9	00:05:43	Staff Management.xlsm › Names	BEFORE: operator types new name "Testy T…" into A140, the first empty row of name_Table
10	00:07:03	Staff Management.xlsm › Names	AFTER: row 140 Testy Tester – Contractor, GPwSI, Active TRUE; calculated O = 0 (no requirements yet), P = 22, Q = Testy Tester. Managed By (B140) left…
11	00:07:33	Staff Management.xlsm › Compliance Timeframes	Compliance_Timeframe_Table columns A–H: Compliance Type, Every x months, Core Requirement, Compliance Group 1–4, Unique Lookup
12	00:09:13	Staff Management.xlsm › Compliance Timeframes	Rows 19–42: staff types end at row 22 (Medic Revalidation, 60 months); Onboarding types from row 23 use 10000 months as a 'never expires' se
13	00:10:03	Staff Management.xlsm › Compliance Timeframes	Formula bar, I7 Unique Lookup 2: =IF(CONCAT([@[Compliance Group 4]],"-",[@[Compliance Group 3]])="-","",CONCAT(…))
14	00:10:23	Staff Management.xlsm › Compliance Timeframes	Formula bar, J2 Missing: =XLOOKUP(CONCAT(add_Requirements_Name,"-",[@[Compliance Type]]),requirements_Table[Unique Lookup],…,"")…
15	00:11:12	Staff Management.xlsm › Compliance Timeframes	Unique Lookup 2 (I), Missing (J), Threshold 1–5 (K–O: 1/0.9/0.8/0.7/0.6; DBS rows 1/0.98/0.9/0.8/0.7) and the side lists Compliance Type (Q, 22 items …
16	00:12:23	Staff Management.xlsm › Roles	Roles table: Role → Clinical/Non-clinical. Header row reads 'Reception | Non-clinical' – a data row has been promoted to the header
17	00:12:53	Staff Management.xlsm › Requirements	requirements_Table (~2,300 rows): Name, Compliance Type, Required (T/F), Core Requirement, Notes, Onboarding, Managed By, Unique Lookup
18	00:13:23	Staff Management.xlsm › Manage Requirements	Manage Requirements opened: name selector add_Requirements_Name (D6) still holds the previous person (Annamarie Webster); buttons Add Requirements, Li…
19	00:13:43	Staff Management.xlsm › Manage Requirements	BEFORE: Testy Tester selected; 'Update Existing Site Requirements' panel (I–L) lists every compliance type with blank Required – nothing set…
20	00:14:03	Staff Management.xlsm › Manage Requirements	Formula bar, T11 helper: =IF(I11<>"",IF(O11<>"",O11,IF(L11<>"",L11,"")),"") – merg
21	00:14:28	Staff Management.xlsm › Manage Requirements	AFTER 'List Missing': the 22 missing compliance types now fill the Add New Requirements grid (C11:C32); status bar shows a copy marquee (&#x…
22	00:15:13	Staff Management.xlsm › Manage Requirements	Required (D11) data-validation dropdown open with TRUE / FALSE for PI Insurance (Requirement column E offers Core / Non Core)
23	00:17:43	Staff Management.xlsm › Manage Requirements	Grid complete down to IR35 Compliant = TRUE; status bar 'Select destination and press ENTER or choose Paste' shows values being copy-pasted …
24	00:18:33	Staff Management.xlsm › Requirements	AFTER 'Add Requirements': 22 Testy Tester rows appended (2299–2320). Managed By returns 0 because Names!B140 is blank
25	00:18:43	Staff Management.xlsm › Requirements	Formula bar, G2299 Managed By: =XLOOKUP([@Name],name_Table[Name],name_Table[Managed By],"")
26	00:19:03	Staff Management.xlsm › Names	Manual fix: Managed By dropdown on Names!B140 set to 'Provide Wellbeing'
27	00:19:13	Staff Management.xlsm › Requirements	Managed By now resolves to Provide Wellbeing for the Testy Tester rows
28	00:19:33	Staff Management.xlsm › Requirements	Formula bar, I2 Compliance Timeframe: =XLOOKUP([@[Compliance Type]],Compliance_Timeframe_Table[Compliance Type],Compliance_Timeframe_Table[Every x mon…
29	00:19:53	Staff Management.xlsm › Requirements	Save in progress ('Saving as Z:\Performance Dashboards and Data\User input\Staff Management.xlsm'); J Active = XLOOKUP into name_Table[Activ
30	00:20:13	Staff Management.xlsm › Manage Compliance	Operator types 'TEST' into the Update Existing Compliance name cell (update_Co…, R5); autocomplete to Testy Tester
31	00:20:33	Staff Management.xlsm › Manage Compliance	Add New Compliance grid, row 9 PI Insurance 01/06/2026; Time Frame (O) auto-fills 12

Fig	Time	File › Tab	Shows
32	00:20:43	Staff Management.xlsm › Manage Compliance	Row 9 complete: Completed, RAG Green, Notes 'Certificate'
33	00:23:02	Staff Management.xlsm › Manage Compliance	Completion Status dropdown values: Completed / Sign-off Required / Scheduled (row 15 Medic Revalidation 01/03/2025, 60 months)
34	00:23:23	Staff Management.xlsm › Manage Compliance	BEFORE 'Add New Compliance': last rows of the entry grid; blue 'existing' panel still empty for Testy Tester
35	00:23:53	Staff Management.xlsm › Manage Compliance	AFTER 'Add New Compliance': entry grid cleared; blue panel now shows PI Insurance 01/06/2026 → Due 01/06/2027, Professional Registration 11/...
36	00:24:03	Staff Management.xlsm › Manage Compliance	Formula bar, V9 Notes: XLOOKUP(CONCAT(R5,"-",R9),'Compliance Data.xlsm'!compliance_Data_Store_Table[Unique Lookup 3], ...[Complian...
37	00:25:03	Staff Management.xlsm › Manage Compliance	Amendment entered in the green update panel: Completion Date X9 = 01/05/2026 and Due Date AB9 = 01/05/2027 typed by hand (Delete flag column AC)
38	00:26:03	Staff Management.xlsm › Manage Compliance	AFTER 'Update Existing Compliance': PI Insurance now 01/05/2026, due 01/05/2027 – record overwritten in place
39	00:27:33	Staff Management.xlsm › Manage Compliance	Data refresh: status bar 'Running background query ... (Click here to cancel)' – Power Query refresh; Excel window is inactive (refresh driven...
40	00:28:53	Staff Management.xlsm › Names	BEFORE deactivation: Testy Tester Active = TRUE; Start Date shows '8/1/2026' (different format to other rows' dd/mm/yyyy); column H &#x
41	00:29:03	Staff Management.xlsm › Names	AFTER: Active = FALSE; workbook saving
42	00:29:23	Staff Management.xlsm › Manage Compliance	On close: 'Microsoft Visual Basic – Run-time error 13: Type mismatch' from the last-updated prompt macro
43	00:29:33	Staff Management.xlsm › Manage Compliance	'Save your changes to this file?' – location Z: › Performance Dashboards and Data › User input; operator chooses Don't Save
44	00:29:38	Compliance Data.xlsm › (taskbar preview)	Brief hover over the Excel taskbar group: a second workbook ('Comp...' = Compliance Data.xlsm) is open, showing a Unique Lookup list prefixed ...

Appendix B. Key timestamped utterances (re-transcribed)

Time	Utterance (lightly cleaned)
00:00:58	I’m going to run you through how we update the data store for the majority of the compliance dashboard.
00:01:14	This particular area of the compliance data is called staff management.
00:02:14	...reads and writes from the data store. So it’s kind of a semi-automated update process.
00:02:54	First of all, to fill in this tab that says Names.
00:04:16	This column here, which is Active... if it’s true, it will be featured in the visuals on the dashboard.
00:05:40	We’ll call the person Testy Tester... a freelance contractor, so they’re not actually managed by anybody.
00:06:56	We haven’t set any compliance areas up yet. There are 22 compliance areas we could conceivably set up.
00:07:16	...set up something called compliance timeframes.
00:08:41	Performance review should be core. Matt very much decided these on his own.
00:10:19	XLOOKUP, CONCAT, add requirements name, compliance type... I think it’s saying is it in the requirements or not.
00:10:40	If it’s above 0.9 it’s going to be green; if it’s below 0.6 it’s going to be red, and shades in between.
00:11:21	When the data is refreshed for the first time, these compliance timeframe tables are built through power queries.
00:14:18	If I click List Missing, Visual Basic will show me for Testy Tester which of these I haven’t set up.
00:17:53	So now I’m going to write this into the requirements data, which is done by Visual Basic.
00:19:00	So actually let’s add Provide Wellbeing in there.
00:19:41	Now I’m going to save that, because when I refresh my data... it can’t if I’ve not saved it.
00:20:45	I can do these all in one go if I want. It doesn’t like it, the Visual Basic, if you miss a row.
00:23:31	We’ll now use the Visual Basic to add the record to the compliance data.
00:24:30	If we input some data and it happens to be wrong... we can use this section here to alter it.
00:25:52	That has now amended that to the 1st of May, so that has overwritten that data.
00:26:49	We would save the data store first...
00:27:35	We would do one refresh, we would then save, we would then do another refresh, and then save.
00:27:57	There is Visual Basic in the tools that should do this automatically... but some of the Visual Basic’s not working.
00:28:46	I’m just going to make Testy Tester not active.
00:29:29	We don’t save at this stage because... it closes the compliance data before... these formulas, which are just lookups, can corrupt.